

Realm Studio

Help File and User Manual

Table of Contents

Welcome to Realm Studio 3
Welcome 3
Support 3
Installing Realm Studio 3
Computer Requirements: 3
Installation: 3
Using Realm Studio to Make Maps 6
An Overview of Realm Studio 6
Realm Studio Features 6
How Realm Studio Works 7
When Realm Studio Starts 7
Configuring Your Realm 7
The Realm Studio Main Window 9
The Title Bar Controls 10
The Main Menu Bar 11
The Application Tools 11
The File Menu 12
The Edit Menu 14
The Realm Menu 15
The Assets Menu 20
The Theme Menu 22
The Options Menu 23
The Help Menu 26
Painting Maps with Realm Studio 27
The Realm Studio Main Window 27
Color Selection in Realm Studio 28
The Realm Studio Tabs 30
The Water Tab 41
The Paths Tab 44
The Symbols Tab 46
Labels Tab 53

Creating Box Frames 56
Overlays Tab 57
Regions Tab 60
Drawing Tab 62
The Description Editor 66
Zooming, Panning, and Mouse Functions 68
Realm Studio Assets and Symbol Collections 69
Creating a Symbol Collection 69
The 3D World Globe View 77
Creating the World Globe Scene 78
Taking a Picture of your Scene 84
Recording AVI Video of the Scene 85
Viewing Other Worlds 85
A Final Note on the 3D World Globe View 85
Copyright and Licensing Information 86
Technical Information 86
Realm Studio Implementation 86
The Layers of a Realm Studio Map 87
File Formats 87

Welcome to Realm Studio

Welcome

Thank you for your interest in Realm Studio. Realm Studio is an open-source Windows application intended for making fantasy maps for gamers, world builders, authors, and anyone else that would like to make nice-looking maps quickly and easily.

Using Realm Studio is completely free of charge. Just download RealmStudio.msi, then double-click it to install the Realm Studio application and a default set of assets (symbols, textures, frames, boxes, and so on) for your maps. More information on map assets is in the sections below.

Support

Because Realm Studio is open source and free, support is limited. This Help file provides all the information needed to create maps using Realm Studio, but if you have questions, feature suggestions, or bugs to report, please email support@brookmonte.com.

Installing Realm Studio

Computer Requirements:

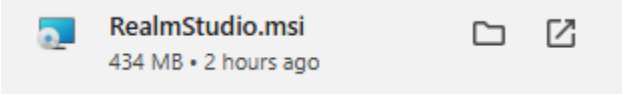
- Realm Studio requires Windows 10 (version 10.0.26100.0) at minimum.
- Realm Studio uses the graphics card (GPU) of the computer to speed up drawing of your Realm Map and 3D models, so your computer should have a good graphics card, equivalent to an Nvidia GTX 1080ti or better. Less capable graphics cards may still work, but with reduced performance.
- Your computer will probably need at least 8GB of memory, and more is better. 16GB is enough for most maps, and 32GB or more is ideal. Large maps with lots of features require more memory.
- The Assets installed with Realm Studio require about 280 MB of hard drive space on the same hard drive as your `Documents` folder. By default, Realm Studio creates a RealmStudio subdirectory within your Documents folder to store assets, maps, and other data used by the program. Depending on the number of Realm Maps that you create, that folder can use a significant amount of space, so you should plan on Realm Studio taking up as much as 1GB of hard drive space.
- If you have previously installed Realm Studio, ***keep a backup copy of all your assets*** (symbol collections, etc.) as they will be overwritten during installation. You can restore them to the Realm Studio Assets folder from your backup.

Installation:

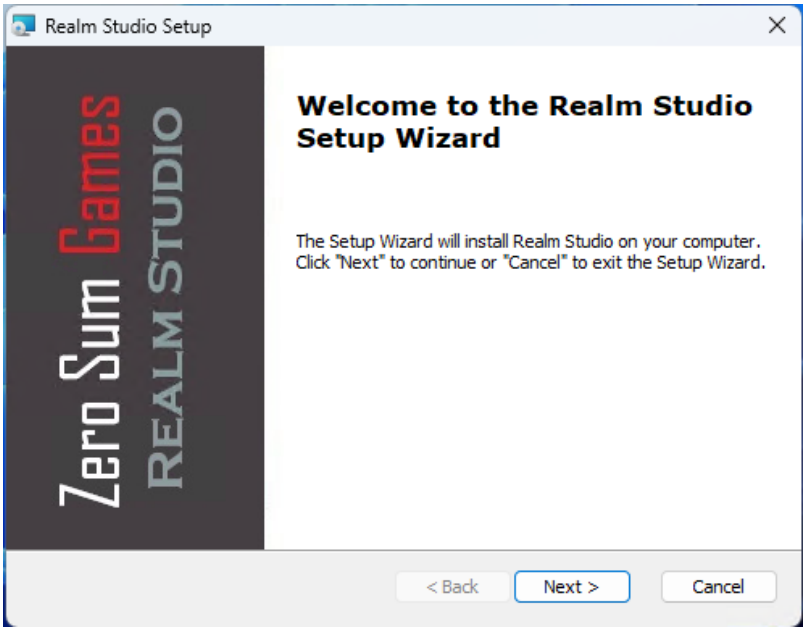
- Download the RealmStudio.msi installer file from Github or from the Microsoft Store. The RealmStudio.msi file is located at https://github.com/PeteNelson372/RealmStudio under Releases or on the Microsoft Store at

https://apps.microsoft.com/detail/xp8lcmjbk0zfxw?hl=en-US&gl=US. Only the latest release is supported.

- The Realm Studio source code is also hosted on Github. If you are a developer and want to download and build Realm Studio yourself, the source code is located at https://github.com/PeteNelson372/RealmStudio. You will need Visual Studio 2022 to build the application.
- If you install Realm Studio from the Microsoft Store, it will be installed immediately. If you download the RealmStudio.msi file from Github, you will need to run the installation program. After the RealmStudio.msi file is downloaded, run it by double-clicking it from File Explorer or from the download dropdown in your browser:



- Since Realm Studio is a new application, if you download the RealmStudio.msi file from Github and run it, it is likely that you will get a security warning from Windows Defender, saying that it has prevented you from running an unknown application. If that happens, you can override Windows Defender to run the RealmStudio.msi installer.
- Use the installation program to install the Realm Studio application and mapmaking assets.



- The Realm Studio installation program (which is launched by the RealmStudio.msi file when you run it) will install Realm Studio and some mapmaking assets to your computer. The mapmaking assets are installed to your Documents folder in a subfolder called RealmStudio/Assets.
- Realm Studio requires Windows .NET 8.0 to run. If your computer does not have the .NET 8.0 libraries installed, the Realm Studio installation program will launch installers to install them for you.

- Once the Realm Studio installation program completes running, you can launch Realm Studio and begin making maps. The rest of this document goes over how to use Realm Studio to make maps, how to install additional assets, and other information you may need. If you have any problems, find bugs, or have suggestions for improvements, please email support@brookmonte.com or open an issue on Github at https://github.com/PeteNelson372/RealmStudio/issues.
- Thank you! Your feedback and support for Realm Studio are very much appreciated. Have fun making beautiful maps!

Using Realm Studio to Make Maps

An Overview of Realm Studio

Realm Studio Features

Realm Studio was designed to make creating beautiful maps easy and fast.

Some of the Realm Studio Features:

- Quickly style your maps using a variety of attractive themes provided with Realm Studio, or define your own themes
- "Paint" landmasses on your map with automatic generation of attractive shoreline textures
- Automatic generation of random continents, islands, archipelagos, atolls, regions, or an entire world
- Draw freshwater lakes and rivers, and automatically generate random lakes
- Set textures for the map background and ocean with scaling, mirroring, and opacity
- Draw paths (roads, tracks, etc.) with a large variety of styles. Pin roads to be straight horizontally or vertically or pinned to any angle in 5-degree increments.
- Place symbols individually or "paint" groups of symbols to create forests, mountain ranges, and so on
- Automatic color randomization of symbols to make them look more realistic
- Use symbols in a variety of raster formats or use SVG vector symbols that can be scaled to any size without looking blocky
- Create attractive map labels using any text font installed on your computer or any of a set of fonts installed with Realm Studio
- Style labels using label presets that are part of a theme or define your own label styles
- Create curved labels that follow an arc or a freeform curve
- Add a frame to your map that can be scaled and tinted
- Add a square or hex grid to your map; change color, size and line thickness
- Easily measure distances and areas on your map
- Draw regions on your map in various styles. Pin region boundaries to landform coastlines easily and instantly
- Easily create grayscale heightmaps of your landforms and view the heightmaps in 3D, with near real-time updates to the 3D view as you paint the heightmap
- View your world map as the 3D globe and record video of it rotating in space, and add a variety of effects: starfield backgrounds, sun, moons, clouds, blur, sepia tone, grayscale, bleach bypass, and more
- Use the built-in 3D viewer to view 3D models in a variety of common formats
- Make maps from 500x500 pixels up to 8192x8192 pixels
- Easily add new assets (symbols, boxes, frames, etc.) to use on your maps

- Use a powerful name generator to create a random name for objects on your map
- Create and save descriptions of objects on your map, with AI integration to assist in creating the descriptions
- Place shapes and bitmaps on any layer of your map and do individual pixel editing of your map with the built-in Pixel Editor
- Automatic checking for new versions of Realm Studio
- **Realm Studio is 100% free and open source. No royalties or payments of any kind are ever required.**

How Realm Studio Works

Realm Studio constructs realms (maps) in many independent layers. The controls to draw or paint each layer are displayed on tabs that represent some aspect of your map – ocean, land, fresh water, symbols for mountains, trees, houses, etc., as well as paths, labels, map grids, and so on. Realm Studio was modeled on Wonderdraft, and many of Realm Studio's features work like those of Wonderdraft, so if you are familiar with that application, you will quickly understand Realm Studio. You will "paint" many of the features of your map using a dotted-line circle cursor. Other features you select or place using the usual Windows arrow or cross cursor. Realm Studio was designed to be easy to use, so figuring out how the controls work won't take long.

Often, clicking a button will change the **Drawing Mode** of Realm Studio. The current **Drawing Mode** is displayed in green text below the map on the left. Make note of the current **Drawing Mode** as you create your map. It will tell you what operation Realm Studio will perform.

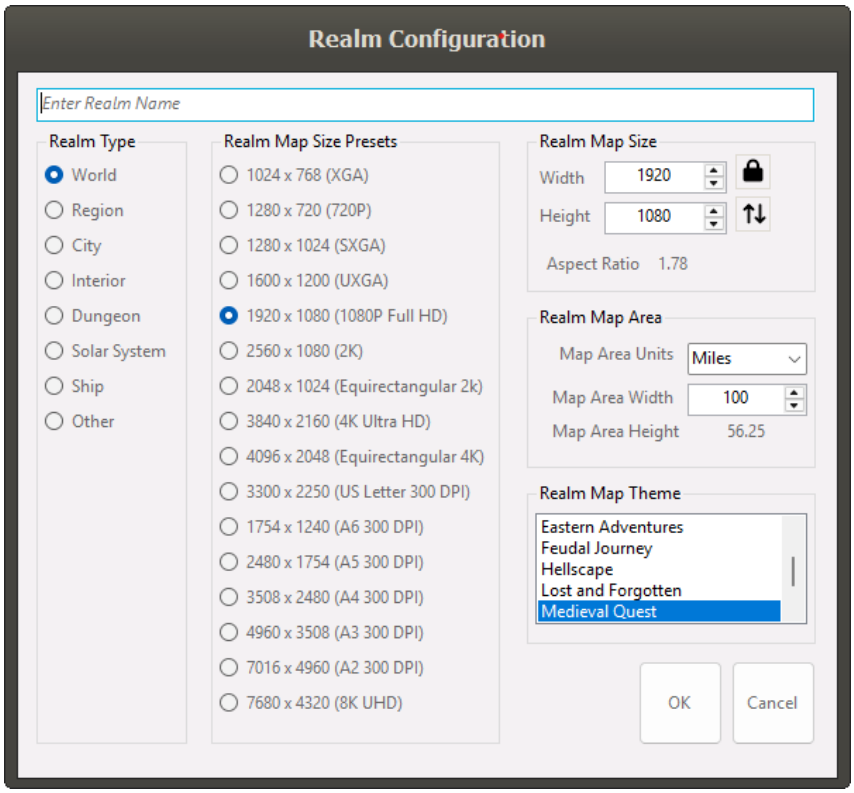
When Realm Studio Starts

Configuring Your Realm

When Realm Studio starts, it displays a "splash screen" for a few seconds (you can dismiss the splash screen by clicking on it, if you don't want to wait). It then displays the main window and loads the assets you can use when creating your maps.

After the assets are loaded, a dialog box titled "Realm Configuration" is shown. The dialog allows you to select the type of realm (map), the size of the map in pixels, the size of the map (map area) in real world units (square miles, kilometers, meters, feet, etc.), and the initial theme for the map. The theme can be changed after it is created. There is more information on themes later.

You should give your realm (map) a name by entering the name in the text box at the top of the Realm Configuration dialog. If you do not give your realm a name, it will be called "Default." You can change the name of your realm later, if you don't give it a name in the Realm Configuration dialog.



At this time, changing the Realm Type only changes the default Map Area Units, but in future versions, changing the Realm Type will cause other changes to the application functionality to support creation of specialized maps for building interiors, dungeons, solar systems, etc.

After you have selected the Realm Type, Realm Map Size, Realm Map Area, and Realm Map Theme on the Realm Configuration dialog, click the OK button to display the Realm Studio Main Window. Clicking Cancel will close Realm Studio.

The Realm Studio Main Window



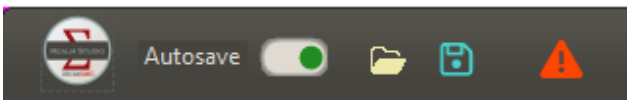
The Main Window is where you will paint your Realm Map. Based on the Realm Map Theme you chose in the Realm Configuration dialog, background texture and vignette type and color, and other textures and colors are selected and applied to your Realm Map. Of course, you are free to choose different texture, colors and other features of your map at any time.

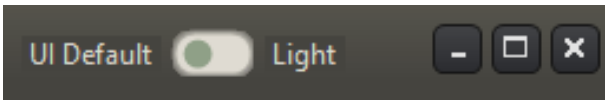
On the far left of the Main Window is a set of tabs labeled Background, Ocean, Land, Water, Paths, Symbols, Labels, Overlays, Regions, and Drawing. Selecting a tab will change the controls displayed just to the right of the tabs and in the panel on the far right of the Main Window. The controls that are displayed allow you to add, edit, and delete the features of that layer of the Realm Map.

The rest of this document describes the controls and features of Realm Studio in detail, so if you have questions about how a feature works or what it does, please refer to the sections below.

The Title Bar Controls

There are several controls in the title bar of the Realm Studio application that give you quick access to some basic functions.





The Autosave Switch

The Autosave switch enables and disables the Autosave function of Realm Studio. When enabled, Realm Studio will periodically save a backup copy of your map. The backup copies are saved in your Documents folder in a subdirectory named "RealmStudio/Realms/autosave." More information on Realm Studio directories and the Autosave functionality is below.

The File Open Icon

The File Open icon opens a Windows File Open dialog to allow you to open a previously saved map.

The File Save Icon

The File Save icon opens a Windows File Save dialog to allow you to save the map you are currently working on. The file name defaults to the name of your realm (map), if it was set in the Realm Configuration dialog or in the Realm Properties dialog.

The New Version Button

If the Automatically Check for Updates user preference is enabled (it is enabled by default), then two minutes after the Main Windows is shown, Realm Studio will check for newer releases in the Realm Studio Github repository. If a newer release is available, the red New Version info button will be shown (the red triangle with the exclamation point). You can click this button to open your web browser to the Github repository to download the RealmStudio.msi installation program to install the new version.

The UI Switch

The UI switch allows you to change the look of the Realm Studio user interface from the default dark borders to light borders. At this time, switching the UI look has not been implemented.

The Main Menu Bar

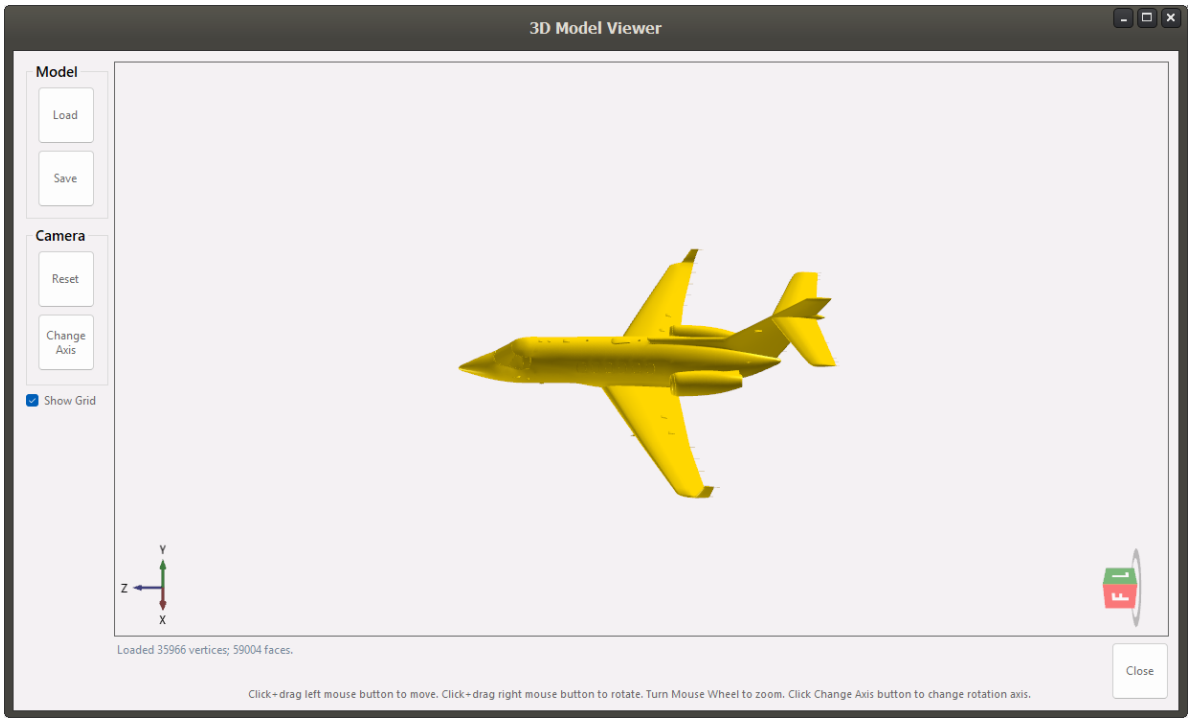
The Application Tools

On the right side of the Main Menu Bar are icon buttons for tools that are common to more than one tab or are independent of the tabs.



The 3D Model Viewer Icon

Clicking the 3D Model Viewer icon (the gold-colored cubes) open the Realm Studio 3D Model Viewer window. The 3D Model Viewer allows you to open and view, rotate, zoom, and pan 3D models in several formats: Stereolithography format (`.stl`), Wavefront OBJ (`.obj`), Object File Format (`.off`), Autodesk 3DS format (`.3ds`), and Lightwave 3D Object format (`.lwo`). Models can be saved in Wavefront OBJ format. The 3D Model Viewer is used when viewing grayscale height maps in 3D and when viewing a world globe.



The Eyedropper Icon

Clicking the Eyedropper icon will cause the `Drawing Mode` to change to ***Select Color*** and the cursor to change to an eyedropper. You can click anywhere on the map to select the color at that location. If you are on the Ocean tab, the Land tab, or the Water tab, the selected color will be set on the Color Brush button.

The Plus Icon

The Plus icon opens the Color Selection dialog. Selecting a color from the dialog will add it to the set of custom colors on the Ocean tab, the Land tab, or the Water tab.

The Area Selection Icon

Clicking the green Area Selection Icon allows you to select a rectangular area on the map. Several functions require that an area on the map be selected, such as Copy/Cut/Paste, creating a Detail Map, and others.

The File Menu

Starting a New Map

Choosing the New option from the File menu allows you to create a new map. If you are working on a map, you will be prompted to save it before creating a new one. When you select this option, the Realm Configuration dialog will be displayed to let you choose the map size, area size, etc. as shown above.

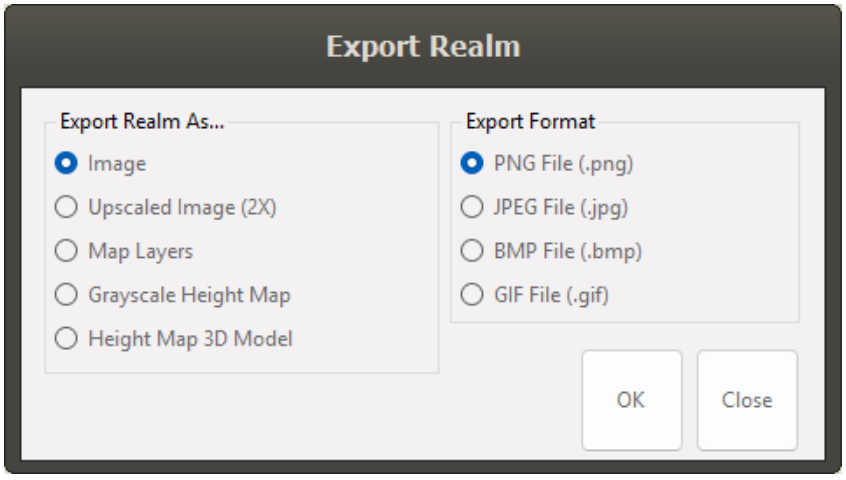
Opening a Map

Choosing the Open option from the File menu will open a Windows File Open dialog to allow you to open a previously saved map. Realm Studio maps are saved with a `.rsmapx` extension, and they are XML files, so you can open a Realm Studio map with Notepad or an XML editor application to view the map structure. Though they are XML files, it is not recommended that you edit the Realm Studio map XML directly, as the map structure could be damaged, rendering it unopenable by Realm Studio.

Saving a Map

Choosing the Save and Save As options from the File menu will open a Windows File Save dialog to allow you to save the map you are working on. By default, your maps will be saved in your Windows Documents folder in a subfolder called “RealmStudio/Realms. You are free to change the location of your saved maps, but it is recommended that you save them to the default location. If future functionality changes require the structure of Realm Studio maps to change, any map upgrade functions will be able to locate your maps more easily if they are saved to the default location.

Exporting a Map



Choosing the Export option from the File menu opens the Export Realm dialog. The Export Realm dialog lets you export your Realm Map in various forms. Maps can be exported as an image file in several formats: Portable Network Graphics (`.png`) file, Joint Photographic Experts Group (`.jpg`) file, Windows bitmap (`.bmp`), or Graphics Interchange Format (`.gif`).

You can also choose to export the Realm Map as an upscaled image at two times the pixel resolution (two times width and two times height) of the Realm Map. You can choose any of the Export Formats as image export.

Currently, export options are limited: the size of the exported file (resolution), color depth, transparency color, and so on cannot be changed. Additional options may be added to the export function in the future.

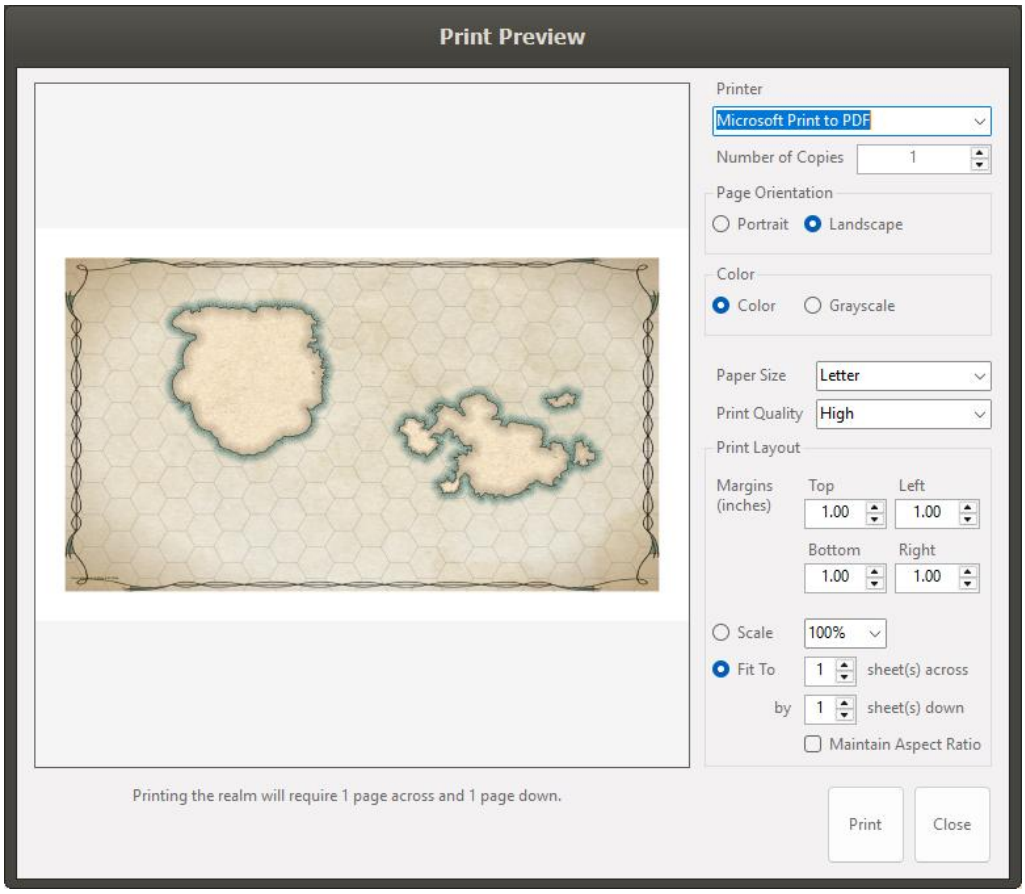
Selecting the Map Layers option will export the Realm Map as a ZIP file (`.zip`) containing a `.png` file for each layer of the map. There are 18 images written into the `.zip` file; however, the number of images included in the `.zip` file may change as application features are added or changed.

Selecting the Grayscale Height Map option allows you to export the Realm Map as a grayscale ("black and white") image. Grayscale height maps are used in many different 3D modeling and graphics applications to create 3D pictures and models of landscapes. There is much more about height maps and 3D models and 3D views of Realm Maps below.

Selecting the Height Map 3D Model option allows you to export your Realm Map in the Wavefront OBJ (`.obj`) 3D model format. This format is one that can be read and viewed by many 3D modeling and 3D viewing applications, like MeshMixer. (You can download MeshMixer here: https://apps.autodesk.com/FUSION/en/Home/Index). Realm Studio also includes a basic 3D viewer that can be used to view your entire map (described above), selected Landforms, or a selected area of your Realm Map as a 3D model. Viewing your map as a 3D model requires you to create a height map for it. These functions are described in much more detail below.

Printing a Map

Realm Studio includes functionality for you to preview your map and send the map to your printer, with options to choose the printer to print the map on, the number of copies to print, to print the map in portrait or landscape, to print in color or grayscale, select paper size, print quality, and margins, and to scale your map or fit it to a selected number of pages. By default, the map will be scaled to fit to one page on the default paper size for the selected printer. Be aware that some printers do not support printing without a margin, so if you want to print your map on multiple pages, it may not be possible to eliminate a border around each page.



The Edit Menu

Undo/Redo

Many (but not all) of the functions in Realm Studio can be undone/redone using either the Edit Undo/Redo menu options or the standard **`ctrl+z`** and **`ctrl+y`** keyboard shortcuts.

Cut/Copy/Paste

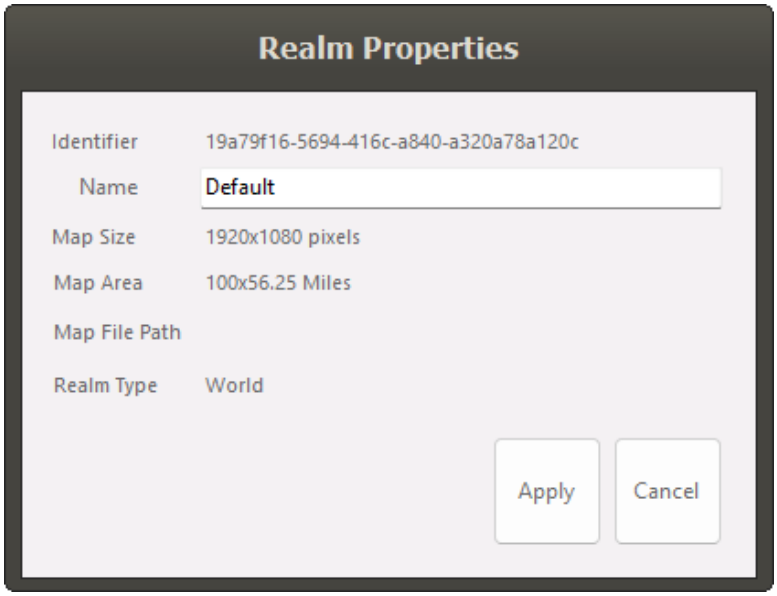
You can use the Area Selection icon to select an area on the map and then copy or cut symbols, paths, and water features (including lakes and rivers) in the selected area. The map components

that are cut or copied can be pasted at a new location. The new location for the components is wherever the mouse cursor is located.

The Realm Menu

The Realm Properties Dialog

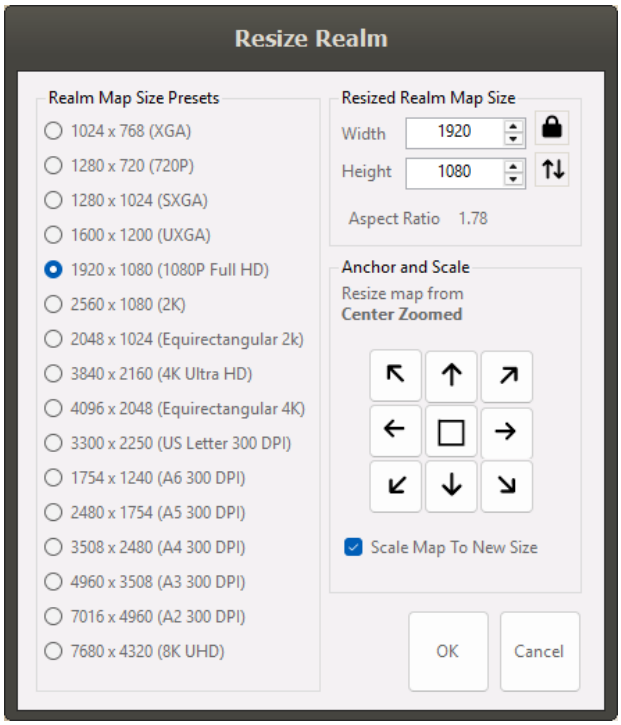
The Realm Properties dialog displays information about the map you are currently working on and allows you to change the name of your realm.



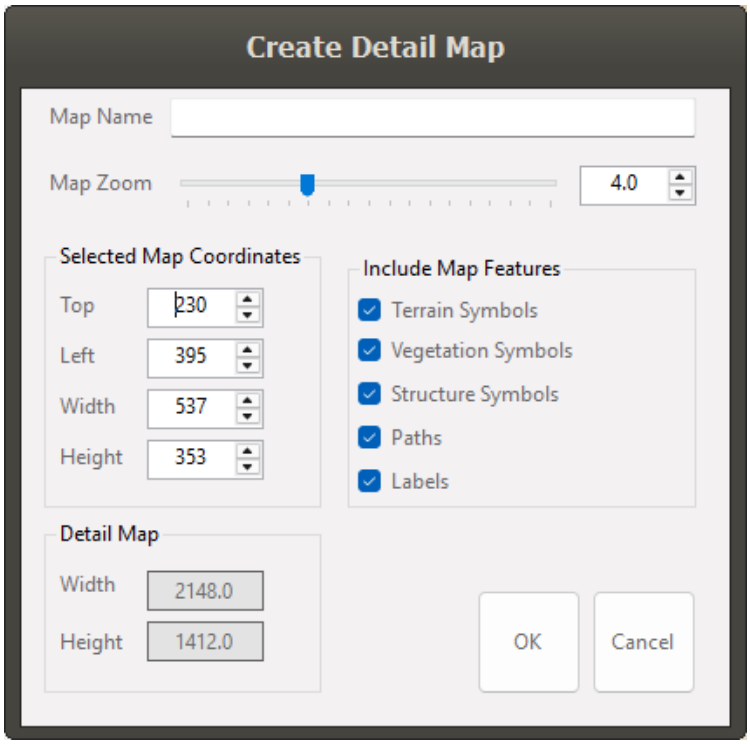
Changing the Map Size

Realm Studio provides functionality to resize your map. The Resize Realm dialog shown below lets you choose a new size for the map. If the new map size is smaller than your current map, the current map is scaled down to fit the new map size. If the new map is larger than your current map size, you can choose to scale the existing map up to fit the new size, or for the existing map to take up a portion of the new map. The Anchor and Scale buttons specify where on the new map the existing map will be placed. The Scale Map to New Size checkbox specifies whether the existing map will be scaled to fit the new size or keep its original size and take up a portion of the new map. The new map will include all the features of the existing map (landforms, water features, symbols, paths, labels, and drawn shapes) of the existing map, scaled proportionally to the new map size.

Before resizing the map, Realm Studio will prompt you to save the existing map so that you can revert to the existing map, if you need to.



Creating a Detail Map

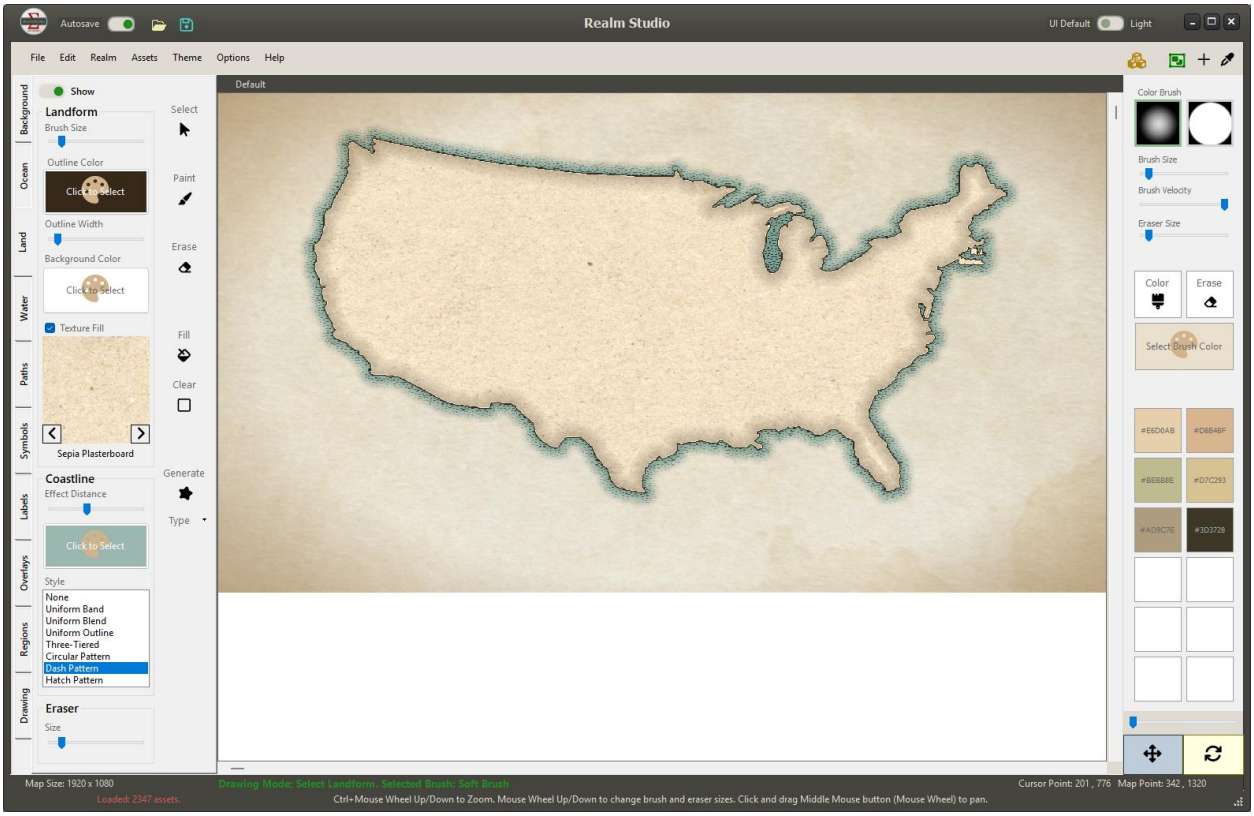


Realm Studio also allows you to create a Detail Map from an existing map. To create a Detail Map, first select a portion of the existing map using the Area Selection button on the right side of the Main Menu Bar. It looks like this: 

After selecting the map area from which to create the Detail Map, select **`Create Detail Map…`** from the Realm option on the Main Menu. The dialog shown above will be displayed. The dialog allows you to change the location and size of the selected area, what features of the existing map to include in the Detail Map, how much to scale the detail map, and to give the Detail Map a name. Once the parameters in the dialog are set as you'd like, click the OK button. If the existing map has not been saved, you will be prompted to save it. After that, the new Detail Map will be created and displayed, with the selected features included, scaled to fit the Detail Map.

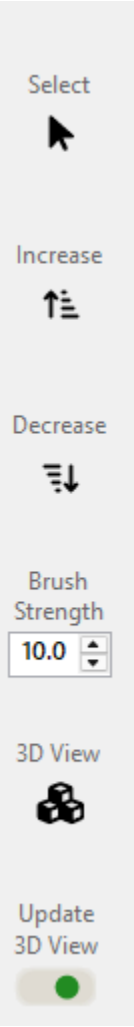
Tracing the Outline of an Image

Realm Studio lets you select any raster image (in the same formats that can be exported), find the perimeter of the objects in the image, and then create a landform from the largest traced object in the image. Because images can vary so much, the process is not reliable. Sometimes, editing the image to increase contrast, remove small artifacts, and remove interior colors and features can improve the results. Here's an example of a map of the continental United States traced and converted into a landform.



Creating and Viewing Grayscale Heightmaps

Realm Studio includes functionality to view and create grayscale heightmaps for the landforms on your map. To create a heightmap, first paint landforms on your map as described in the sections below. Once you've created your landforms, select `Display Height Map` from the Realm option on the Main Menu. When selected, the Land tab will be displayed, but with controls for painting and viewing a grayscale heightmap.



The process for creating a grayscale heightmap is like painting color on a landform. Clicking the Increase button will set the `Drawing Mode` to ***Increase Map Height***. When in this mode, holding the left mouse button while moving the mouse will lighten the color of the heightmap, increasing the height of the landform at that location. Conversely, clicking the Decrease button will set the `Drawing Mode` to ***Decrease Map Height.*** Holding the left mouse button and moving the mouse will darken the color of the heightmap, decreasing the landform height at that location. The Brush Strength Up/Down control value indicates the rate at which the color is changed as you move the mouse. A smaller value changes to heightmap shade more slowly. Grayscale painting of heightmaps is clipped to the landform boundaries.

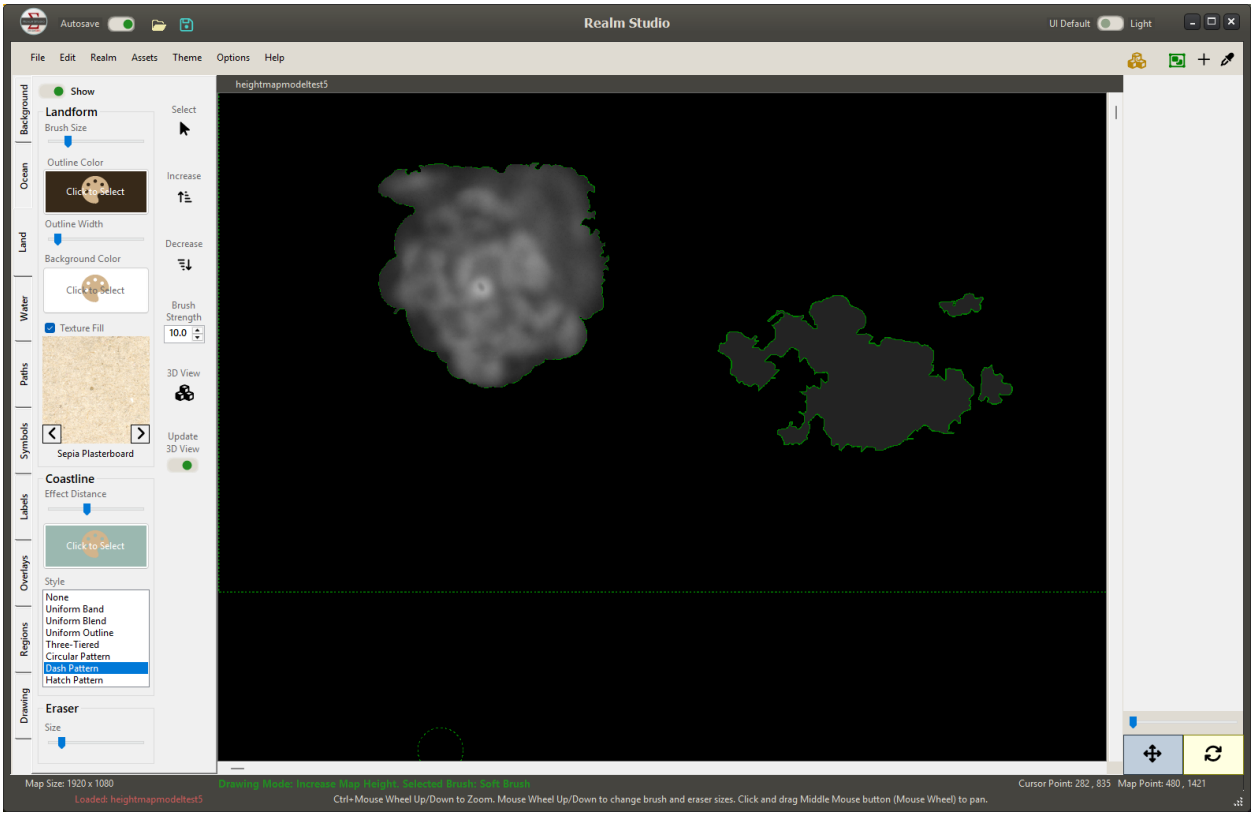
Clicking the 3D View button will open the 3D Viewer window with a 3D view of the landforms on your map derived from the grayscale heightmap. You can zoom, pan, and rotate the 3D view of your map.

When the Update 3D View switch is enabled, every time you release the left mouse button, the 3D view is updated. However, near-real-time updates impact performance while painting the height map, so you can disable updates if performance is too slow. Another way to improve performance while viewing the 3D model is to select the landform or use the Area Select button to select an area of your map. The 3D view will only show the selected landform or area.

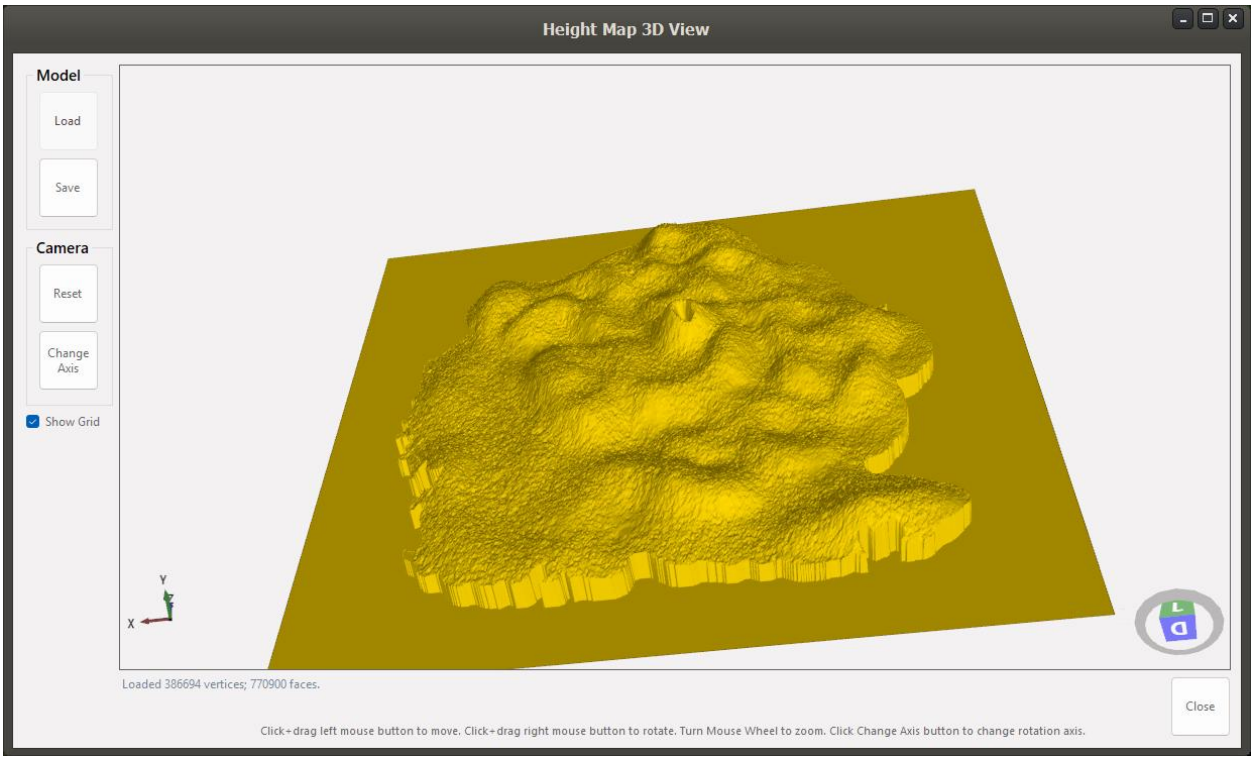
Landforms on your map and the map boundaries are outlined in a green dotted line to help you see them more easily.

To go back to the normal map view, uncheck the `Display Height Map` menu option from the Realm main menu option.

The Heightmap View



A 3D view of a landform heightmap.



The World Globe View

Realm Studio allows you to view your map as a world globe floating in space using the 3D Viewer and to record AVI video of the globe rotating. The 3D World Globe View has a lot of different options, so it is covered in detail in the section titled "The 3D World Globe View."

The Assets Menu

There are several types of "assets" that Realm Studio loads and uses to allow you to create maps.

- **Boxes** are used to outline and emphasize labels that are placed on the map.
- **Brushes** are used to paint colors onto ocean areas, landforms, and water features (lakes, rivers, and so on).
- **Frames** are used as borders around your map.
- **Icons** are used in the user interface of Realm Studio.
- **Label presets** are XML files that describe preset formats for labels (font, size, color, outlining, and glow). You can create your own label presets in Realm Studio in addition to the default presets provided. The label presets are associated with the selected Theme, so each theme has its own presets, and label presets for each Theme do not conflict, even if they have the same name.
- **Name Generators** (and name bases) are files that drive the generation of random names that can be used in labels. If you can't think of a name for a continent, river, island, lake, or anything else on your map, the Name Generator function can generate one for you. Hundreds of thousands of unique names can be generated.
- **Symbols** are small images of objects that can be placed on your map (houses and buildings of all types, trees, bushes, grass, and other kinds of vegetation, mountains, hills, volcanoes and other landscape features, as well as other kinds of objects). Much more information on symbols is in the sections below.
- **Stamps** are bitmaps in `.png`, `.jpg`, or `.bmp` format that can be placed anywhere on your map from the Drawing tab. You can put bitmaps that you use frequently into the Stamps folder for easy access.
- **Textures** are bitmaps that are used as the background for the map, for ocean areas, for landforms, and for 3D and World Globe views.
- **Themes** are XML files that define the look of a map: textures, colors, frames, label styles, and more. You can define your own themes in addition to those that are provided. Themes are described in more detail below.
- **Vectors** are used to define some user interface elements, like some path styles.

Creating Symbol Collections

To use symbols in Realm Studio, they must be part of a Symbol Collection. A Symbol Collection is defined by an XML file within a subdirectory of the Assets/Symbols directory created when Realm Studio is installed. Realm Studio provides a dialog for creating Symbol Collections, which is described in the Realm Studio Assets and Symbol Collections section below.

Importing Assets

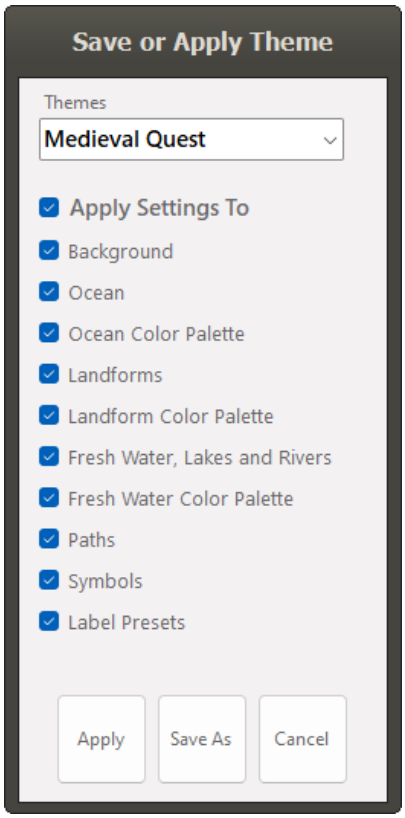
Realm Studio can use most assets (symbols, frames, boxes, and so on) that were developed for Wonderdraft. You can download Wonderdraft assets from the Cartography Assets site and then import them into Realm Studio. Realm Studio provides dialogs for importing downloaded assets. The dialogs are described in detail in the Realm Studio Assets and Symbol Collections section below.

Reloading Assets

When new assets are imported or new symbol collections are created, Realm Studio must reload the assets to be able to use them. Selecting `Assets -> Reload All Assets` from the main menu will load all the assets from the Realm Studio Assets folder.

The Theme Menu

Choosing the Theme menu option on the main menu opens the Save or Apply Theme dialog. This dialog allows you to select a theme, select which settings the theme will change, apply the selected theme to the map settings, and create a new theme from the currently selected map settings in the Realm Studio user interface.



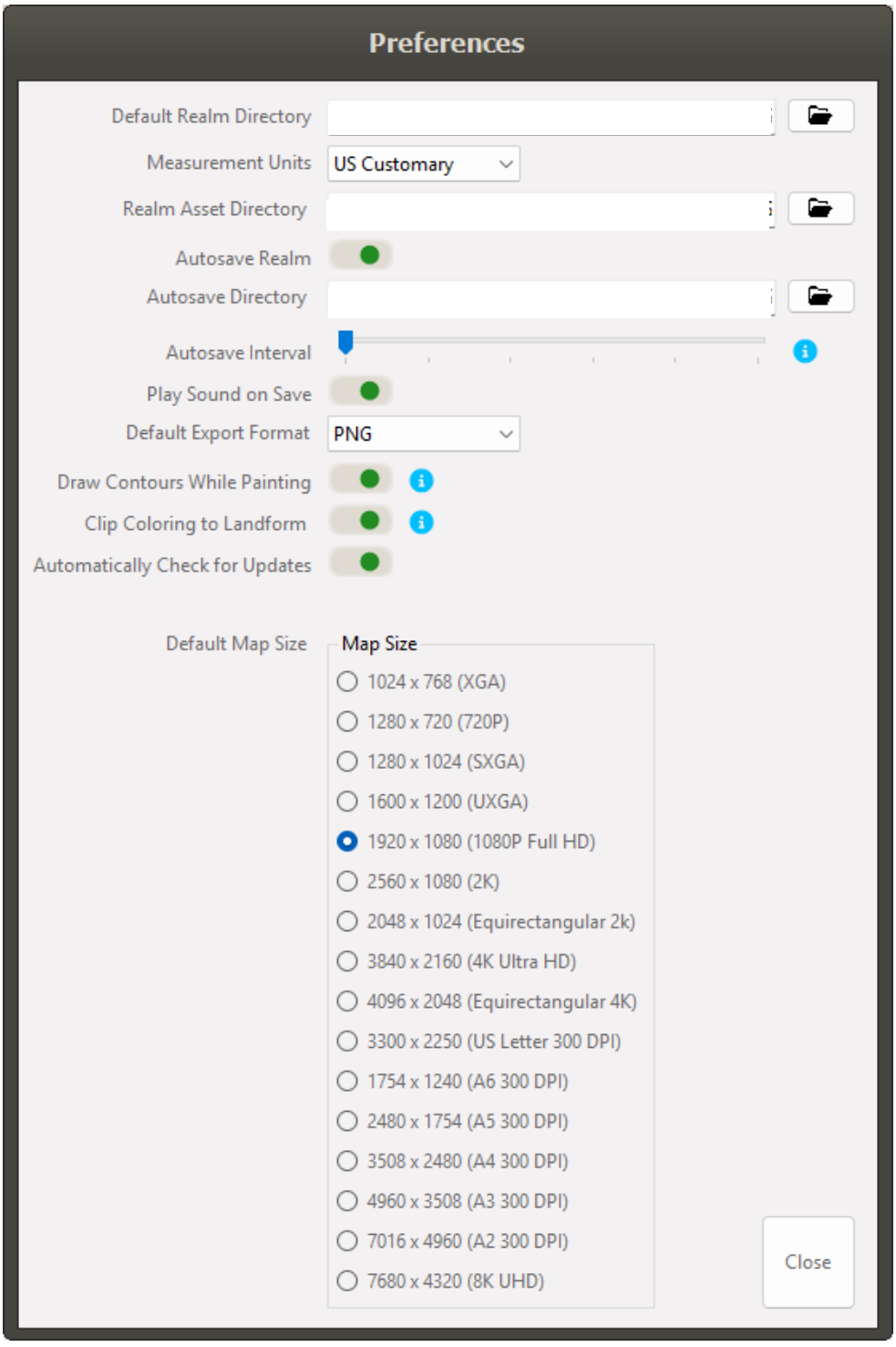
To apply a theme, select it from the dropdown, select which settings you want to change, then click Apply. The settings from the theme will be applied to the associated user interface controls and to the current realm map.

To create a new theme, set all the controls in the Realm Studio user interface to the values you want saved in the theme – colors, textures, label, coastline pattern, and so on. Then click the Save As button on the dialog. Another small dialog will be displayed to let you enter a name for the new theme. Enter the name, then click OK. The new theme will be created with the name you entered. You must reload the Realm Studio assets before the new theme will be available.

The Options Menu

Changing User Preferences

There are several preferences that you can set that change how Realm Studio works and where it finds and saves files. The preferences that you set are saved, so Realm Studio will use the value that you set whenever it is opened. The user preference dialog looks like this:



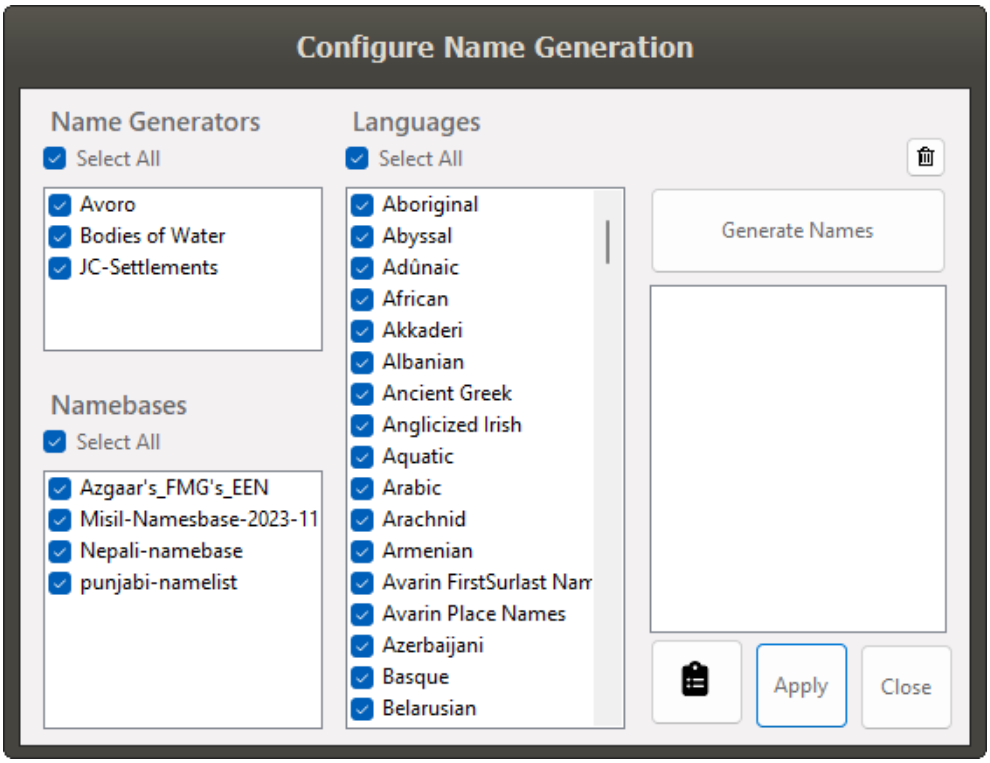
- The **Default Realm Directory** preference tells Realm Studio where to save your maps. When Realm Studio is installed, this value is set to a subfolder of your Windows Documents folder called `RealmStudio/Realms`. If you change this value, Realm Studio will set the default location of saved maps to the folder that you select.
- The **Measurement Units** preference allows you to set the default measurements to either US Customary units (inches, feet, miles) or metric units (centimeters, meters, kilometers).
- The **Realm Assets Directory** preference sets the location of Realm Studio assets folder. When Realm Studio is installed, this value is set to a subfolder of your Windows Documents folder called `RealmStudio/Assets`. If you change this value, Realm Studio will load assets from the folder you select. Be careful when changing this value. If Realm Studio cannot locate the assets it requires, it will not function.
- The **Autosave Realm** switch enables or disables automatic saving of backup copies of your map. When enabled, Realm Studio periodically saves a backup copy of your map. The interval between automatic backups is set by the Autosave Interval trackbar. A maximum of five backup copies are kept; although, at times six may be retained for a short period of time.
- The **Autosave Directory** preference sets the folder in which backup copies of your map are saved. When Realm Studio is installed, the folder is set to a subfolder of your Windows Document directory called `RealmStudio/Realms/autosave`. It is not recommended that you change the folders in which your maps are saved or the backup location, as future functionality will allow you to restore your map from a backup in case of an application error or a problem with the map.
- The **Autosave Interval** trackbar allows you to set the time between automatic backups from five minutes up to thirty minutes in five-minute increments.
- The **Play Sound on Save** switch allows you to enable or disable playing of a sound indicating that your map has been successfully saved.
- The **Default Export Format** preference allows you to indicate your preferred format for maps that are exported as an image file.
- The **Draw Contours While Painting** switch is important when creating large maps (larger than 4K Ultra HD or 3840x2160 pixels). By default, Realm Studio calculates the boundaries of landforms while you are painting the landform. You can see this happening as you paint. On very large maps, calculating landform boundaries can cause landform painting to lag and affect performance. By turning this feature off, calculation of landform boundaries is delayed until you release the left mouse button, which improves performance somewhat.
- When painting colors on a landform, Realm Studio clips the color brush to the outline of the landform (preventing you from accidentally painting land colors onto ocean and water areas. The **Clip Coloring to Landform** switch turns off the clipping, so landform coloring can overlap ocean and water areas. This may improve performance somewhat.
- The **Automatically Check for Updates** switch enables or disables the Realm Studio functionality to check for updates soon after the application starts.

- The **Default Map Size** radio buttons allow you to set the default size of maps that you create.

Setting the Name Generator Configuration

Realm Studio has a built-in random name generator function that uses "name generators" and "name bases" that are in the same format as those used by Wonderdraft, and you can download additional name generators and name bases from Cartography Assets and install them in the Realm Studio Assets folder to be used by Realm Studio. You can also easily create your own name generator or name base files using a simple text editor (like Notepad) and save them into the Realm Studio Assets directory. The format of these files is described in the Technical Information section below.

The set of name generators and name bases installed with Realm Studio allows hundreds of thousands of different names to be generated in dozens of real and fantasy languages. In some cases, you may want to limit the random names generated to one or more specific languages or only for bodies of water, or only from specific name generators or name bases. The Name Generator Configuration dialog (opened from the `Options -> Name Generator Configuration…` menu option) lets you do that.



You can select specific name generators or name bases to use or select specific languages. Clicking the Generate Names button will generate ten random names based on the selections you've made. A rolling list of the last thirty random names generated is kept. Selecting a name and clicking the Apply button will copy the selected name to a label being created. Clicking the

Clipboard button will copy the selected name to the clipboard so you can paste it wherever you like. If no name is selected, all the names in the list are copied to the clipboard.

Generating Names for Bodies of Water

To generate names only for bodies of water, deselect all Name Generators and Namebases. In the Name Generators list, select only the "Bodies of Water" Name Generator. Select the Languages for the names of the bodies of water you would like to generate, then click Generate Names. Note that only the names of the bodies of water are in the language(s) you select. The type of the body of water is in English, e.g. "ocean," "lake," "swamp," and so on. So, for example, an ocean name generated with Chinese selected might be "Huangzhou Ocean." You can select as many languages as you like when generating bodies of water, but at least one must be selected.

The Help Menu

Displaying this Help File

Selecting the Help -> Contents menu option opens this Help file in your browser from the RealmStudioDocs Github repository web pages created for the Realm Studio documentation. The URL is: https://petenelson372.github.io/RealmStudioDocs/.

Check for New Release

If you select this menu option, Realm Studio will check the Github repository to see if a new version of Realm Studio has been released. If so, a message will be displayed letting you know. If not, a message will be displayed telling you that you are using the latest version. If a new version is available, a link to the Github repository is the About Dialog will open your web browser to the Github repository where you can download the RealmStudio.msi installation program.

The About Dialog

Selecting the Help -> About menu option opens the About dialog for Realm Studio. The About dialog displays important implementation, copyright, and licensing information, as well as credits for libraries and images used to create Realm Studio. Links to the Realm Studio source code on Github are shown, as are links to the Microsoft Store page for Realm Studio, release notes, privacy policy and to create an email to the Realm Studio support email address (which is support@brookmonte.com).

Painting Maps with Realm Studio
The Realm Studio Main Window



- At the center of the Realm Studio main window, the Map Area is where the map is displayed as you draw/paint it.
- On the left side of the main window, the Tab Controls organize the Realm Studio functions according to the features of the map.
- At the top of the main window is the application Title Bar and the title bar controls.
- Just below the title bar is the application Main Menu. On the right side of the Main Menu bar is a set of icon buttons giving access to application tools and functions that are used on many tabs or are independent of the functions on the tabs.
- To the right of the Map Area is a panel in which additional drawing controls are displayed. The controls displayed on the Additional Drawing Controls panel change when tabs are selected.
- At the bottom-right of the Main Window are controls that control the zoom level of the map. The map can also be zoomed by holding down the Control Key and rolling the mouse wheel.
- At the bottom of the Main Window are controls showing the status of the map, the current **Drawing Mode**, and the current selected drawing layer.

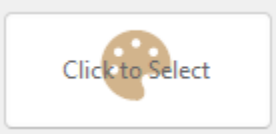
- Just underneath and on the right of the Map Area are scrollbars that allow you to pan the map right/left and up/down. The map can also be panned by holding down the middle mouse button (often the mouse wheel).

Selecting a tab causes Realm Studio to display a set of controls on the left and right sides of the main window (in the Tab Controls and Additional Drawing Controls areas) that give you access to functions for creating maps. Realm Studio maps are organized in layers, and drawing on a layer does not affect the other layers. There are many more layers than just the ten implied by the tabs. For display, each of the layers is composited into a final image that is displayed in the map area of Realm Studio.

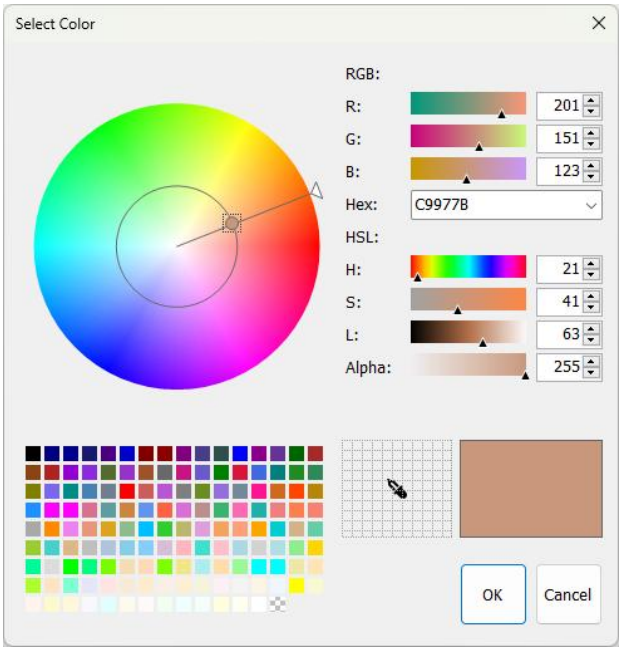
The Tab Controls area on each of the tabs is divided into two sections. On the left is a set of controls for selecting colors, textures, and other options that affect what will be drawn or painted on the map for the selected map layer. On the right side of the Tab Controls area, immediately next to the Map Area is a set of buttons that (usually) either change the **Drawing Mode** of Realm Studio or apply a selected value (texture, color, etc.) to the map. Each of the controls and buttons is described in more detail below.

Color Selection in Realm Studio

On many of the tabs and dialogs of Realm Studio, Palette buttons are shown that look like this (with some variation in size and background color):

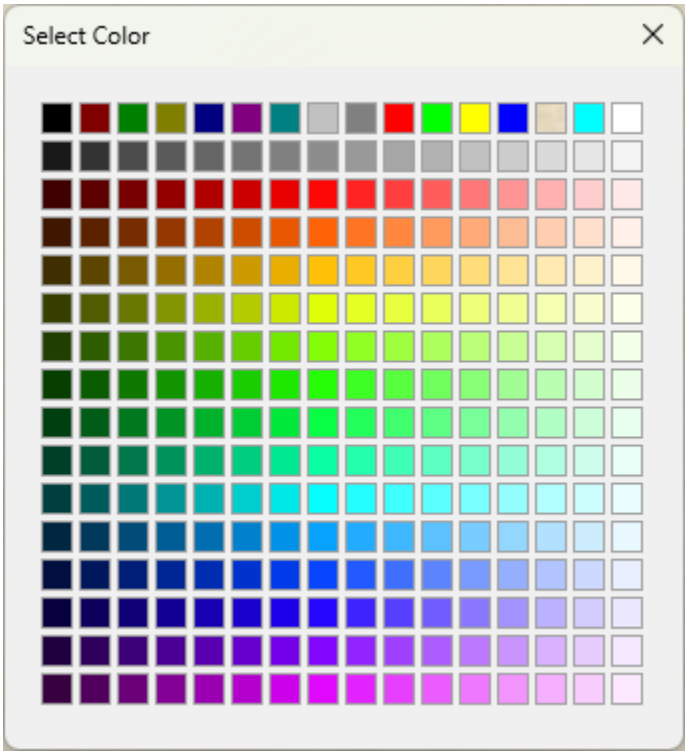


Clicking the Palette button with the left mouse button opens the Color Selection dialog.



The controls on the Color Selection dialog allow you to select a color using the color wheel, a set of pre-defined colors, or RGB or HSL and Alpha channel sliders. Clicking and holding the left mouse button on the eyedropper control changes the cursor to an eyedropper that allows you to select a color from anywhere on your computer screen, even outside the Realm Studio window. Note that it only affects the color in the Color Selection dialog and does not affect brush colors on the Ocean, Land, or Water tabs. Use the Eyedropper button in the Menu Bar to select brush colors.

On most Palette buttons, right clicking the button will open a quick Color Selection dialog that presents a set of pre-defined colors. Simply click a color to select it.



The Realm Studio Tabs

The Background Tab

Realm Studio allows you to set a background texture for your map. Usually, this is not necessary, as your map will have an ocean texture or be filled with a landform that has a texture. However, the option is available if you choose to use it.

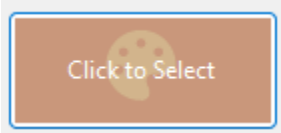
Setting a Background Texture

To set a background texture, scroll through the available background textures using the arrow buttons at the bottom of the texture preview window until you find the one you want to use, then click the Fill button. To clear the background texture, click the Clear button. You can scale and mirror the background texture to achieve the desired effect. Setting the background texture scale to zero will remove it.

The Map Vignette

The map vignette is the shaded area around the outer edges of the map. You can change the color of the vignette and the strength of the vignette (how dark it is and how much of the map it fills). The controls for the vignette are shown on the Background tab, but it is painted on top of most of the other map layers, so the vignette color and strength may affect the appearance of other things painted near the edges of your map.

To change the color of the vignette, left or right click the Color Palette button in the Vignette box.



A Color Selection dialog or a quick Color Selection dialog will be displayed.

Use the controls to select a color. The selected color will be shown as the background of the Color Palette button and applied to the map vignette. In general, buttons with the palette icon allow you to select a color for something to be drawn or painted on your map, and the process for selecting a color is the same throughout Realm Studio.

The “strength” of the map vignette (how dark it is and how much of the map area it fills) can be changed by adjusting the trackbar slider in the Vignette box. The appearance of the vignette changes as you move the slider. If you don’t want a vignette for your map, adjust the slider all the way to the left to make the vignette strength zero.

The vignette shape can be changed using radio buttons under the Strength slider. The vignette can be oval or rectangular. An oval vignette shades the corners of the map, leaving an oval area in the center of the map unshaded. A rectangular vignette shades the outer edges of the map,

leaving a rectangular area in the center of the map unshaded. Try different variations of vignette color, shape, and strength to get the look you want.

The Ocean Tab

Clicking the Ocean tab displays the controls for setting the look of the ocean areas on your map. The ocean area of your map can have a texture applied and/or a color applied. In addition, you can "paint" on the ocean to make it look however you want.

The Ocean layer is on top of the map Background layer, but under all other layers, so when you paint land areas and create other objects on the map (grids, map scales, labels, etc.) they are show "on top" of the Ocean layer.

Setting an Ocean Texture

Applying a texture to your Ocean layer works the same way as it does for the Background. Select a texture from the texture preview box on the Ocean tab, then click the Apply button with the Paint Roller icon. Remove the ocean texture by clicking the Remove button. You can change the opacity of the ocean texture using the Opacity trackbar. You can also scale and mirror the Ocean texture to get the look you want.

Lowering the opacity of the Ocean texture (increasing its transparency) lets the Background texture show through, if one has been set. This has the effect of combining the textures, which can create interesting effects.

Setting an Ocean Fill Color

You can fill the entire Ocean area with a selected color. Select the color you want to fill the Ocean area with by clicking the Color Palette button in the Color box. It works the same way as the Color Palette button on the Background tab. When you've selected the color, click the Fill button. The entire Ocean area will be filled with the color you've selected. Clear the color by clicking the Clear button.

The ocean fill color can be combined with the ocean texture. By selecting a color that is more transparent (the Alpha value in the Color Selection dialog is low), the ocean texture will show through the ocean fill color, in effect letting you apply a color overlay "on top" of the texture. This technique is especially effective when used to apply color to a black-and-white or grayscale texture.

Painting on the Ocean Layer

In addition to the controls displayed in the tab control on the left side of the main window, clicking the Ocean tab will display controls in the Additional Drawing Controls panel on the right side of the main window. These controls can be used to paint on top of the ocean texture and on top of the ocean fill color. To paint on top of the ocean layer, click the Color button with the paint brush icon. The **Drawing Mode** of Realm Studio will be changed to ***Ocean Paint*** (as shown in the green text at the bottom left of the main window in the Map Status and Drawing Mode area), and the cursor will be changed to a dotted-line circle with crosshairs at the center.

The Ocean Tab
Painting Controls



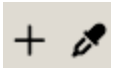
At the top of the Painting Controls, you can select the soft-edged brush, the hard-edged brush, or one of the patterned brushes. The selected brush is outlined in green.

The three sliders control the size of the brush (reflected in the size of the dotted-line circle cursor), the "brush velocity" and the eraser size.

The Color and Erase buttons set the **Drawing Mode** to either paint on the ocean or erase the painted color from the ocean. Erasing painted color does not affect the ocean background texture or color. As you paint or erase, you can undo/redo what you've painted by using the Undo/Redo menu options or using the usual `crtl+z` and `ctrl+y` keys.

The Color Palette button lets you select the color to be painted on the ocean. Clicking it will display a Color Selection dialog as shown earlier.

The set of buttons here are color preset buttons. There are four predefined color presets. The white boxes are for preset colors that you can define. At the top right of the main menu bar (just above the Ocean Tab Painting Controls) are two buttons:



Clicking the "+" button opens the Color Selection dialog. Choosing a color in the Color Selection dialog and clicking "OK" will assign the selected color to the first color preset button that doesn't already have a color assigned. Clicking a color preset button that has a color assigned will set the selected color (shown by the background color of the Select Brush Color button changing to the preset color). Color presets that you define are not saved with the map, so you will have to define them again if you close and re-open the map.

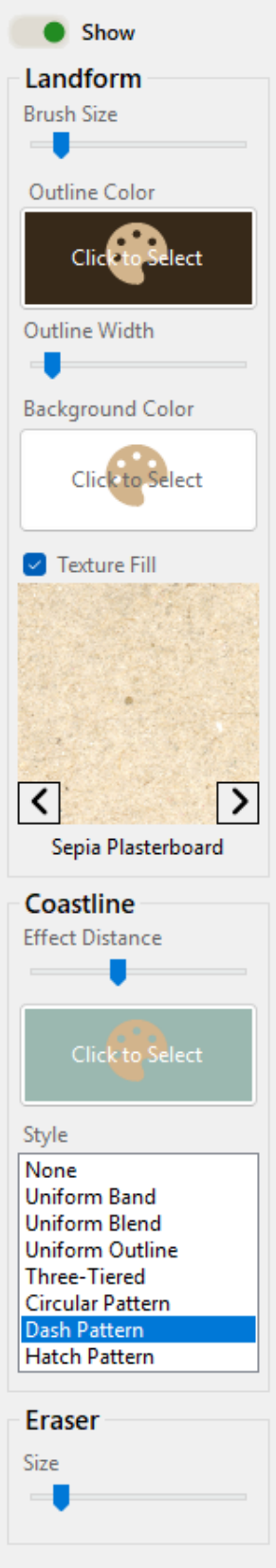
Clicking the eyedropper button changes the cursor to an eyedropper icon and allows you to click anywhere on the map to select the color at that place on the map and assign it as the brush color.

Creating Wind Roses

A wind rose is a set of lines radiating from a point on a map. Wind roses were used long ago by cartographers and navigators to indicate directions to sail to get from one place to another (usually from city to city). Realm Studio allows you to create, style, and place wind roses on your map.

You can change the color and style of your wind roses using the controls in the Wind Rose box. Place wind roses by clicking the Windrose button, then clicking anywhere on your map. Each time you click on the map, a wind rose will be placed, centered at the location where you clicked. Click the Windrose button again to stop placing wind roses. You can remove all the wind roses you've placed by clicking the Remove button.

Future versions of Realm Studio may add more options for styling wind roses.



The Land Tab

Painting Landforms

To create landforms on a Realm Studio map, you "paint" them like colors are painted onto the ocean layer.

To paint a landform, click the Paint button with the paintbrush icon. Like when painting color on the ocean, the cursor will change to a dotted-line circle with crosshairs at the center. The circle area is the area that will be painted. Holding the left mouse button down and dragging the circle will paint the landform. As you paint the landform, Realm Studio fills the landform area with the selected background texture, paints the coastline with the selected style and size, and paints a gradient around the borders of the landform. It also (by default) calculates the contour (border) of the landform.

You can paint each landform you create with a different background texture, border color, coastline color and style, and coastline effect distance. This is one difference between Realm Studio and Wonderdraft. In Wonderdraft, all landforms have the same style and colors, and changing the style or colors changes all landforms that have been painted. In Realm Studio, all landforms are independent objects with their own style and colors.

You can also erase landforms by clicking the Erase button (with the eraser icon).

Landform Brush Size

You can change the size of the landform brush by using the Brush Size trackbar slider or by rolling the mouse wheel forward or backward when the **Drawing Mode** is set to ***Landform Paint***.

Landform Outline Color

Clicking the Color Palette button under the Brush Size trackbar allows you to select a color for the landform outline. Clicking the button will display a Color Selection dialog. Select the color you want to use for the landform outline. Changing the landform outline color also changes the color of the shading around the outer edges of the landform (just inside the outline).

Outline Width

You can change the width of the outline of the landform using the Outline Width slider.

Background Color

If you don't want your landform filled with a texture, you can select a color for the landform by clicking the Background Color button. To fill the landform with the selected color, the Texture Fill button must be unchecked.

Landform Texture

You can change the texture used as the background of landforms. Scroll through the textures using the buttons at the bottom left and bottom right of the texture preview to find the one you want to use. The texture that is displayed in the texture preview will be used as the background texture of the landforms you paint.

If you don't want the landform you paint to be filled with a texture, uncheck the Texture Fill checkbox. If the checkbox is not checked, the landform will be painted with the color selected using the Background Color button.

Coastline Effect Distance

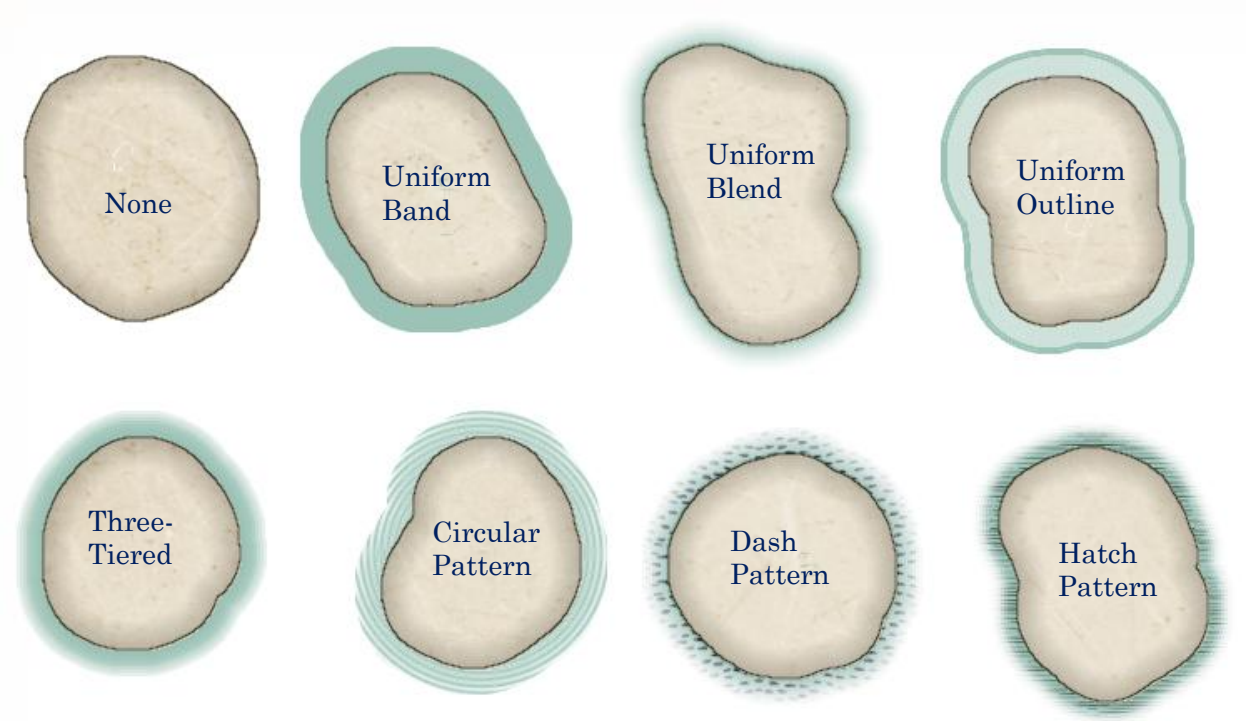
Realm Studio paints a coastline effect (or you might think of it as a shallow water effect) around the landforms you paint. The Coastline Effect Distance trackbar changes how wide the coastline effect is. Setting the Coastline Effect Distance to zero will result in no coastline effect being painted. The slider also governs how wide the shaded area around the outer edge of the landform is, so setting it to zero also causes no shading to be painted.

Coastline Color

You can change the color of the coastline effect by clicking Color Palette button in the Coastline box and selecting a color in the Color Selection dialog that is displayed. As for the other Color Palette buttons, the color you select is shown as the background color of the Color Palette button, so you always know what color is selected.

Coastline Style

There are several different patterns that can be selected from the Style list to paint the coastline effect. If you select "None" in the list, then no coastline effect is painted, but the shading gradient around the outer edge of the landform will still be painted. The picture below shows an example of each of the patterns.



Merging Landforms

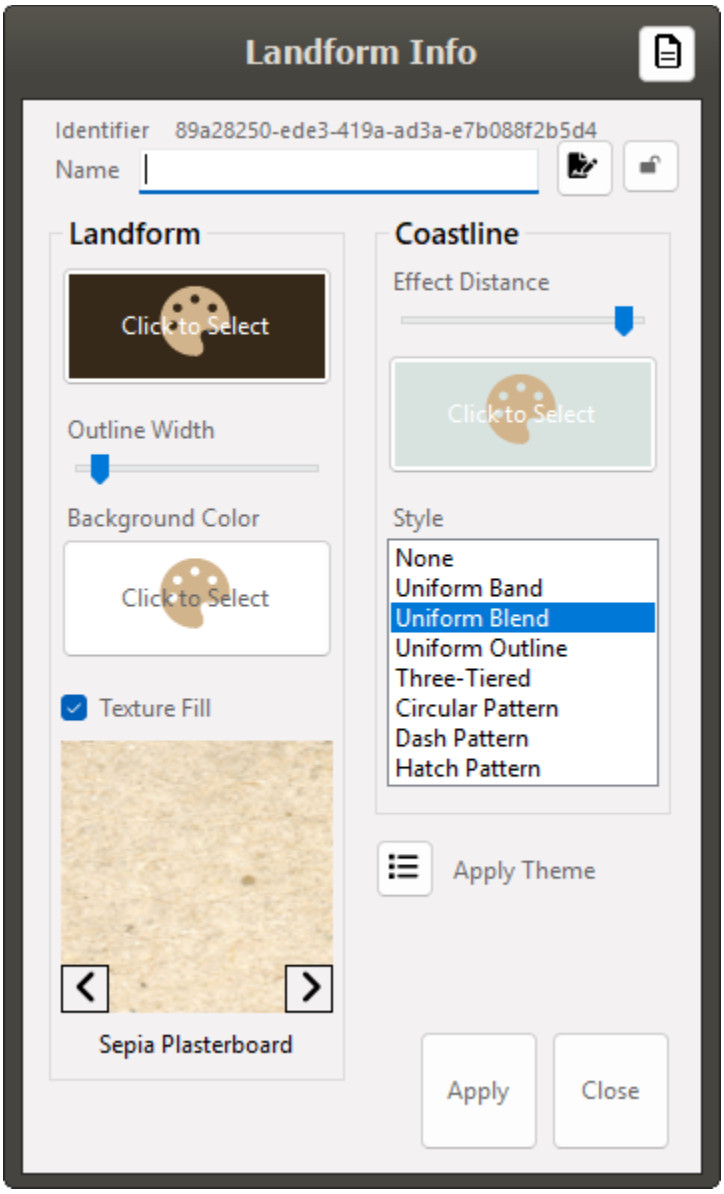
When you are painting your landforms, if the area you paint overlaps with another landform that you previously painted, Realm Studio "merges" the landforms when you release the mouse button; that is, the two landforms are merged into a single landform. The merged landform takes on the characteristics of the *previous* landform that was painted. You don't have to take any action to merge overlapping landforms; Realm Studio merges them automatically.

Erasing Landforms

You can erase landforms (or parts of a landform) by clicking the Erase button (with the eraser icon). As for painting landforms, the cursor changes to a dashed circle with crosshairs. Hold down the left mouse button and drag it to erase. The size of the eraser can be changed by using the Eraser Size trackbar slider. A useful technique for creating landforms is to paint the rough shape you want for the landform, then "sculpt" the landform by using a small eraser to make more detailed changes to the shape,

Changing Landform Attributes

Clicking the Select button (with the arrow icon) will change the **Drawing Mode** to ***Landform Select***. When in this mode, clicking on a landform will select it. The selected landform will have a dashed box drawn around it. When a landform is selected, you can make changes to it or delete it entirely. When the **Drawing Mode** is ***Landform Select***, right-clicking on the landform will open a dialog showing the current data for the landform. The dialog looks like this:



In the dialog, you can give the landform a name and change the look of the landform. Clicking the Apply button will apply the changes to the selected landform (and only the selected landform). Clicking the button next to the Name textbox will use the Realm Studio Name Generator (see the previous section) to generate a random name for the landform. If you like the name you've entered or that was generated, you can click the Lock button to prevent accidentally generating a new name that replaces the existing name.

In the title bar of the Landform Info dialog, the Edit Landform Description button  allows you to open an AI-assisted Description Editor dialog to enter a description for the landform. See the section below on the Description Editor.

Filling the Entire Map Area with a Landform

Clicking on the Fill button (with the fill icon) will fill the entire map with a single landform. Before filling the map, Realm Studio verifies that no landforms have been painted. It will prompt you to clear the existing landforms, if any have already been painted.

Once the map has been filled, you can sculpt the edges of the using the landform Eraser to create coastlines.

Clearing Landforms

Clicking the Clear button (with the empty square icon) will clear all painted landforms from the map. Before clearing the landforms, Realm Studio prompts you for confirmation.

Generating Random Landforms

Realm Studio includes functionality to generate random landforms of various kinds. Clicking the Type dropdown under the Generate button allows you to select what kind of landform(s) you want to generate. Clicking the Generate button will then generate a random landform of that type. The generated landform(s) will have the border color, background texture, coastline effect width, color, and style currently selected.

Realm Studio can generate the following types of landforms:

- **Region**: a Region is a large area of a world not necessarily surrounded by water.
- **Continent**: a Continent is a large area surrounded by water, that often has one or more islands nearby.
- **Island**: an Island is a relatively small landform surrounded by water.
- **Archipelago**: an Archipelago is a group of islands relatively near one another.
- **Atoll**: an Atoll is a ring-shaped group of low islands, usually found in tropical regions.
- **World**: a World usually contains many landforms of different types and may have polar icecaps or continents.

Using the Random Landform Generation Function

The Landform Generation function uses simplex noise and a randomly selected "shaping function" (a bitmap found in the Assets/Shaping Functions folder) of the chosen type to generate random landform shapes. Once the shape of the landform is generated, Realm Studio calculates the boundary of the landform and then applies shading and textures to create the landform as it appears on the map. When generating continents, archipelagos, and entire world maps that may include many individual landforms, the calculations can take up to several minutes. Realm Studio displays a progress bar when generating world maps so you can track the progress of the landform generation.

Occasionally, the randomly generated landform shape will be such that the boundary calculation fails. When that happens, the boundary may be displayed as a long straight line or will not completely enclose the entire landform. If that happens, you can either clear all the landforms and re-generate, select the landform with the incorrect boundary and delete it, or attempt to fix the boundary by erasing the part of the landform that has a boundary and let Relam Studio re-calculate the boundary. Attempting to fix the landform using the Eraser may not work, so you may have to delete the generated landform and try again.

If you click the Area button (under the Generate button), you can select an area of the map by clicking and holding the left mouse button and dragging the mouse. The selected area will be surrounded by a green dotted-line box. If you then select a landform type (continent, island, archipelago, or atoll) and click Generate, Realm Studio will generate a landform of the selected type *within the selected area*. By using the Area button to select areas to fill with generated landforms, you can quickly create an entire world, but with more control over the results than randomly generating a world using the World type.

When you select the World type, Realm Studio will fill the entire map area (any area selected is ignored) with a random number of generated landforms placed randomly on the map. Northern and southern polar icecaps/continents may also be generated (you will have to paint them to make them look like ice; initially, they are generated looking like the other generated landforms).

Once random landforms of any type are generated, they can be modified like landforms you create by painting.

Future versions of Realm Studio will expand on landform generation functionality and include automatic generation of cities and other map features.

Painting on Landforms



Just as for the Ocean layer, you can paint and shade colors on top of landforms. Painting on landforms works the same as it does on the Ocean layer.

Select a Color Brush, then click the Color icon to change the **Drawing Mode** to ***Landform Color***. The cursor will turn into the usual dashed circle with crosshairs. Paint color on top of your landforms by holding down the left mouse button and dragging. You can change the brush size and brush velocity using the sliders. Brush velocity controls how often color is painted as you move the mouse. This allows you to control the density of the color that is painted, allowing colors to be blended. Reducing the opacity of painted colors (making them more transparent) allows color to be blended more effectively also, often giving a better appearance.

Erase color that you have painted by clicking the Erase button. Change the painted color by clicking the Color Palette button and selecting a color from the Color Selection dialog.

You can also select a preset color by clicking one of the color presets. There are six default preset colors. As on the Ocean layer, you can create your own preset colors by clicking the "+" button on the right side of the main menu bar.

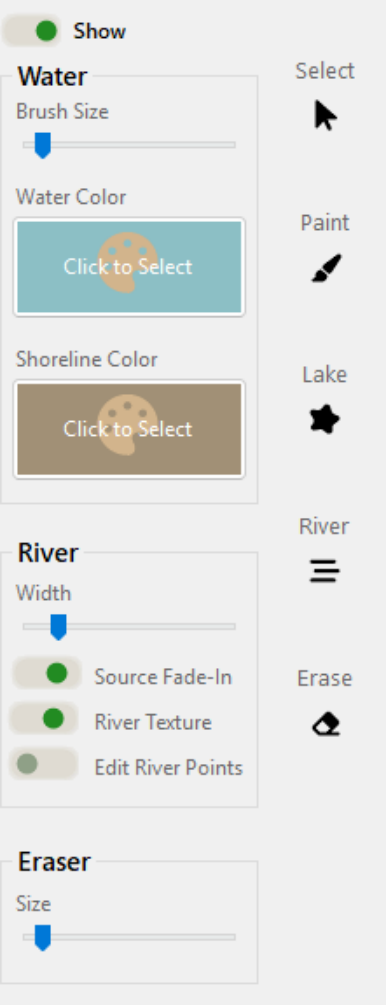
You can clear a preset color you have selected and holding the **`ctrl`** key and clicking the preset button. You can only clear preset buttons that you have defined; the default presets cannot be changed or cleared.

Clicking the eyedropper button changes the cursor to an eyedropper and lets you click anywhere on your map use the color at that location as the selected color for painting.

Colors painted on landforms are clipped to the outer boundary of the landform; that is, you cannot paint landform colors outside of the landforms. There is a setting on the Preferences dialog to disable clipping of painted colors.

The Water Tab

Painting Water Features



Realm Studio allows you to paint water features (lakes, rivers, and so on) on top of the landforms that you paint. Clicking the Paint button (with the paintbrush icon) changes the **Drawing Mode** to ***Water Feature Paint***. When in this mode, the cursor changes to the usual dotted-line circle with crosshairs. Holding down the left mouse button and dragging the mouse paints a water feature on top of a landform.

Water features are clipped to the boundaries of the landform, so you can't paint a water feature on top of ocean or background areas of your map.

Water features are painted with the selected Water Color and are shaded with a gradient derived from the color. The shoreline of the water feature is painted with the Shoreline Color and is shaded with a gradient derived from the Shoreline Color. You can change the Water Color and the Shoreline Color in the usual way by clicking on the Color Palette button.

Like with landforms, overlapping water features that you paint are merged into a single water feature (except for Rivers).

You can select a water feature by clicking the Select button, then clicking on the water feature. When selected, you can change or delete the water feature like you can with landforms. Right-clicking will open a dialog allowing you to give the water feature a name or change its colors.

Erasing Water Features

Clicking the Erase button with the eraser icon changes the **Drawing Mode** to ***Water Feature Erase***. When in this mode, the cursor changes to the usual dotted-line circle. Holding the left mouse button down and dragging it will erase painted water features and lakes. The Eraser Size trackbar slider adjusts the size of the eraser. As with landforms, the eraser can be used to sculpt painted water features that have already been painted to give them the detailed shape that you would like.

Painting on the Water Layer



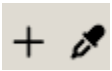
As with the ocean layer and landforms, you can paint on top of water features. The controls for painting on top of water features work the same way as they do for the ocean layer and for landforms.

Select the brush you would like to use (the gradient "soft" brush, the "hard" brush, or one of the pattern brushes). Click the Color button with the paintbrush icon to set the **Drawing Mode** to ***Water Feature Color***. Use the dotted-circle cursor to paint on top of water features (painted water features, lakes, or rivers). You can use painted color on top of water features to blend rivers with lakes and painted water features, to indicate areas of fast-flowing water, create swampy areas, and so on.

Like for landforms, color painted on water features is clipped to the boundaries of water features you have created, so you cannot paint water feature colors on top of landforms, ocean, or the map background.

You can erase color that you have painted on top of water features by clicking the Erase button.

You can make use of color presets and set your own preset colors in the same way as you do for ocean colors and landform colors.



Use the "+" button to choose a preset color and assign it to a button. Use the eyedropper to pick a color anywhere on the map to assign it as the paint color.

Creating Lakes

A lake can be generated with a random shape by clicking the Lake button, then clicking on a landform at the location you want a lake. Lakes are water features, like those that you paint, so overlapping lakes are merged into a single lake, and lakes that overlap water features that you paint (or have already painted) are merged with the painted water feature. The water brush size indicates the size of the lake that is generated (the boundaries of the cursor circle are roughly the boundaries of the generated lake).

Creating Rivers

In Realm Studio, rivers can be painted by clicking the River button, which changes the **Drawing Mode** to ***River Paint***. Rivers are water features; however, rivers **do not** merge with painted water features or lakes. You paint a river by clicking at the location of the source of the river, then holding down the left mouse button and dragging the cursor to the end point (mouth) of the river. As you are dragging, you can move the mouse to paint curves in the river.

You can taper the river from source to end by enabling the Source Fade-In switch (it is enabled by default). When the switch is enabled, the source of the river will start as a point, then gradually widen to the value set by the width trackbar slider, simulating how most rivers start as small streams, then gradually widen as more streams and rivers join. If you don't want your river to taper, disable the switch. Your river will be painted with the width set by the trackbar slider for its whole length.

By default, rivers are painted with a texture to simulate waves and ripples on a flowing river. You can disable the texture by disabling the Texture switch. The effect of applying the texture is subtle and might be difficult to see, especially against a textured landform background.

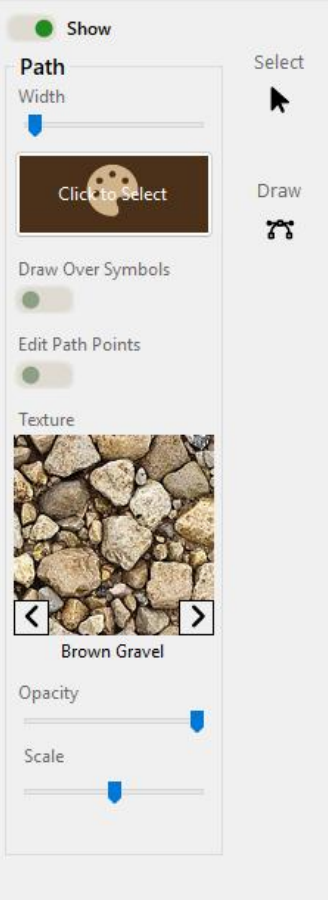
Rivers are painted with water color gradient shading and shoreline gradient shading. You can change the colors of the shading in the same way as for painted water features and lakes.

When a river is selected, (by clicking the Select button, changing the **Drawing Mode** to ***Select Water Feature***, then clicking on a river) enabling the Edit River Points switch will change the **Drawing Mode** to ***Edit River*** and display control points along the path of the selected river. Putting the mouse over a control point will highlight it. When a control point is highlighted, you can click and drag it to a new position, changing the shape of the river.

The Paths Tab

Drawing Paths

The controls on the Paths tab allow you to draw paths and walls (roads, trails, footprints, animal tracks, city and castle walls, and so on) on top of landforms. There are several different styles of paths that can be drawn, from simple lines to paths drawn with a border and filled with a texture. The are two styles of walls, differing only in whether the towers in the wall are round or square. Paths and walls can be drawn under symbols that are placed on the map (this is the default), or they may be drawn on top of placed symbols, if you need to achieve a particular effect on the map.

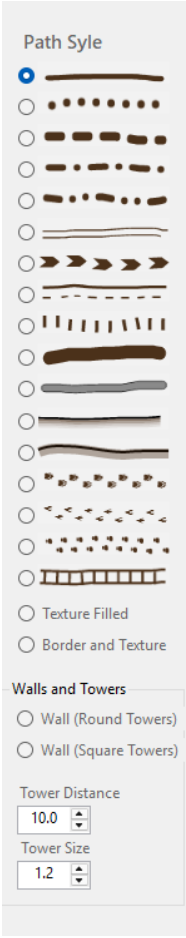


To create a path or wall, select the style you want from the set of styles on the right side of the Realm Studio main window (the path controls and the path styles are displayed when you click on the Path tab).

Select the width of the path or wall and the color using the trackbar and the Color Palette button. Use the Draw Over Symbols switch to set whether the path or wall will be drawn on top of placed symbols or under them.

Click the Draw button to begin drawing paths. Hold down the left mouse button and drag the mouse to draw the path or wall.

For Texture Filled and Border and Texture path styles, select the texture from the Texture preview. You can change the scale and opacity of the texture using the sliders to get the look you want for the path.



Walls with towers are created in the same way as Texture Filled or Border and Texture paths. Choose the texture for the wall from the Texture Preview. The look of the wall is often improved by scaling the texture down. The Tower Distance and Tower Size controls change the spacing of towers along the wall and their size. The values are relative to the width of the wall so that they remain proportional. A tower is always drawn at the start of the wall and then drawn spaced along the wall based on the Tower Distance value.

If you hold down the **`shift`** key while drawing a path, you can draw a straight-line path. The angle of the path is limited to 5º increments (0º, 5º, 10º, 15º…355º). Holding down the **`ctrl`** key

will draw a straight-line path that is either horizontal or vertical. The direction of the straight-line path is determined by the direction you drag the mouse cursor.

Editing Paths

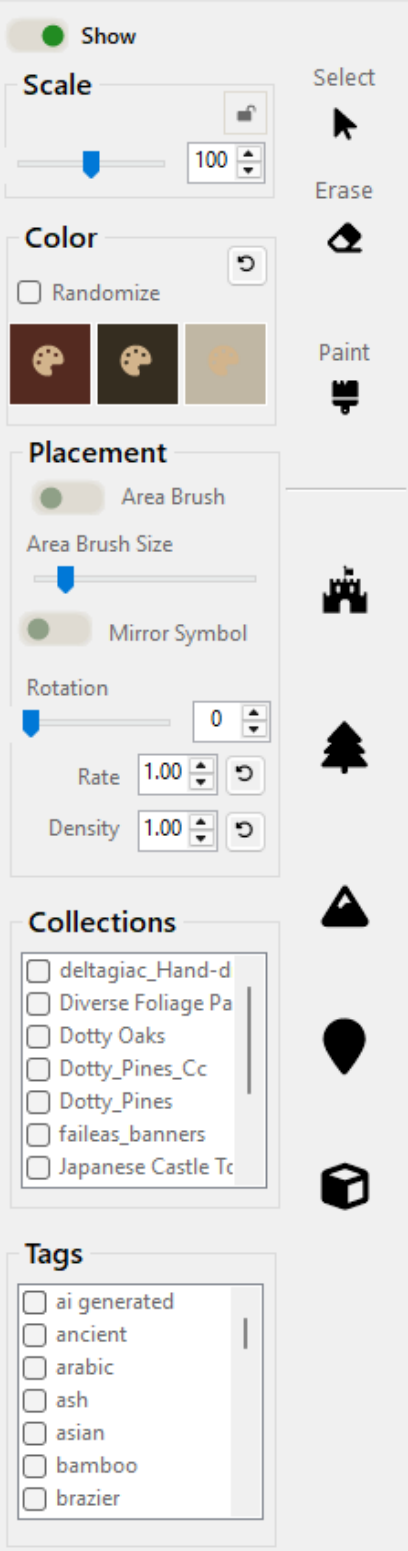
After you have drawn a path, you can delete it, move it, or edit individual points on the path.

Click the Select button to change the **Drawing Mode** to ***Select Path***, then click on a path to select it. A box will be drawn around the selected path. When selected, you can delete the path by pressing the Delete button. You can also move the path by holding down the left mouse button and dragging it to a new location.

If the path you have drawn is not quite the right shape, you can display control points along the path by selecting it and then enabling the Edit Path Points switch. Doing so will display control points as small white circles. Moving the mouse cursor over a point will highlight it in purple. You can click and drag a control point to a new location, changing the shape of the path. Pressing the Delete button will delete a highlighted control point.

The Symbols Tab

The Symbols tab controls allow you to place symbols from your Symbol Collections on your map.



The Scale trackbar slider and up/down control allow to scale (resize) symbols. The lock icon allows you to lock the scale, so all symbols you place have the same scale.

The Color Palette buttons allow you to choose three colors with which to paint the symbols you place. Checking the Randomize button will cause painted symbols to have slightly different colors than the selected color, giving groups of symbols (like a forest of trees) a little more interesting look. There is more on painting symbols below.

The controls in the Placement box allow you to enable the Area Brush (a dotted circle cursor) and set its size so that large areas of the map can be filled or painted quickly.

The Mirror Symbol switch causes the symbol to be flipped horizontally. The Rotation trackbar slider and up/down control rotate the symbol around its center. The Rate and Density up/down controls change how often symbols are placed and how close together they are placed.

The symbols you can choose from to place on your map are displayed in the Symbol List on the right of your map.

The Collections list shows all the Symbol Collections from which you can select symbols to place on your map. By clicking the checkboxes, you can filter the symbols displayed to just those from the selected collections.

The Tags list shows all the tags available for tagging symbols. The checkboxes next to the tags filter the symbols to just those with the selected tags.

The buttons below the divider display all the symbols of the associated Symbol Type (Structure, Vegetation, Terrain, Marker, and Other), select the first displayed symbol of that type, and let you start placing symbols. The symbols displayed are filtered by any collections or tags you have selected. You must select a Symbol Type to display and place symbols.

On the right of your map, the filtered Symbol List is displayed. If you haven't selected any Symbol Collections from the Collections list or tags from Tags list, all the symbols for the Symbol Type (Structure, Vegetation, Terrain, Marker, or Other) you click are displayed (from the Symbol Collections you have available).



Click a symbol to select it and place it on your map. When you select a symbol, it is highlighted in blue, and the mouse cursor changes to an image of the symbol. This is the Primary Selected symbol. You can change the size of the symbol by rolling the mouse wheel forward and back or by using the Scale trackbar slider. Place the cursor on your map by clicking where you want to put it. If the scale is locked, you cannot change the size of the symbol.

Holding the mouse button down and moving the mouse will place copies of the symbol as you move it. How close together and how often symbols are placed can be changed using the Rate and Density up/down controls.

When creating forests, mountain ranges, villages, or whatever, you may want to place many related symbols near each other. Realm Studio allows you to do this easily. First select a symbol by clicking on it. Then, hold down the **`shift`** key and click on the related symbols you want to place. They will be highlighted in a lighter shade of blue. The light blue selected symbols are the "Secondary Selected" symbols. Hold down the left mouse button and move the mouse over your map where you want to place symbols. Realm Studio will randomly choose one of the symbols you've selected and place it each time it places a symbol. To clear the list of selected symbols, just click on another symbol.

You can use the Area Brush switch to enable the Area Brush when painting Vegetation or Terrain symbols. When the Area Brush is enabled, the dotted-line circle cursor is displayed. As you hold the left mouse button and move the mouse, symbols are painted within the area of the circle cursor. This works best with small symbols to fill large areas (like forests). You can also use the Area Brush to paint large areas filled with grayscale symbols with a color. The Area Brush also works with Secondary Selected symbols to place randomly selected symbols..

You can search for a symbol (or symbols) by typing in the Search box above the Symbol List. You must type at least three characters to filter the list.

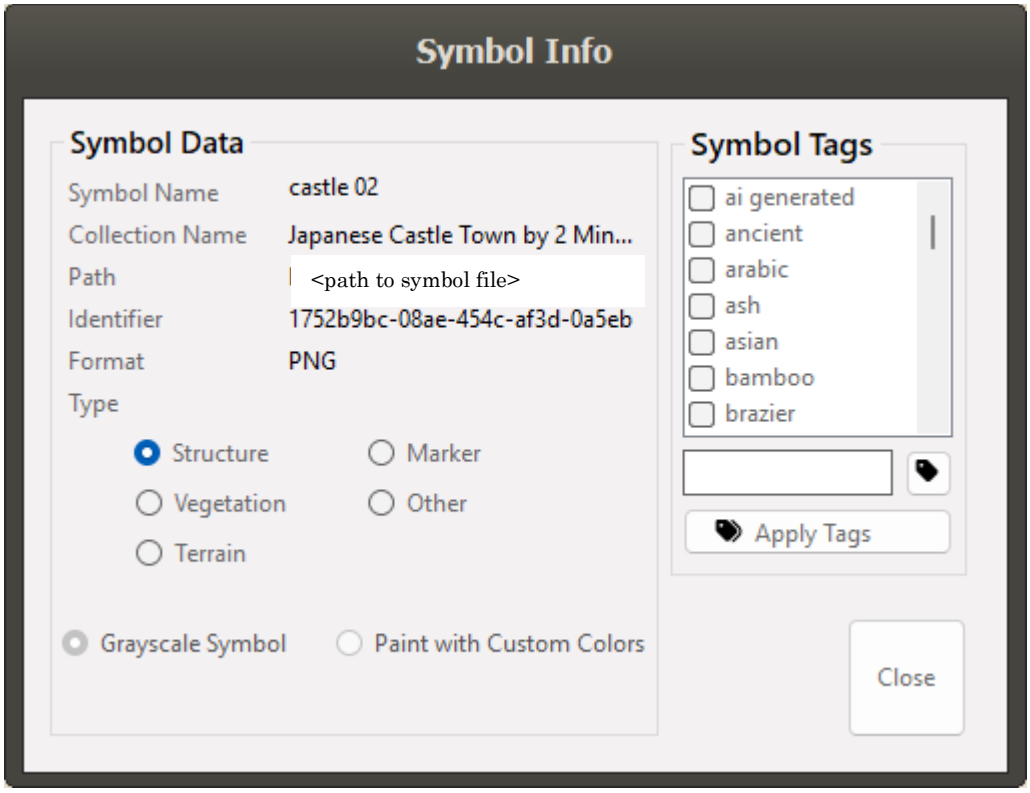
Selecting Symbols

To select a symbol that has been placed on your map, click the Select button (with the arrow icon). The **Drawing Mode** will change to ***Select Symbol***. Click on the symbol to select it. Realm Studio draws a green dashed box around a selected symbol

Displaying Symbol Information

Realm Studio will display information about symbols in two ways.

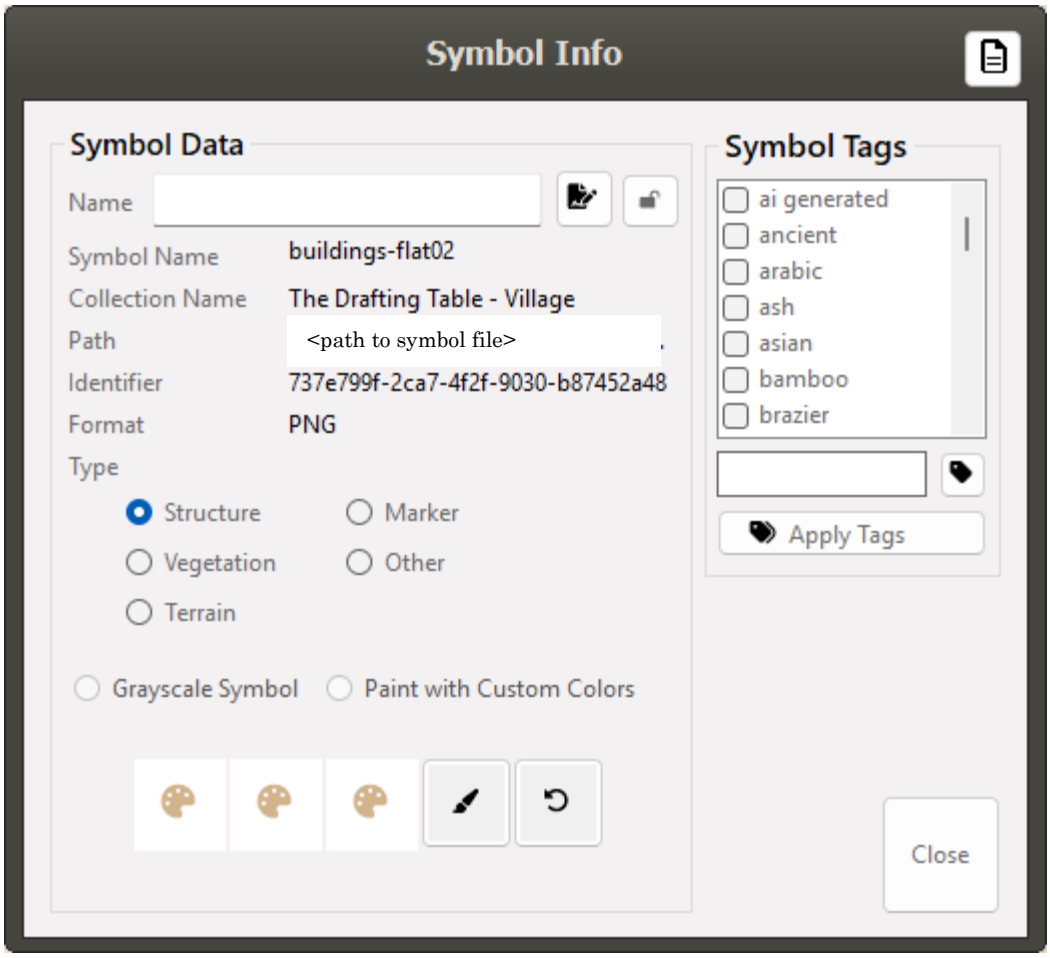
First, if you right-click on a symbol in the Symbol List, a dialog box will be shown with some information about the symbol.



You can change the tags for the symbol by selecting them in the Symbol Tags box, then clicking the Apply Tags button. You can also add new tags to the list of available tags by entering a tag in the text box below the Symbol Tags list, then clicking the Tag icon button next to it.

You can also change the Type of the symbol (but this should almost never be necessary).

Second, you can display information about a symbol that has been placed on the map. To do so, click the Select button, then right-click the symbol on your map. A dialog with information about the selected symbol will be shown:



This dialog allows you to name the symbol (for example, if the symbol indicates the location of a tavern, you can put the name of the tavern in the Name field). The button next to the Name field will generate a random name for the symbol using the Name Generator. The Lock button next to it will lock the name, preventing it from being accidentally changed. You can also change the tags for the selected symbol, create a new tag (as described above), and if the symbol is grayscale or paintable with custom colors, paint the symbol with the selected color(s). To paint the symbol, select colors with Color Palette button, and then click the Paintbrush button.

You can add a description for the symbol by clicking the button in the Symbol Info dialog title bar. Clicking the button opens the Description Editor for the symbol and allows you to use the AI Assistant to help you generate a description of the map object represented by the symbol. For example, if the symbol represents a tavern in your world, you can use the AI Assistant to help you write a description of the tavern.

Erasing Symbols

To erase symbols that have been placed on your map, click the Erase button (with the eraser icon). Like other erasing functions, the cursor will change to the dotted-line circle cursor. Hold down the left mouse button and move the mouse over your map. All symbols within the circle

cursor will be removed from the map. You can also delete a symbol by selecting it and pressing the **`Delete`** key on your keyboard.

Moving Symbols

You can move a symbol that has been placed on the map by selecting it, then holding down the left mouse button and moving the mouse to move the symbol to a new location. The symbol will move along with the mouse cursor. Release the left mouse button to place the symbol. You can also move a selected symbol using the right/left/up/down arrow keys on your keyboard.

Painting Symbols

Grayscale symbols placed on your map can be painted with a single color. The color of symbols paintable with custom colors can also be changed, but in a different way, using the three colors from the Symbol Tab Color Palette buttons.

Grayscale symbols can be painted individually or using the Area Brush; however, only Vegetation and Terrain grayscale symbols can be painted with the Area Brush. Structure and Other symbols cannot be painted with the area brush.

To paint a grayscale symbol, select it, then either right-click one of the three Color Palette buttons on the Symbol tab, or select it and click the Paint button (with the paintbrush icon). Right-clicking one of the three Color Palette buttons will paint the grayscale symbol with the color of the right-clicked button. Clicking the Paint button will paint the selected symbol with the color of the first (leftmost) Color Palette button. You can also paint grayscale symbols quickly by selecting the Symbol Type button corresponding to the symbols that you want to paint, selecting the color you want to paint the symbols in the first Color Palette button, then clicking and holding the left mouse button and dragging the crosshairs cursor over the symbols.

Just above the Color Palette buttons is a checkbox labeled “Randomize.” By checking this checkbox, when you paint symbols with the crosshair cursor or with the area brush, the color that is painted onto the symbols will vary by 10% from the selected color. This gives features like forests and rows of houses some variation in color, improving their appearance. Future versions of Realm Studio will improve this feature.

Symbols that are paintable with custom colors start out looking like this:



As you can see, colorable symbols are painted black, white, green, blue, and red. This is how paintable symbols will appear on the Add/Update Symbol Collection dialog when you create the symbol collection. In the Symbol List and when the symbol is placed on your map, the red, blue,

and green colors are replaced by the three colors shown on the three Symbol tab Color Palette buttons, so it will look something like this:



Red is mapped to (replaced by) the color of the first (leftmost) Color Palette button. Green is mapped to the second (middle) Color Palette button. Blue is mapped to the third (rightmost) Color Palette button.

You can change the colors of a custom colorable symbol either by selecting it and changing the colors on the Color Palette buttons or by selecting and right clicking the symbol to open the Symbol Info dialog for the symbol and changing the colors on the Color Palette buttons in the Symbol Info dialog. When you change the symbol colors from the Symbol Info dialog, the colors of the symbol are changed when you close the dialog.

Changing the Rendering Order of Symbols

When you place symbols on your map, they sometimes overlap, so a symbol is "in front" of another symbol. Often, you want the symbols to overlap, but you want the symbol currently in front of another symbol to be behind it.

Here is an example:



In this case, you might want the tall green tree to appear as if it is behind the ruined building. By selecting the tree and pressing the `Page Up` and `Page Down` keys on your keyboard, you can change the order that the symbols are rendered (drawn by Realm Studio), making it appear that the tree is behind the ruined building (or you can select the building and change its position).

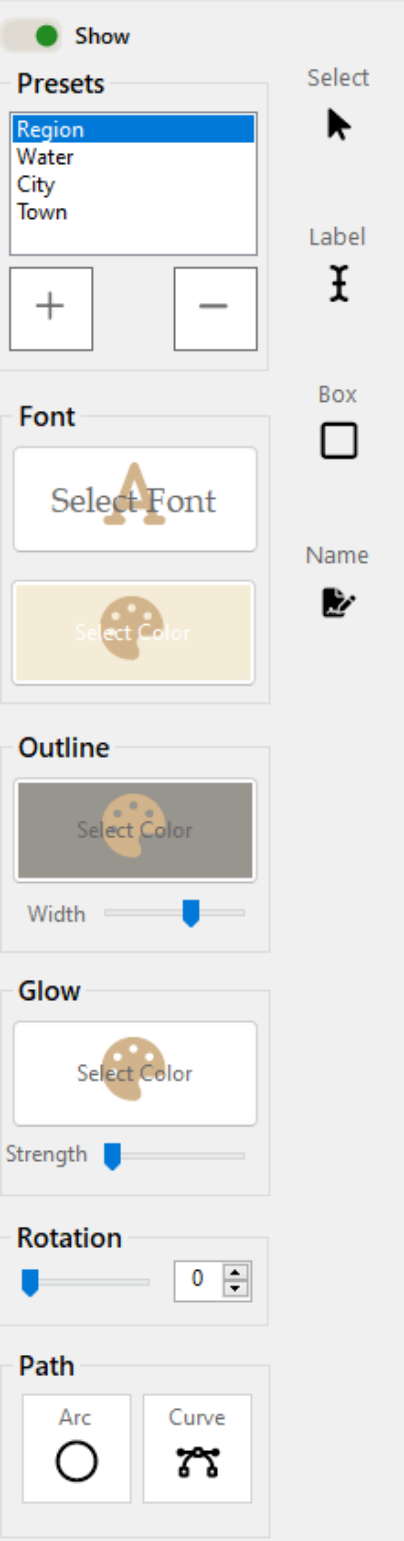


Symbols that are rendered later will appear to be in front of symbols that are rendered earlier. The `Page Up` key moves the selected symbol one position later in the rendering order; the `Page Down` key moves it one position earlier. Holding down the `ctrl` key while pressing `Page Up` or `Page Down` moves the symbol five positions later or earlier. If you have a lot of symbols on your map (like, for example, with a forest), it can be tedious to get the symbol positioned in the right place in the rendering order. To help with that, you can press the `Home` and `End` keys while a symbol is selected. Pressing the `Home` key moves the symbol to the top of the rendering order (behind all other symbols). Pressing the End key moves the symbol to the bottom of the rendering order (in front of all other symbols).

Future versions of Realm Studio will enhance the controls for setting the symbol rendering order.

Labels Tab

The controls on the Labels tab allow you to place labels and boxes on your map.



Label Presets allow you to quickly set the font, font size, outline, and glow for your label.

Clicking the "+" button allows you to create a named preset from the currently selected font, font size, outline, and glow from the current settings. Clicking the "-" button allows you to delete a Label Preset. The pre-defined default Label Presets cannot be deleted.

Clicking the Select Font button opens the Font Selection panel. Select a font and font size from the drop-down lists and the select font style (bold or italic) using the buttons. Clicking the large "A" button increases the font size. Clicking the small "A" button decreases the font size. Click the OK button to set the label font. The label of the Select Font button changes to use the font you selected for the text of the button label.

Use the Font box Color Palette button to select the color of the label text.

Use the Outline box Color Palette button to select the color of the outline around your label text. Use the trackbar slider to set the outline width.

Use the Glow box Color Palette button to select the color the "glow" around your label. Use the Strength trackbar slider to set the area of the glow and its opacity. Experiment with this to get the appearance you want, as it can be difficult to see the glow against some backgrounds.

The Rotation trackbar slider and up/down control are used to rotate a selected label.

A label can follow an arc or curve you draw. Clicking the Arc and Curve buttons sets the **Drawing Mode** so that you can draw an arc or curve for a label to follow.

Creating a Label

To create a label, click the Label button (with the I-beam icon). The **Drawing Mode** is set to ***Place Label***. Click on the map where you want to place your label. (First, be sure to set the label font, size, outline, and glow). A text box will be placed on your map where you clicked, with the prompt "…Label…" in it. Type the text of your label in the text box (what you type will replace the prompt), then press the **Enter** key. The text box will be replaced with your label, and the label will have the style (font, size, etc.) that you selected. The label will be selected (have a dotted line box) drawn around it. When the label is selected, you can move it by dragging it with the mouse to get it to the exact position you want.

Selecting a Label

To select a label, click the Select button (with the arrow icon) on the Label tab. The **Drawing Mode** will change to ***Select Label***. Click on the label. The selected label will have a dotted line box drawn around it.

Changing the Appearance of a Label

You can change the appearance of a label that has been placed on the map. To do so, select the label, then use the controls (as shown above) to change the font, font size, color, outline, and/or glow styles. The appearance of the label will change to reflect the styles you choose.

Deleting a Label

To delete a label, select it then press the **Delete** key on your keyboard.

Rotating a Label

You can rotate a label around its center by selecting the label, then using the Rotation trackbar or the Rotation up/down control to rotate it. The label can be rotated from 0º to 359º in 1º increments. The label will rotate as you move the slider or change the rotation value in the up/down control.

Creating a Label that Follows an Arc

To create a label that follows an arc, you must first draw the arc that the label will follow. Click the Arc button at the bottom of the Label tab. The **Drawing Mode** changes to ***Draw Arc Label Path***. Move the cursor to approximately where you want the label to be placed, then click and hold the left mouse button down while moving it to the right and down. You can also move the mouse to the right and up to create an arc that curves up. An arc will be drawn with a dotted line as you move the mouse. Release the left mouse button when the arc is the size and curvature that you want.

Next, click the Label button (with the I-beam icon) and create a label as described above. The label will be created when you press the **Enter** key, but rather than being placed wherever the label text box was placed, the label will follow the arc that you drew, like this:



Just as when a "normal" label is created, the label following an arc is initially selected and can be moved, rotated, deleted, or have its font and style changed.

A label that is placed on an arc that curves tightly will probably not look very good, so keep that in mind when creating curved labels.

Creating a Label that Follows a Freeform Curve

Clicking the Curve button allows you to draw a freeform curve that your label will follow. The process is the same as when creating labels that follow an arc: Click the Curve button. The **Drawing Mode** will change to ***Draw Curve Label Path***. Draw the curve by holding down the left mouse button and moving the mouse. Release the mouse button when you are done drawing the curve. If you don't like the curve you've drawn, press the `Escape` key. Pressing the `Escape` key cancels all pending operations. Once you have a curve you like, proceed to create the label as described above. Just as when creating a label that follows an arc, when you press the `Enter` key in the label text box, the label will be created following the curve you drew, like this example.



For both labels that follow arcs and labels that follow freeform curves, when the label is created, it is left-justified on the arc or curve. It can take some practice to create an arc or curve that fits the size and shape of the label you want to create. Like for arcs, tight curves will probably not look very good, especially for larger fonts.

Creating Box Frames



Realm Studio allows you to place boxes of various styles onto your map as frames around labels. There are several styles of boxes that you can choose from, and additional styles can be downloaded from Cartography Assets. Boxes that are usable with Wonderdraft can be used with Realm Studio. Like for other types of assets, Wonderdraft box assets must be imported into Realm Studio before they can be used. When the box asset is imported into Realm Studio using the Import Wonderdraft Assets dialog (as discussed above), an XML file is created that describes the box asset to Realm Studio. Once the XML file is created, the box asset can be loaded by Realm Studio.

To create a box frame on your map, select the box style that you want to use from the list on the right of your map. It will be highlighted in blue. Click the Box button (with the square icon), then hold down the left mouse button and move the mouse down and right to draw the box. Release the mouse button when the box is the size and shape you want.

Boxes can be "tinted;" that is, their background color can be combined with a selected color. You can select the tint color by clicking the Color Palette button above the list of box styles and selecting a color from the color selection dialog. You can change the tint of a box that has already been placed by selecting it, then clicking the "Select Tint Color" Color Palette button and choosing a color from the dialog.

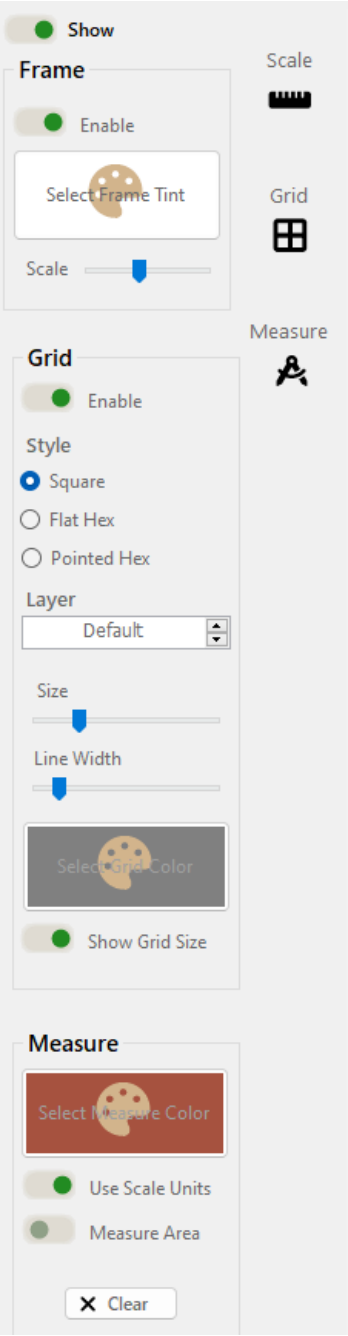
A box that has been placed can be moved by selecting it, then holding down the left mouse button and moving the mouse. The selected box will follow the mouse cursor as it moves.

A box that has been placed can be deleted by selecting it and pressing the **`Delete`** key on your keyboard. You can undo the deletion by selecting the **`Undo`** menu item from the **`Edit`** menu or by pressing **`ctrl+z`** on your keyboard.

Labels on your map will be drawn in front of any boxes that are placed, so boxes can be used (and are intended to be used) as frames around labels to create titles, legends, etc. for your maps. Simply drag any labels you want to position them over the box that will act as the frame for the labels.

Overlays Tab

The Overlays tab includes controls for creating frames around your map, placing grids (square and hexagonal), generating a scale for your map, and for measuring distance and area.



Map Frames

At the top of the tab in the Frame box, the Enable switch enables and disables rendering of the frame surrounding your map. The Frame Tint Color Palette button allows you to apply a tint to the frame you select (in the same way that boxes are tinted). The Scale trackbar changes the size (width) of the frame. The frame will always surround your map, even as it is scaled larger or smaller. You select the style of the frame you want to place on your map from the list of frame styles in the panel to the right of your map.

Grids

In the Grid box, you can enable/disable and change the type, size, and color of a grid placed on your map. The Layer up/down control allows you to select where your grid will be rendered relative to other objects drawn or your map. By default, the grid will appear under boxes, labels, and the map frame, but over everything else. However, the grid can be placed so that it appears over the background and ocean layers, but under everything else, or above landforms and water features, but under symbols, paths, boxes, labels, and the frame. When the Show Grid Size switch is enabled, Realm Studio will print text in the lower-left corner of your map indicating the size of each square or hexagon in the grid using the map area units. To place the grid on your map, click the Grid button. You can change the frame style after it has been placed on your map.

Measuring Distance and Area

You measure distance in your map using the Measure function. To measure distance, click the Measure button., then click the left mouse button where you want to start measuring and move the mouse to the end of where you want to measure. A line will be drawn in the color of the Measure Color Palette button. At the endpoint of the line, text is drawn indicating the length of the line. If the Use Scale Button switch is enabled, distance will be given in map area units (e.g. miles).

You can continue measuring by clicking the left button again and moving the mouse. The cumulative length of the lines is drawn at the end of lines. Right click the mouse to stop measuring.

You can also measure approximate area using the Measure function. First, enable the Measure Area switch. After enabling the Measure Area switch, the process is the same as for measuring distance, except as you move the mouse, an area will be highlighted, and text will be drawn giving area rather than distance. You must have set at least two points before area will be displayed.

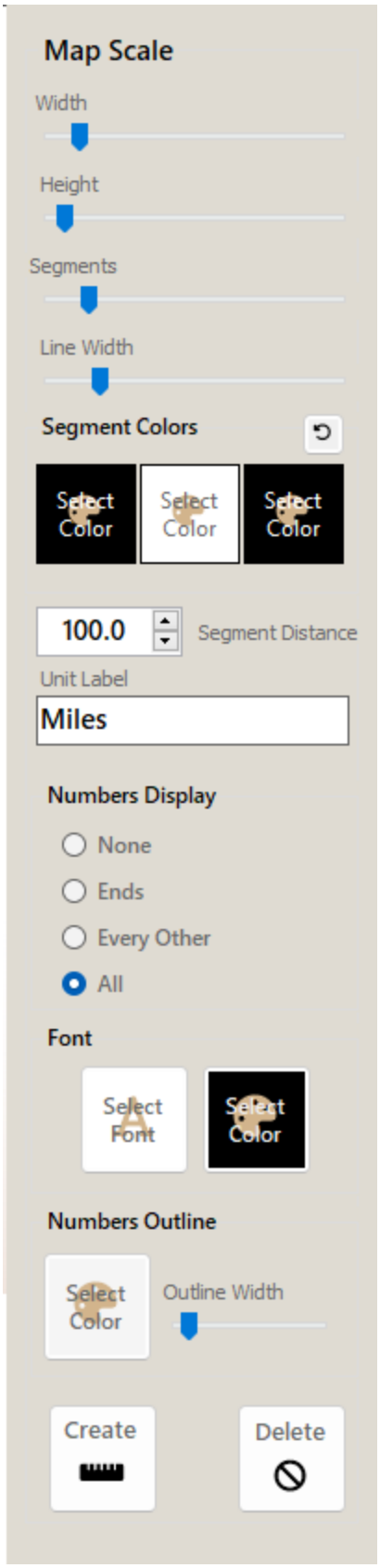
Clear the measuring lines or area by clicking the Clear button.

Choosing a Frame

When the Overlay tab is selected, a list of frame styles you can choose is displayed to the right of your map. To display a frame on your map, just select the frame style you like from the list. Once the frame is placed, you can change the scale and tint or disable the frame using the controls in the Frame box.

As with symbols and other kinds of assets, more frame styles are available for download at Cartography Assets, some free and some for purchase. Frames that can be used with Wonderdraft can (usually) be used with Realm Studio. Click this link to take you to a list of Wonderdraft frames on Cartography Assets that you can download for use with Realm Studio. Zip files containing frame styles downloaded from Cartography Assets must be imported using the Import Wonderdraft Assets dialog described earlier. More detailed information on the process for importing assets created for Wonderdraft into Realm Studio is provided in the Technical Information section at the end of this document.





Creating a Map Scale

Clicking the Scale button displays a panel with controls for creating a scale indicating distance on your map. A map scale looks something like this:



There are several controls for changing the appearance of map scales. The Width and Height trackbar sliders change the size of the map scale. The Segments slider changes the number of distance segments in the scale. The Line Width slider changes the width of the map scale outline.

The Color Palette buttons change the color of the alternating segments and the outline.

The Segment Distance up/down changes the distance labels displayed above the map scale. The text in the Unit Label text box is drawn as the label below the map scale. The default text for this box comes from the map scale units chosen in the Realm Configuration dialog, but you can change the label by entering it in the Unit Label text box.

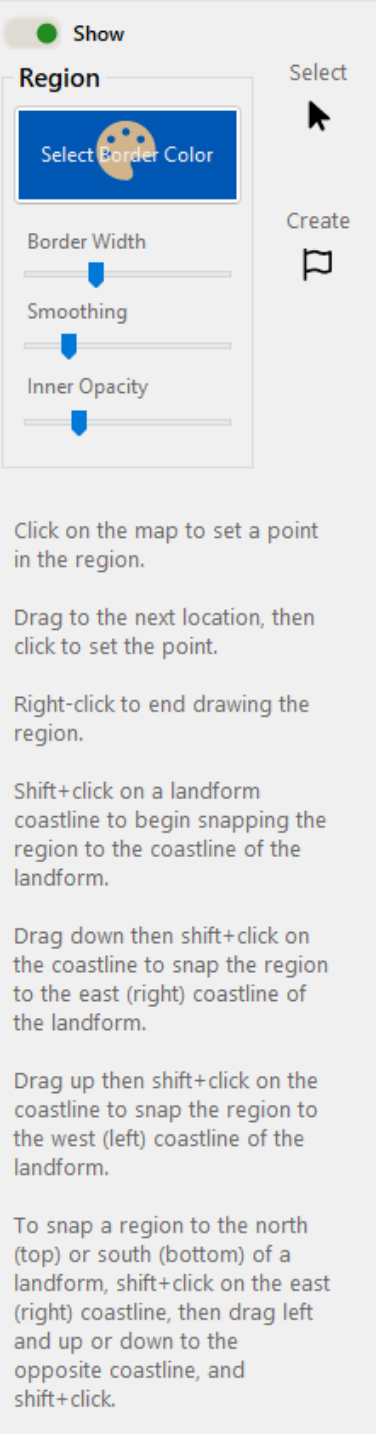
The Number Display radio buttons control where the map scale distance segment labels are displayed.

You can change the font, font color, outline color, and outline width of the numbers and unit labels using the Select Font button, Font Color Palette, Numbers Outline Color Palette button, and Outline Width slider.

Once you have set the map scale style as you like, click the Create Button. A map scale will be created and placed at the lower left corner of your map. You may have to pan the display of your map using the scroll bars or by holding down the middle mouse button and moving the mouse to be able to see the map scale. You can move the map scale by clicking on it and moving the mouse while holding down the left mouse button. The map scale will follow the mouse cursor.

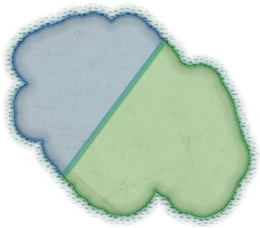
Regions Tab

The controls on the Regions tab allow you to create highlighted and bordered regions on your map. A Region could be a geographical region, a country or some other political entity, or whatever else you like. Creating Regions is relatively simple, but there are some operations that are a bit complex, so there is some text on the Regions tab to help remind you how to do them.



To create a Region, set the border color, border width, smoothing (how rounded the corners of the region are), and how clear or opaque the inner area of the region is. Click the Create button (with the flag icon). The **Drawing Mode** will change to ***Draw Region***. Click at a point where you want to place the Region. Drag the mouse to the next point in the Region and click the left mouse button. Continue clicking and dragging to set each point of the region. As you click and drag, the region will be drawn on the map with the style you selected. Right click to end drawing the region.

You can snap a Region to the coastline of a landform. This operation can take a little bit a time to learn how to do. The process is described in the text on the Regions tab. Unlike Wonderdraft, which requires you to wait for a lengthy calculation to complete before you can snap a Region to a coastline, Realm Studio lets you do so immediately. Hold down the `shift` key and click on the coastline. When you get near to the coastline, a small circle is displayed, indicating that the cursor is near enough to the coastline to snap the region to the coastline. If you drag ***up***, then hold down the `shift` key and click on the opposite coastline, the border of the Region will snap to the west (left) coast of the landform. If you `shift+click` on the coastline, then drag ***down*** and `shift+click` on the opposite coastline, the border of the region will snap to the east (right) coast. Snapping to the north (top) or south (bottom) coast of a landform uses the same process, as described in the text on the Regions tab.



Above is an example of two regions snapped to a coastline.

Once a Region has been placed, it can be moved or edited. Click the Select button, then click on the Region. Control points on the border of the Region will be displayed. You can move the entire Region by dragging it with the mouse. You can also move a control point by highlighting it (by moving the mouse cursor over it), then dragging it. Clicking on the border of the Region creates a new control point at that location.



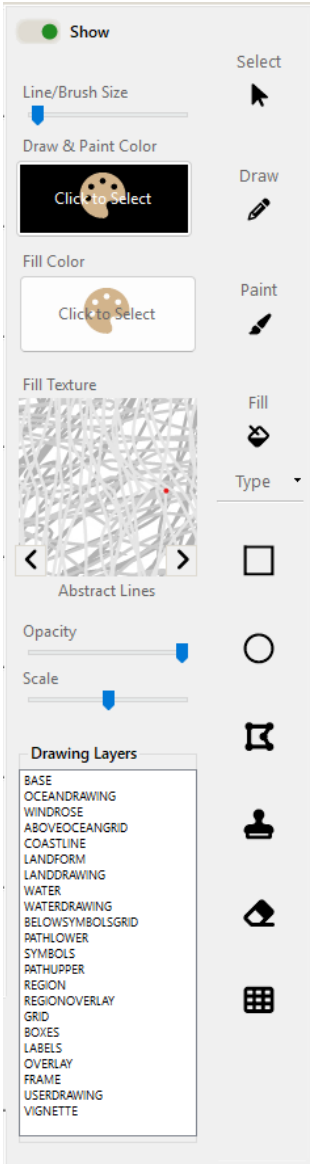
You select the border style of the Region using the radio buttons on the panel to the right of your map.

Delete a Region by selecting it and pressing the `Delete` key.

Right clicking a selected region will open a dialog allowing you give the region a name, change the color, smoothing, inner opacity, and border style

Drawing Tab

The Drawing tab includes controls to allow you to draw lines, paint lines with a brush, place shapes and bitmap stamps anywhere on your map. In addition, a Pixel Editor is included. The Pixel Editor shows a selected 64x64 pixel section of the map allowing changes to individual pixels of the map. The drawn and painted line, and the placed shapes can be added to nearly all the layers of the map. Drawing, painting, placing shapes, and pixel edits can be undone/redone like other drawing operations on the map.



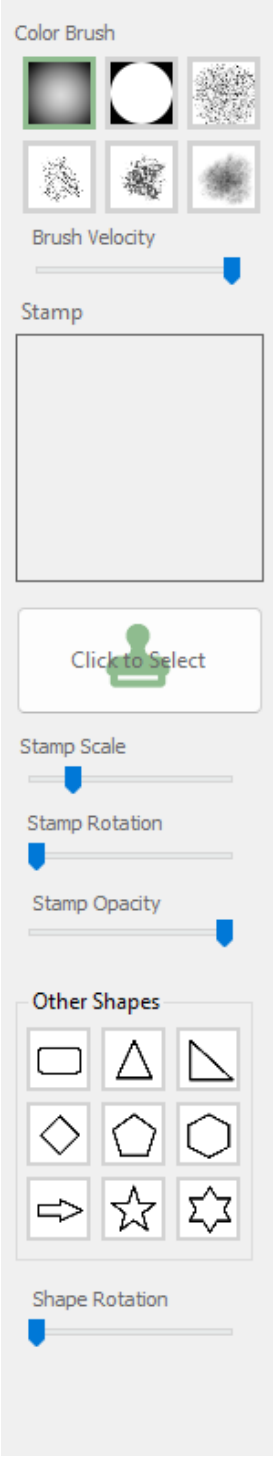
On the left-hand side of the Drawing tab, the main drawing controls are displayed, as on other tabs.

Shapes are drawn or painted with an outline color and can be filled with a color or a texture. Clicking the Fill button will cause the drawn shape to be filled. Clicking the Type dropdown button will display a small dropdown that allows you to select either Color fill or Texture fill. If you select Texture fill, the shape will be filled with the Fill Texture displayed. If you are painting a line (by selecting the Paint button) and have selected Texture fill, the line will be painted with that texture. The opacity and scale of the texture can be changed using the associated sliders. You can draw a shape using only the selected fill color or texture (without an outline) by selecting a Draw & Paint Color and setting the Alpha channel value to 0 (transparent).

The Drawing Layers list allows you to select the map layer that the drawn lines, painted lines, and placed shapes will be placed on. By default, drawn and painted lines and shapes are placed on the USERDRAWING layer, which is above all other layers, except the map vignette. Clicking on a layer in the Drawing Layers list will select that layer for drawing. The selected layer for drawing is displayed on the bottom right of the Realm Studio window in light blue text. Layers are rendered in order from the top of the list to the bottom, so the BASE layer appears under all other layers, and the VIGNETTE layer is drawn above all other layers.

Under the Fill button are basic controls for drawing shapes, placing stamps, erasing, and pixel editing. Clicking the square button lets you draw squares and rectangles. Holding down the **`Ctrl`** key forces the shape to be square. Clicking the circle button lets you draw circles and ellipses. Holding down the **`Ctrl`** key forces the shape to be a circle. Clicking the polygon button (under the circle button) allows you to draw a polygon with an arbitrary number of sides. Clicking the left mouse button sets the vertex of a polygon side. Click the right mouse button to stop drawing the polygon. Clicking the stamp button allows you to place a bitmap stamp. The stamp is selected from controls on the right side of the Realm Studio window. The eraser button allows you to erase anything drawn on the selected layer, even from map objects drawn using

controls on other tabs. The grid button changes the cursor to a 64x64 pixel square. Placing the square cursor over a section of the map and left clicking opens the Pixel Editor with that section displayed for editing. More information on the Pixel Editor later.



On the right-hand side of the Drawing tab is a panel displaying additional controls for drawing and placing shapes, painting lines, and placing stamps.

At the top of the panel are buttons for selecting the brush to be used when painting lines. The selected brush is outlined in green and is displayed at the bottom of the window in green text next to the Drawing Mode.

Clicking the green Stamp button allows you to select a bitmap (in .png, .jpg, or .bmp format) to be placed anywhere on the map. The File Selection dialog defaults to the `Stamps` subfolder of the Realm Studio `Assets` folder. The selected stamp bitmap is displayed in the Stamp picture box. The scale, rotation, and opacity of the stamp bitmap can be changed using the sliders; however, scale and rotation changes are not displayed in the Stamp picture box. The stamp as it will look when it is placed is shown as the application cursor when the Stamp button on the main Drawing Tab area is clicked.

The Other Shapes buttons allow additional shapes to be selected and drawn. They work in the same way as the square and circle button on the left side of the window. Click the button for the shape you want to draw, then click and drag the left mouse button to draw the shape.

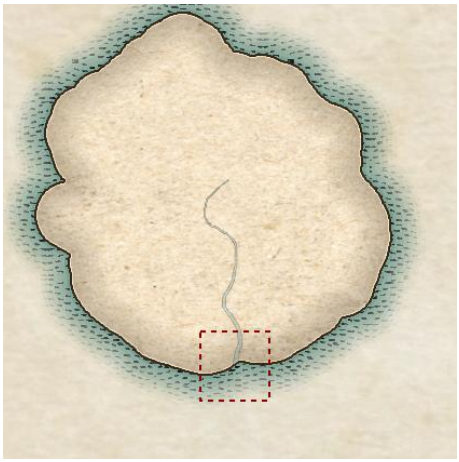
A drawn shape that has been placed can be selected when the Select button on the main Drawing Tab area is clicked, setting `the Drawing Mode` to ***Select Drawn Object***. To select a drawn or painted object, make sure the same layer is selected as the layer that the object is placed on. Once a drawn object is selected, it can be moved by holding down the left mouse button and dragging, rotated by using the Shape Rotation slider, or deleted by pressing the Delete key. Note that Stamp Rotation and Shape Rotation are different operations. A stamp can be selected and moved or deleted, but it cannot be rotated or have its opacity or scale changed once it has been placed.

Placed stamps are not the same as Symbols placed from the Symbols tab. You cannot add a name or description to a Stamp, nor can their colors be changed. Stamps are displayed just as they are, except for changing scale, rotation, and/or opacity. The same is true for placed shapes. You cannot add a name or description to a placed shape. Because of this limitation (and for

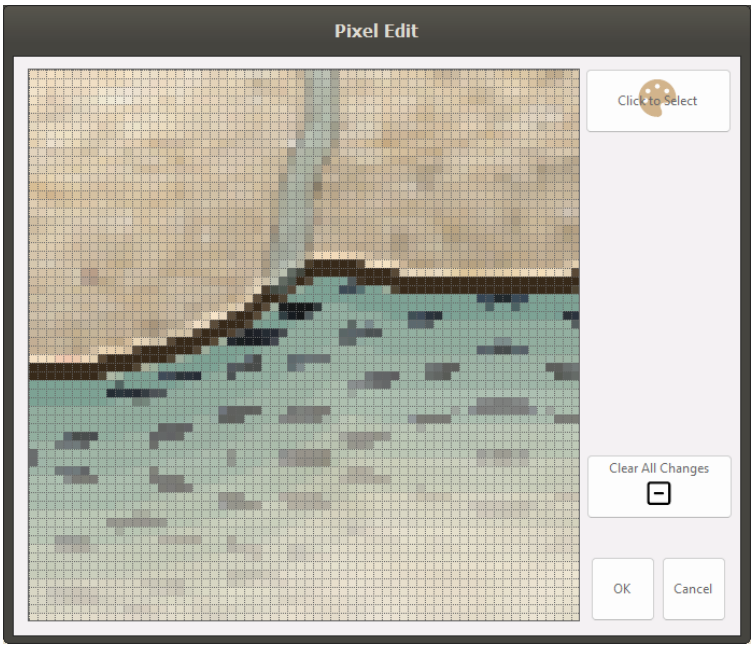
future integration), landforms, bodies of water, regions, paths, and symbols should be implemented using the controls on the associated tabs. Drawn lines, painted lines, placed shapes and stamps are intended to be used to enhance the appearance of your map where other objects do not allow you to achieve the look you want.

Using the Pixel Editor

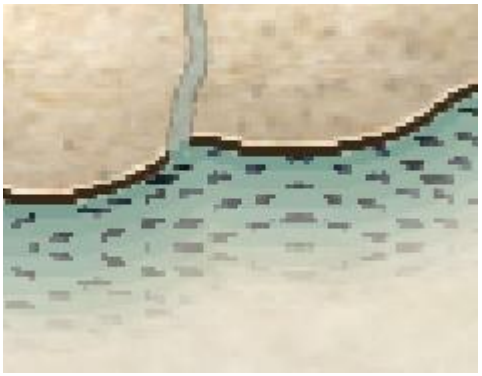
The Pixel Editor allows you to edit individual pixels of your map. Clicking the Grid button on the main Drawing Tab displays a 64x64 pixel square cursor that you can move to the place on your map where you would like to edit individual pixels. For example, say you have drawn a landform with a river that flows into the ocean, and you would like to improve the transition from the river into the ocean. It might look something like this:



Placing the square cursor over the mouth over the river and clicking will open the Pixel Editor showing the selected area.



Use the Palette button to select a color. You can change the color of individual pixels by left clicking the pixel or by holding down the left mouse button and dragging over the pixels. Right clicking a pixel or holding down the right mouse button and dragging over pixels reverts them to their original color. After the pixel edits are made, click the OK button in the Pixel Editor to display the result in your map.



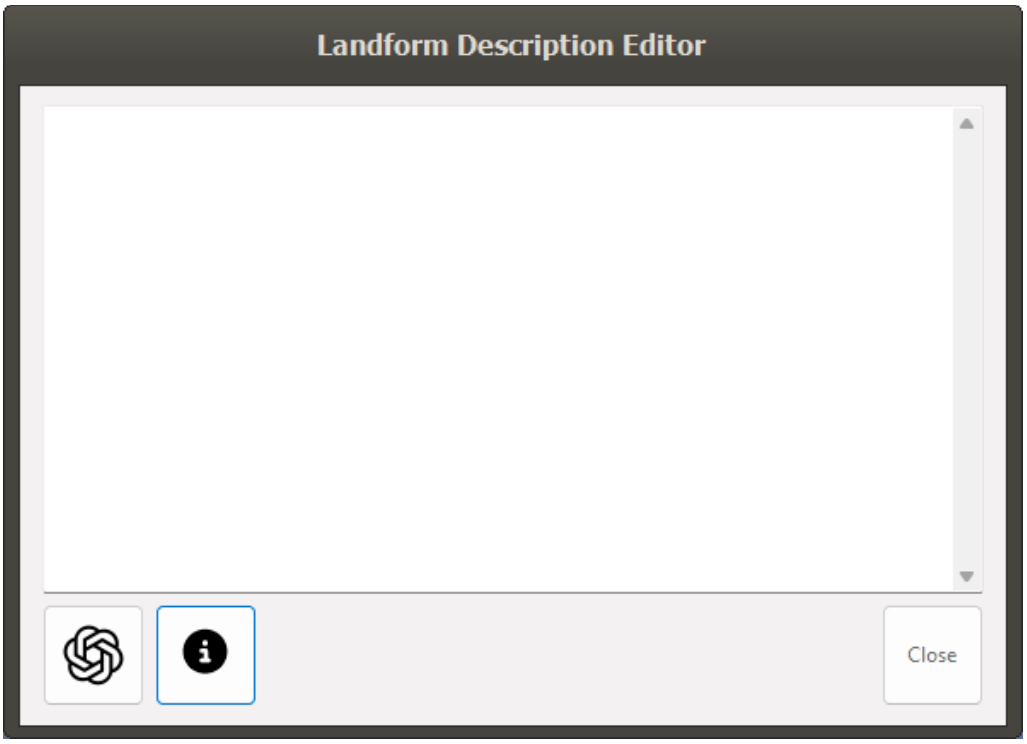
If you don't like the pixel edits you have made, you can revert all the changes using the Clear All Changes button.

Like other changes made using the controls on the Drawing tab, pixel edits can be undone/redone using the usual **ctrl+z** and **ctrl+y** keyboard and the Undo and Redo menu commands.

The Description Editor

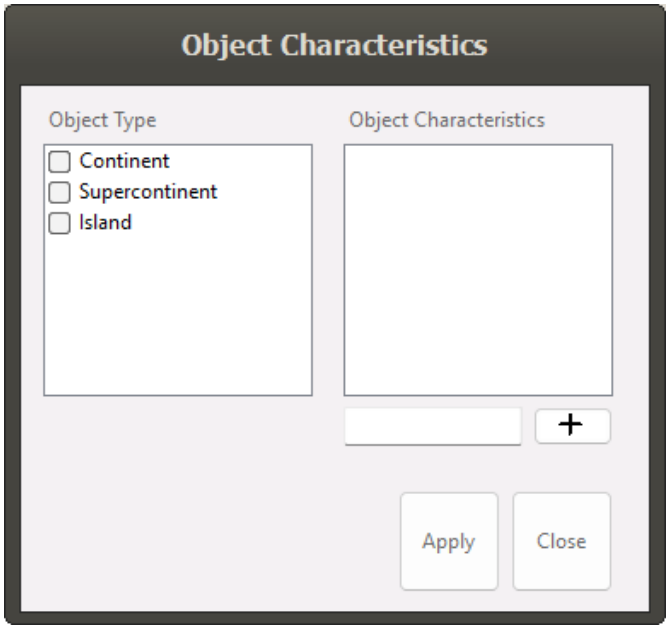
Landforms, water features (lakes, etc.), rivers, paths (roads, trails, streets), symbols, and regions all allow you to enter a description for them. The Description Editor dialog is opened from the button on the title bar of the Info dialog for the map object (landform, water feature, path, region, etc.). The editor is a simple plain text editor, but it also includes AI assistance for writing the description of the object.

The AI Assistant is aware of the kind of map object for which it is generating a description, so if you want assistance with a description for a landform, it will generate a description of a landform, and if you want assistance with a description for a river, it will generate a description of a river, and so on. However, you can give the AI hints on what you want in the description, so that the generated description is more likely to match what you have envisioned for the object. The Description Editor looks like this. Note that the title bar text changes depending on the type of object being described.

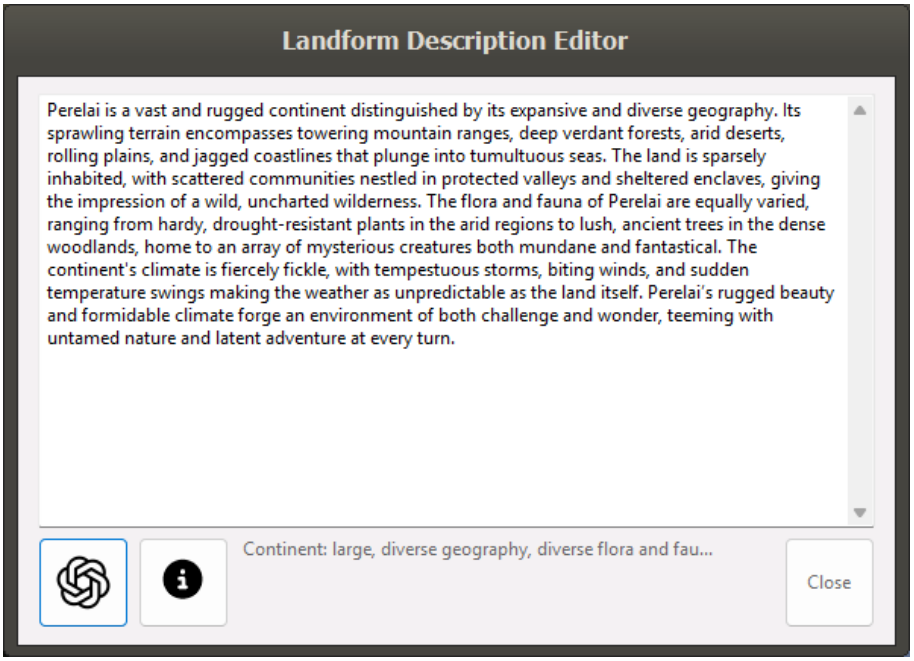


Clicking the Info button (the round "i" button) opens the Object Characteristics dialog, shown below. On the Object Characteristics dialog, you can select an option to further refine the object type. For example, rather than describing a generic "Landform," you can tell the AI you want a description of an island or a continent. The list of Object Types changes depending on the type of the map object (Landform, Water Feature, River, Symbol, or Region). In addition to the Object Type, you can give the AI Assistant more hints to help it in generating a description you like. By typing a descriptive adjective or phrase (e.g. hot, cold, humid, temperate, crowded, etc.) in the text box next to the "+" button, then clicking the "+" button or pressing the **Enter** key, the

adjective will be added to the list of Object Characteristics. Clicking the Apply button will send the selected Object Type and Characteristics to the Description Editor.



When you are satisfied with the object type and characteristics you've selected, clicking the button with the OpenAI logo will make a request to the AI Assistant to generate a description that you can use as inspiration (or use as-is). Here's an example of what the AI Assistant can generate:



You can edit the generated description so that it fits your vision for the object or click the button again for a new AI-generated description.

Zooming, Panning, and Mouse Functions

Realm Studio generally responds to the mouse in the same way as other Windows applications do.

You select objects by first changing the **`Drawing Mode`** to select the object type you want to select, placing the mouse cursor over the object and clicking the left mouse button. When an object in Realm Studio is selected, a dotted-line box is drawn around it.

You can "drag" some objects by selecting them, then holding down the left mouse button while moving the mouse.

Some objects, such as landforms, water features, lakes, rivers, symbols, paths, and regions will display informational dialogs when you set the **`Drawing Mode`** to select the object, then click the right mouse button on the object. The dialogs allow you to change attributes of the selected object (object name, textures, colors, description, and so on).

While painting landforms and painting color on ocean, landforms, and water features, and some other operations, you can quickly change the size of the dotted-circle cursor by rolling the mouse wheel forward and backward (up and down).

You can zoom the map using the controls at the bottom right of the application window:



The trackbar zooms the map. The button with cross arrows zooms the map so that the entire map fits into the map display. The button with circular curved arrows resets the zoom level to 100% (the default).

You can also zoom the map by holding down the **`ctrl`** key and rolling the mouse wheel forward and backward (up and down).

You can pan the map back and forth and up and down using the scrollbars or by holding down the middle mouse button (which is usually the mouse wheel) and dragging the mouse.

Pressing the **`escape`** or **`esc`** key resets the **Drawing Mode** to ***None*** and cancels any pending operation.

Realm Studio Assets and Symbol Collections

Creating a Symbol Collection

As mentioned above, symbols are small images (usually `.png` files or `.svg` files) of various types of objects that can be placed on your map as representations of those objects – mountains, hills, houses, taverns, trees, rocks, banners, mermaids, clouds, and many, many others.

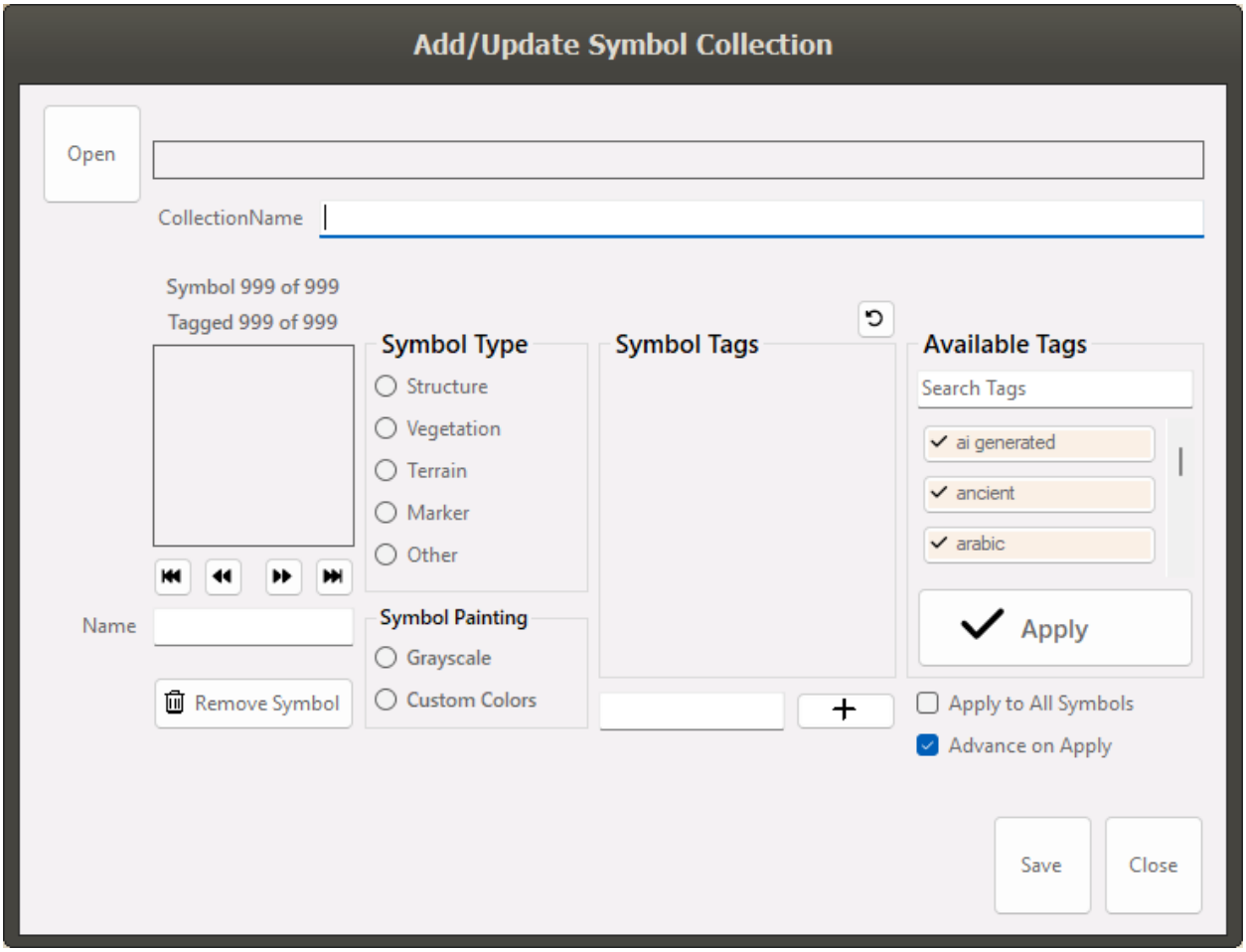
You can create your own symbols and include them in Symbol Collections if you like, but most map makers use collections of symbols created by artists. Websites like Cartography Assets (https://cartographyassets.com/) have packs (usually zip files) of many kinds of assets (symbols and others) that can be downloaded, both for free and for sale. Realm Studio can import and use assets created for Wonderdraft (https://www.wonderdraft.net/). More information on downloading and importing symbols created for Wonderdraft is below. Unlike Wonderdraft, Realm Studio can also use vector files in the Scalable Vector Graphics (`.svg`) format as symbols. The SVG Repo site (https://www.svgrepo.com) has thousands of SVG files that you can download and use in Realm Studio Symbol Collections that you create. Unlike symbols in other image formats, symbols created from SVG files do not become pixelated (blocky) when they are resized.

Future versions of Realm Studio may be able to import symbols created for other mapmaking software.

To use symbols on your map, they must be added to a Symbol Collection. A Realm Studio Symbol Collection has two parts: 1) a subfolder for the collection found in your Windows Documents folder under the RealmStudio/Assets/Symbols subfolder that contains all the image files for the symbols, and 2) a `collection.xml` file that describes the collection and all the symbols in the collection. The `collection.xml` file is stored in the same folder as the symbol images for the Symbol Collection.

Since you may (probably will) have several thousand different symbols to use when making maps, Realm Studio uses Symbol Collections to organize your symbols so that you can find the ones you want to use more quickly. In addition, Realm Studio allows you to tag the symbols so that all the symbols with selected tags can be quickly retrieved for use on your map.

Symbol collections originally created for use in Wonderdraft do not have a `collection.xml` file that Realm Studio can read. However, Realm Studio can load the `wonderdraft_symbols JSON` file provided with the Wonderdraft collection. The Add/Update Symbol Collection dialog allows you to tag and categorize the symbols in the collection and then create the Realm Studio `collection.xml` file for the collection. The dialog looks like this:



Creating a Symbol Collection for Realm Studio can be a tedious process, but Realm Studio does its best to make it quick and easy. First, it examines the file name and name of the symbol and of the collection and applies tags to the symbol based on them. So, if the symbol contains the word "house," for example, it will apply the "house" tag to the symbol. Similarly, Realm Studio will try to derive the Symbol Type (Structure, Vegetation, Terrain, or Other) based on the tags it applies. There are synonym files in the Assets folder that are used to associate tags with symbol types.

You must select a Symbol Type for each symbol in the collection, but you don't have to apply any tags to the symbols in the collection. However, symbol tags can help you find symbols more quickly. For example, if you are creating a village on your map, you might want to gather all the symbols of houses to see which ones you want to use. By tagging your symbols, you can easily do that by filtering the symbols by tag on the Symbols tab. There is more information on how to filter symbols below.

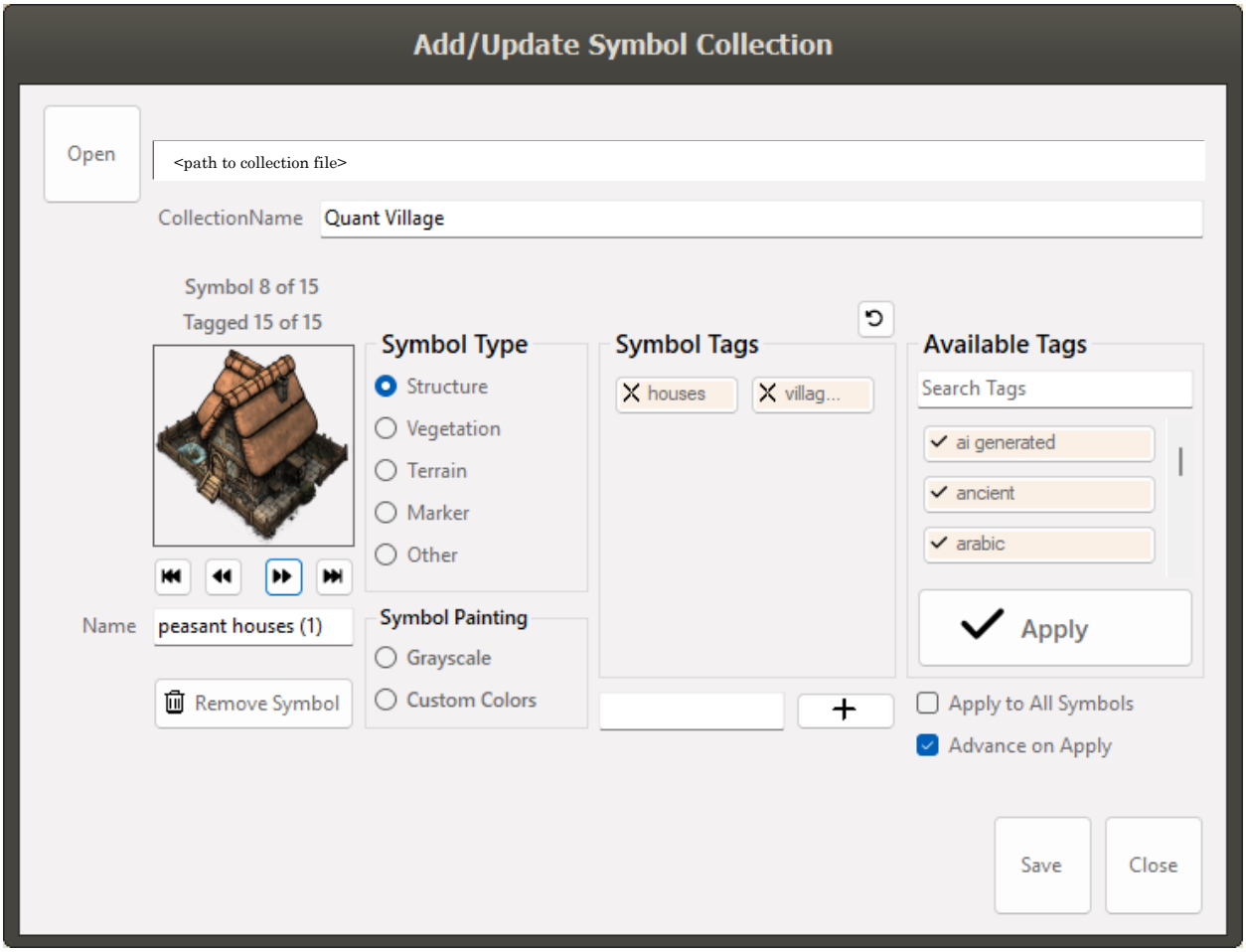
There are many symbol collections provided when Realm Studio is installed, but not all have had a `collection.xml` file created for them. If you want to use the symbols in the collection,

use the process outlined in the next section to categorize and tag the symbols and create the `collection.xml` file.

IMPORTANT! If you download symbols from Cartography Assets and create Realm Studio symbol collections, ***keep a backup copy of the assets and the `collection.xml` file***. When you install a new version of Realm Studio, the assets are also re-installed, and you will need to restore the assets and `collection.xml` files that you have created. In a future version, the Realm Studio installer will not overwrite or remove `collection.xml` files

Using the Add/Update Symbol Collection Dialog to Create a Symbol Collection

1. Click the Open button at the top-left of the dialog. This will open a Windows Folder Selection dialog. Select the folder under the Windows Document/RealmStudio/Assets/Symbols folder that the collection is in. The dialog will show the collection path and name in the boxes in the dialog, and the first symbol in the collection will be displayed, as shown:



   Use the buttons under the image of the symbol to navigate back and forth through the set of symbols in the collection.

Often, the artist that created the symbol collection will have included preview images or other images in the collection that are not symbols to be included in a map. If such an image is shown in the preview window, you can click the Remove Symbol button, and that image will be excluded from the Realm Studio symbol collection.

As you can see in the image above, Realm Studio has figured out the symbol type and has automatically tagged the symbol. At this point, you can change the name of the symbol, change the symbol type, and add or remove symbols tags. You can also add new tags to the list of available tags, and the new tags you add will be saved.

2. If Realm Studio was not able to figure out the symbol type, select a type using the Symbol Type radio buttons.
    a. The **Structure** type is for things that have been made by people: buildings, bridges, fences, signs, and so on.
    b. The **Vegetation** type is for plants of all kinds.
    c. The **Terrain** type is for mountains, hills, cliffs, ravines – anything that is part of the natural landscape that doesn't grow,
    d. The **Marker** type is for symbols that indicate locations or points of interest on the Realm Map.
    e. The **Other** type is for miscellaneous symbols that don't fit into the previous categories.
3. In the Symbol Painting group, you can select if the symbol image is grayscale (black, white, and shades of gray) or can be painted with custom colors.

    Grayscale symbols can be painted with a single color using the controls on the Symbols tab.

    Symbols that can be painted with custom colors are initially displayed in the Add/Update Symbol Collection dialog using red, green, blue, black, and white. When they are displayed on the Symbols tab and on your map, the red, green, and blue areas of the symbol are replaced by the three custom colors selected on the Symbols tab. There is more information on coloring symbols below.

    Very rarely will you have to change what is automatically selected here. Realm Studio can almost always figure out which symbols are grayscale and which can be colored.

4. The tags currently applied to the symbol are shown in the Symbol Tags box. The tags that are available to apply to the symbol are shown in the Available Tags box. To add a tag to the symbol, click on the tag in the Available Tags box. To remove a tag from the symbol, click on the X next to the tag in the Symbol Tags box. When you click the X, a small dialog with buttons to confirm or cancel removal of the tag is shown. Clicking the green button confirms removal of the tag; clicking the red button cancels removal.



5. Once you've assigned a Symbol Type and any tags you want to apply to the symbol, click the Apply button. Clicking the Apply button commits the changes you've made (all the values shown on the dialog) to that symbol. If the "Advance on Apply" checkbox is checked, the next symbol in the collection will be shown in the dialog. If you click the Apply button without assigning a Symbol Type, a warning will be displayed, and the dialog will not advance to the next symbol in the collection.
6. Once all the symbols in the collection have been assigned a Symbol Type, and they have been tagged, click the Save button. When the Save button has been clicked, the `collection.xml` file for the Symbol Collection will be created.

   If you open a collection that already has a `collection.xml` file, Realm Studio will read the file to determine the Symbol Type and tags for each symbol. You can then modify the Symbol Type and tags, as well as the symbol name, and then save the changes by clicking the Save button.

   If you click the Save or Cancel button before all the symbols have been assigned a Symbol Type, a warning will be displayed to remind you to assign a type to all the symbols in the collection.

Importing Assets

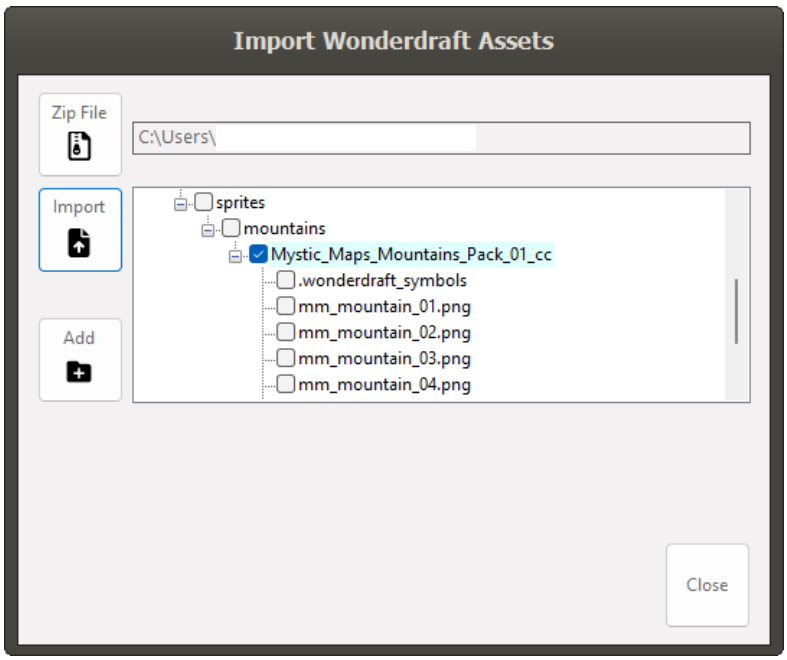
Importing Wonderdraft Assets from a Zip File

Wonderdraft assets downloaded from the Cartography Assets website are packaged as zip files. Realm Studio provides a dialog to import the assets from the zip files into your Documents/RealmStudio/Assets folder. The dialog is opened from the `Assets -> Import Assets From -> Wonderdraft Assets Zip File...` menu option. It looks like this:



To import assets from a downloaded zip file, follow these steps:

1. Click the Open button. Realm Studio will display a File Open dialog. Navigate to the folder containing the zip file containing the assets (usually they will be in your Downloads folder) and click Open in the File Open dialog. The path to the zip file will be shown in the box next to the Open button.
2. Click the Import button. Realm Studio will read and analyze the zip file to find the folders in the zip file containing assets of any kind that Realm Studio uses (textures, frames, boxes, symbols, and so on). The folders and assets are displayed in a tree structure in the box next to the Import button with the folders containing assets highlighted with a color indicating the type of asset in the folder:

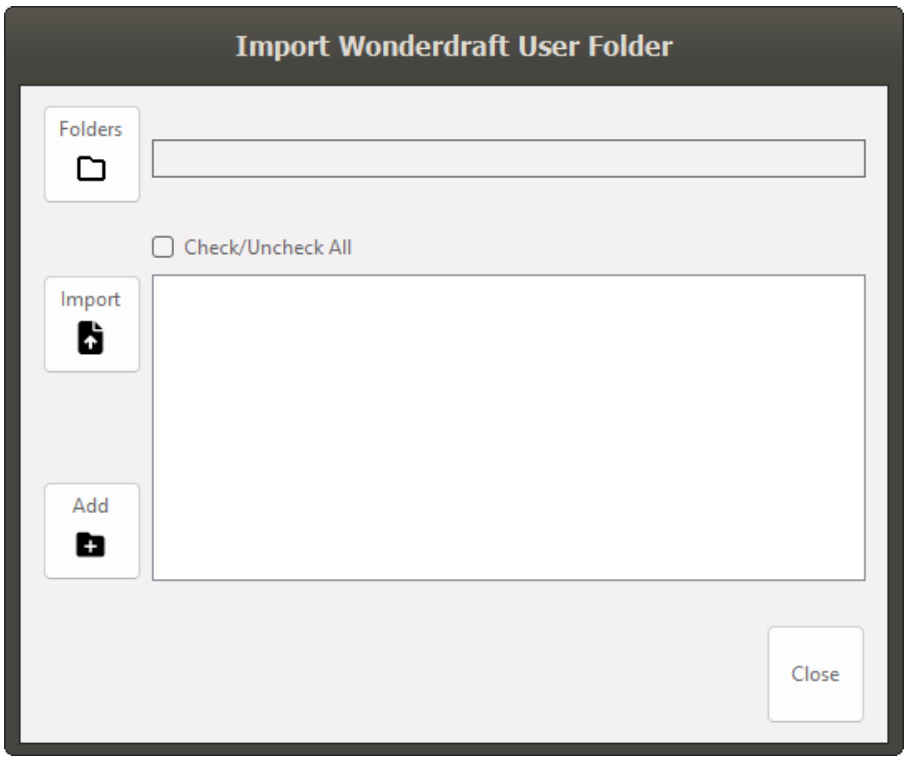


You can select and deselect the assets you want to add to Realm Studio by clicking the check box next to the highlighted folders.

3. Click the Add button. Realm Studio will display a message asking you to confirm that you want to copy the assets from the zip file into the Realm Studio Assets directory. When you click OK. Realm Studio will copy the assets. When the assets have been copied, a message is displayed to let you know. Important Note: copying Symbol assets from the asset zip file into the Realm Studio assets directory does not create the collection.xml file for the symbols. You must use the Add/Update Symbol Collection dialog described previously to create the collection.xml file before the symbols are available to be used by Realm Studio.

Importing Assets from the Wonderdraft User Folder

If you have already purchased and installed Wonderdraft, Realm Studio is able to import assets from the Wonderdraft user folder. Selecting the **`Assets -> Import Assets From -> Wonderdraft User Folder...`** menu option will open a dialog similar to the Import Wonderdraft Assets dialog described in the previous section.



This dialog operates in almost the exact same way, also.

1. Click the Folders button to open a Windows Folder Selection dialog that allows you to select the subfolders of the Wonderdraft user folder from which you want to import assets. Realm Studio automatically locates the Wonderdraft user folder and sets the default location of the Windows Folder Selection dialog to that folder.
2. Select the Assets subfolder of the Wonderdraft user folder, then click the Select Folder button on the Windows Folder Selection dialog.
3. Back in the Realm Studio Import Wonderdraft User Folder dialog, click the Import button. Realm Studio will find all the folders containing assets, and display them in a tree structure, as in the Import Wonderdraft Assets dialog.
4. Select the folders holding assets you want to import, then click the Add button. Realm Studio will import the assets from the selected Wonderdraft folders into the Realm Studio assets folder. For imported symbol collections, you must still use the Create Symbol Collection dialog to complete importing Wonderdraft symbol collections into Realm Studio. Other types of assets don't require any additional steps to make them available in Realm Studio.

The 3D World Globe View

Realm Studio allows you to use the 3D Viewer to construct a dramatic scene and view your map as a world globe in space, animate the globe showing it rotating around its polar axis, and take a snapshot or record an AVI video of the scene with the world globe rotating in the scene. The 3D World Globe View has a lot of options to customize the scene, including adding clouds, displaying a local star (a sun or other main object), moons, a planetary ring, an atmospheric effect, and several video effects (blurring, bleach bypass, bloom, sepia toning, and grayscale). The video effects can be combined to create the look you want.

To display the World Globe View, select Display World Globe from the Realm option on the Main Menu. This is how it looks when it is initially opened.



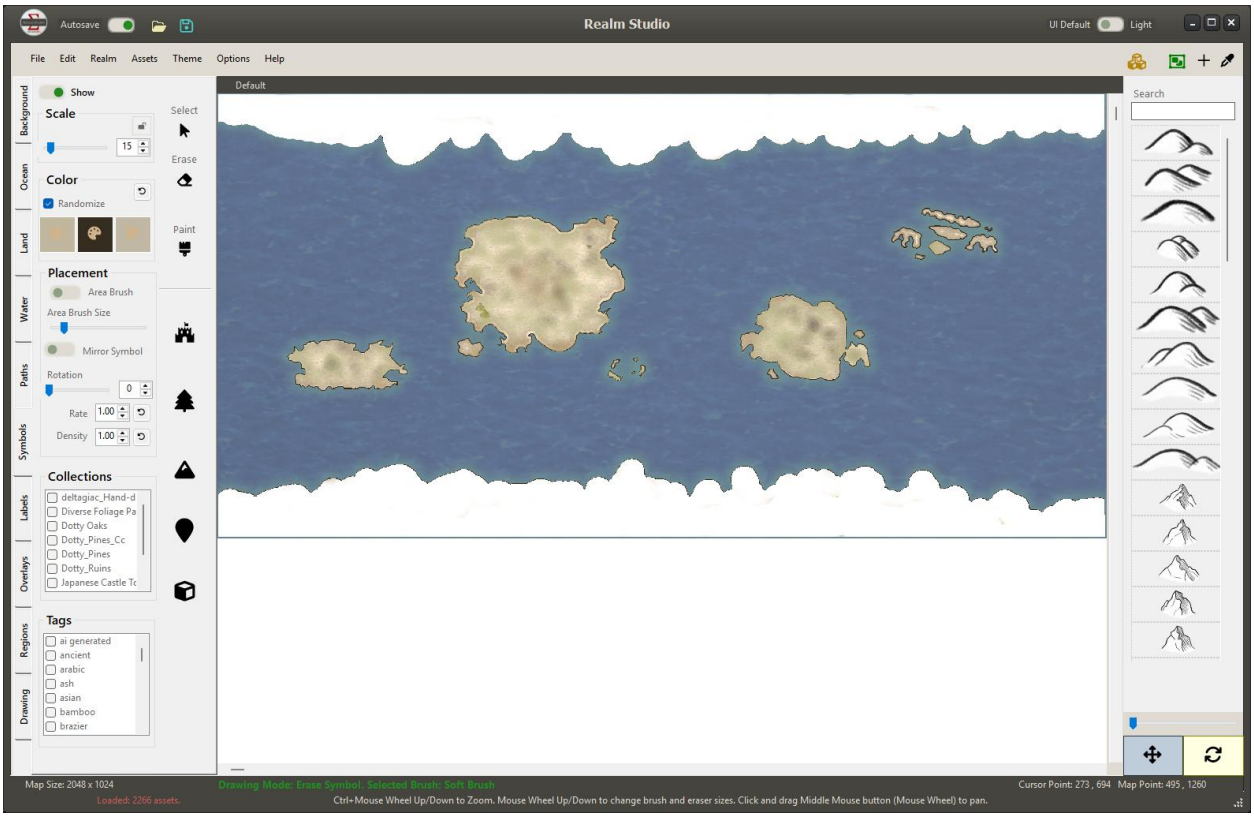
When the World Globe View is displayed, your world map is converted to a texture (bitmap picture) and "wrapped" around the sphere that represents your world. The math used to wrap a rectangular picture (your map) onto a sphere works best when the picture is "equirectangular;" that is, when it is twice as wide as it is high. The Realm Configuration dialog has a couple of map size options (2048 x 1024 and 2048 x 4096) that are equirectangular, so that map you create will wrap onto a sphere minimizing visible problems. Maps that are not equirectangular will typically leave an obvious seam when wrapped onto the globe.

The controls in the Model, Camera, and Scene panels work the same way as they do in the 3D Viewer. The controls in the World Animation panel are specific to the World Globe view, and that panel is only displayed when the World Globe view is shown.

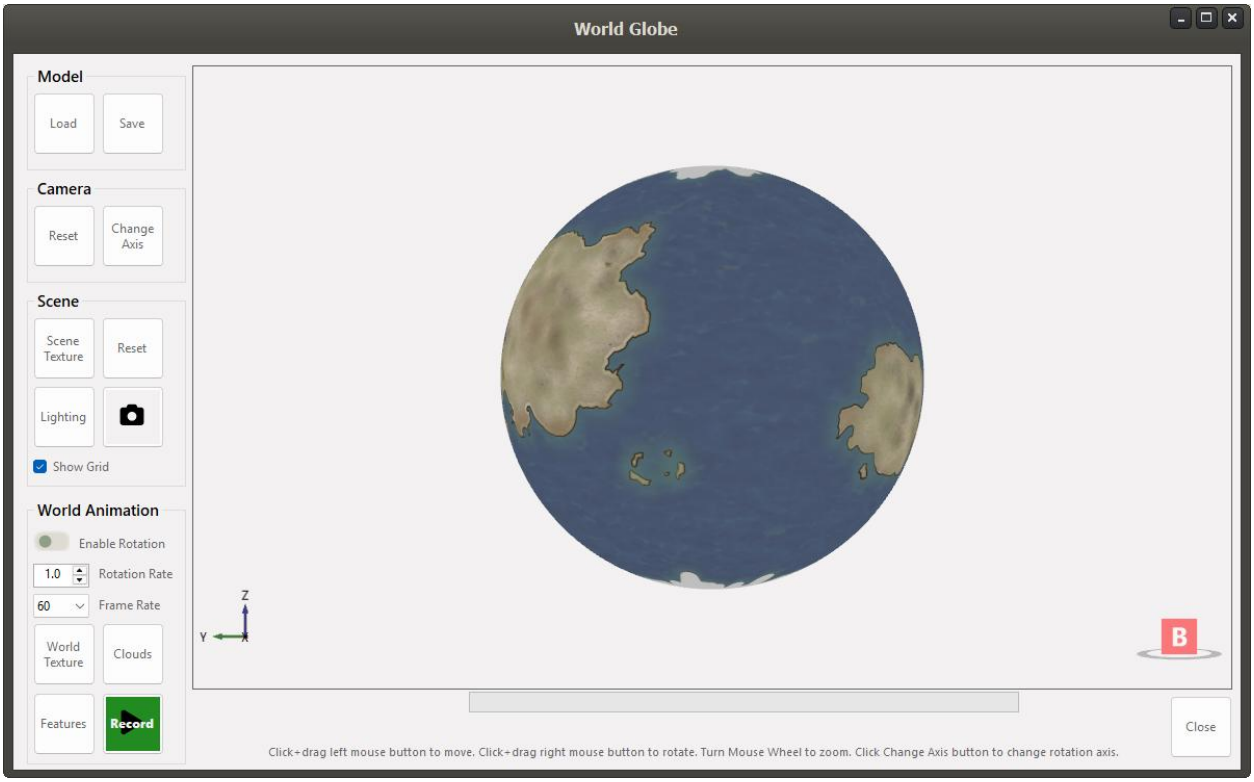
Creating the World Globe Scene

To begin creating a World Globe scene, you start with your map. Set the background texture and color, the ocean texture and color, and paint your landforms. You can use the landform generation function to quickly create landforms or modify the landforms you've painted. Apply shading and color to the landforms to create areas of mountains, vegetation, and snow and ice. Placing symbols on the map will usually not look very good, as they are 2D images, and don't look right on a 3D globe. Use color and shading to indicate areas of mountains, deserts, forest, etc. rather than symbols.

After opening the World Globe View, you can begin creating the scene. Starting from this basic map:



The World Globe View will start out looking like this:

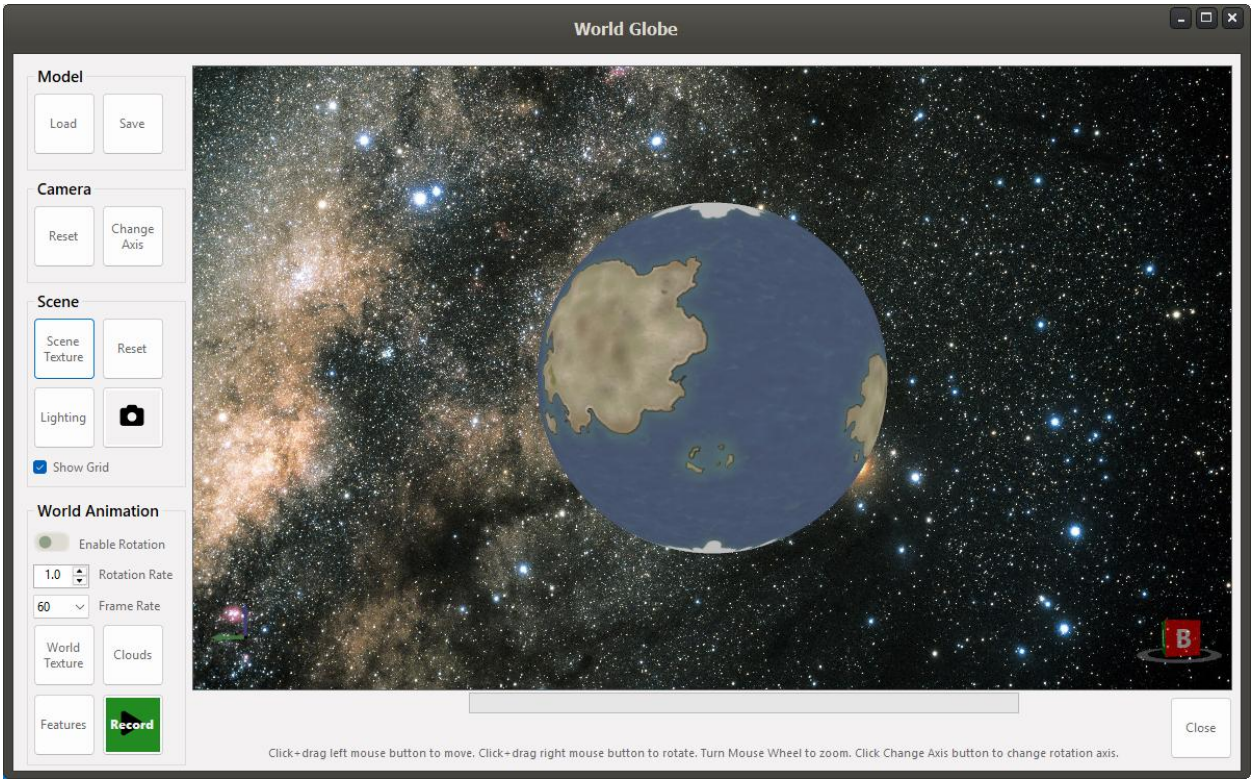


If you want to rotate the World Globe or see it from other angles, you can use the mouse to click and drag the globe within the view. Click the Change Axis button to change the axis of rotation. When the Z axis is pointing up, holding down the right mouse button and dragging the mouse will rotate the globe around its center (the polar axis).

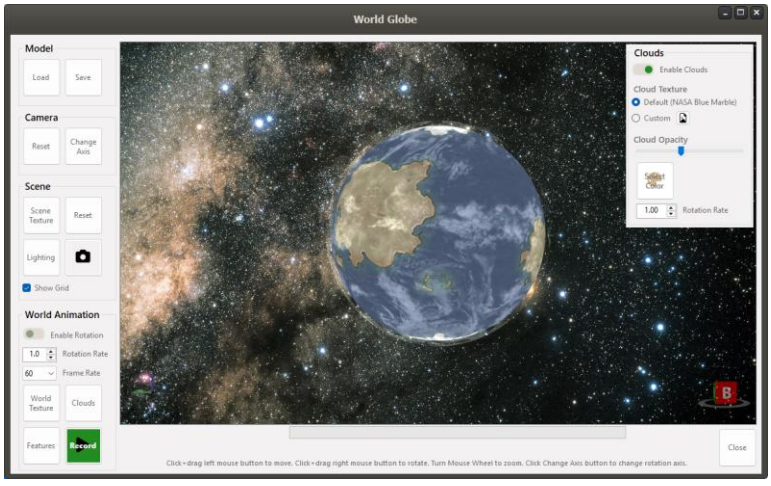
Note that in the initial rectangular map, the polar ice caps appear much larger than they do on the globe. This is caused by the distortion required to display a spherical globe as a rectangular map. You can see the same effect when viewing a rectangular map of the Earth. Greenland, since it is close to the North Pole, appears much larger than its actual size, which you can see when looking at a globe of the Earth. Keep that in mind when painting your landforms. Ice caps or land areas close to the northern and southern poles of your world will have to be drawn much larger than you might think to appear correctly on the World Globe View.

Once you have the World Globe looking the way you want, you can create a dramatic scene of your world floating in space. Start by adding a background to the scene.

1. Click the Scene Texture button in the Scene panel.
2. A File Open dialog will open that lets you select a background texture (picture).
3. Realm Studio includes several starfield textures in the Assets/Textures/Starfields directory. You can choose one of those, download one, or paint your own.
4. Select the texture for the background that you like, and it will be displayed behind your world globe. It might look like the following:



After setting the background texture of the World Globe scene, you can click the Clouds button to add a cloud texture to the world. You can choose the default cloud texture, which are from the NASA Blue Marble photos, or choose custom cloud textures in the **`Assets/Textures/Clouds`** folder, or cloud textures that you download or paint yourself. The Clouds panel lets you change the opacity and tint/color of the cloud texture, so if you want to create the look of toxic green clouds of poisonous gases, you can do that! You can also can the rotation rate of the clouds, so they can appear to be moving faster or slower than the world is rotating. The following picture shows the World Globe with NASA Blue Marble clouds applied.

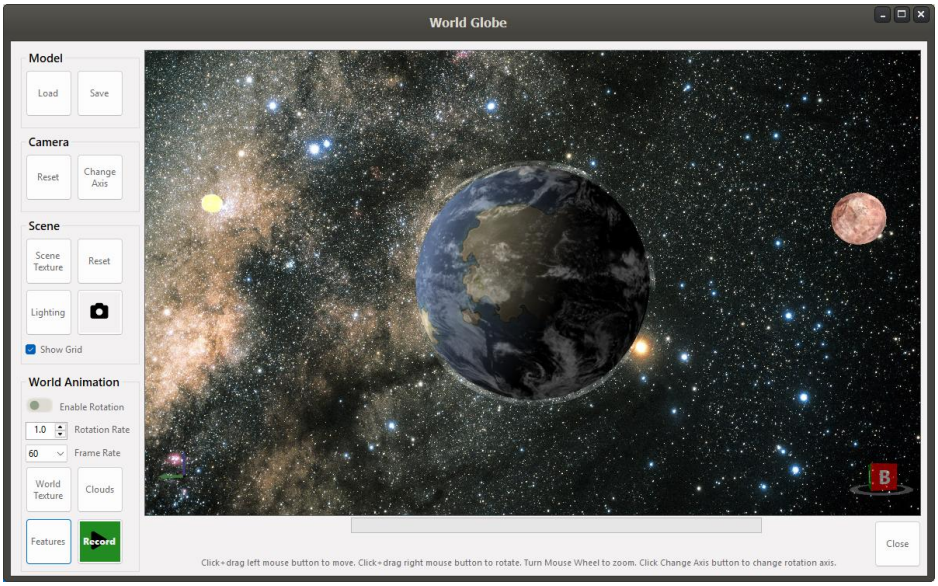


Click the Clouds button again to close the Clouds panel.

Clicking the Features button will open the Features panel. The Features panel has tabs you can click to display controls for adding various features to the scene, like a local star, an Atmospheric effect, several visual effects implemented by GPU shaders (blur, bleach bypass, bloom, sepia tone, and grayscale), and to add one or more moons orbiting your world.

Rather than going over all the Features in detail, you are encouraged to experiment to understand what they do and how they affect the look of your scene. Here are some pictures of the previous scene with a local sun and a moon added and various effects applied.

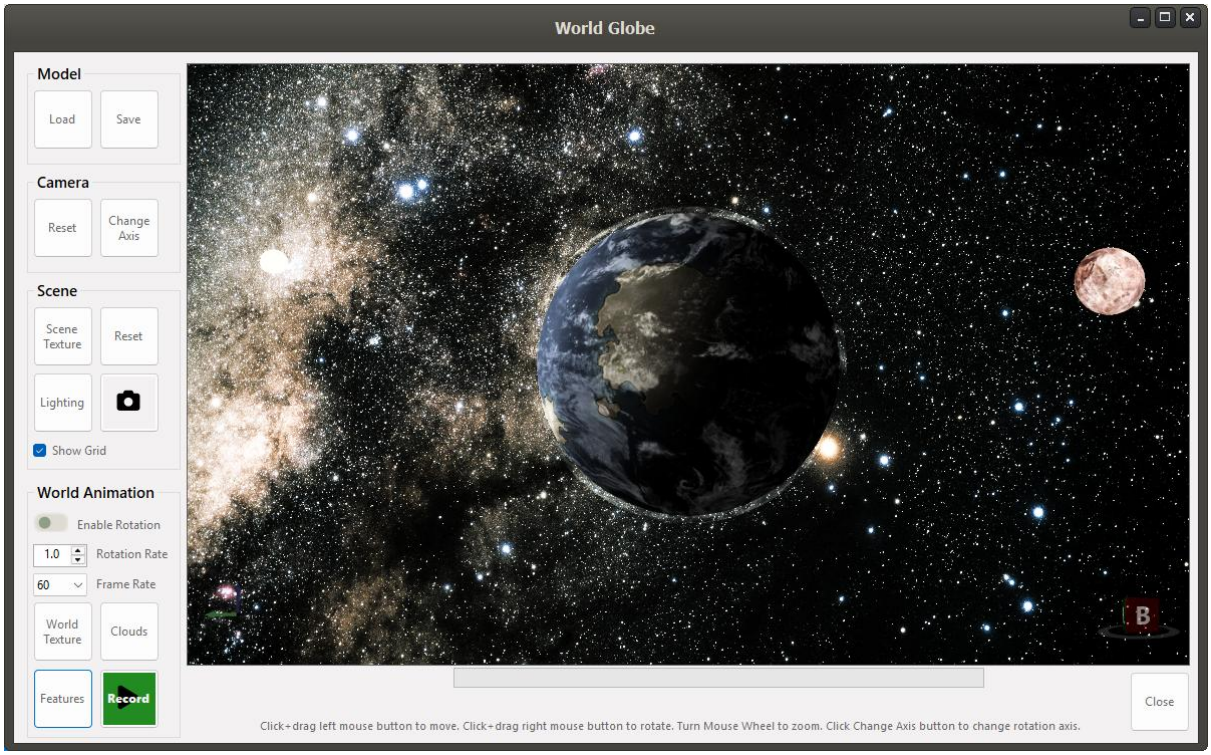
The scene with no effects applied:



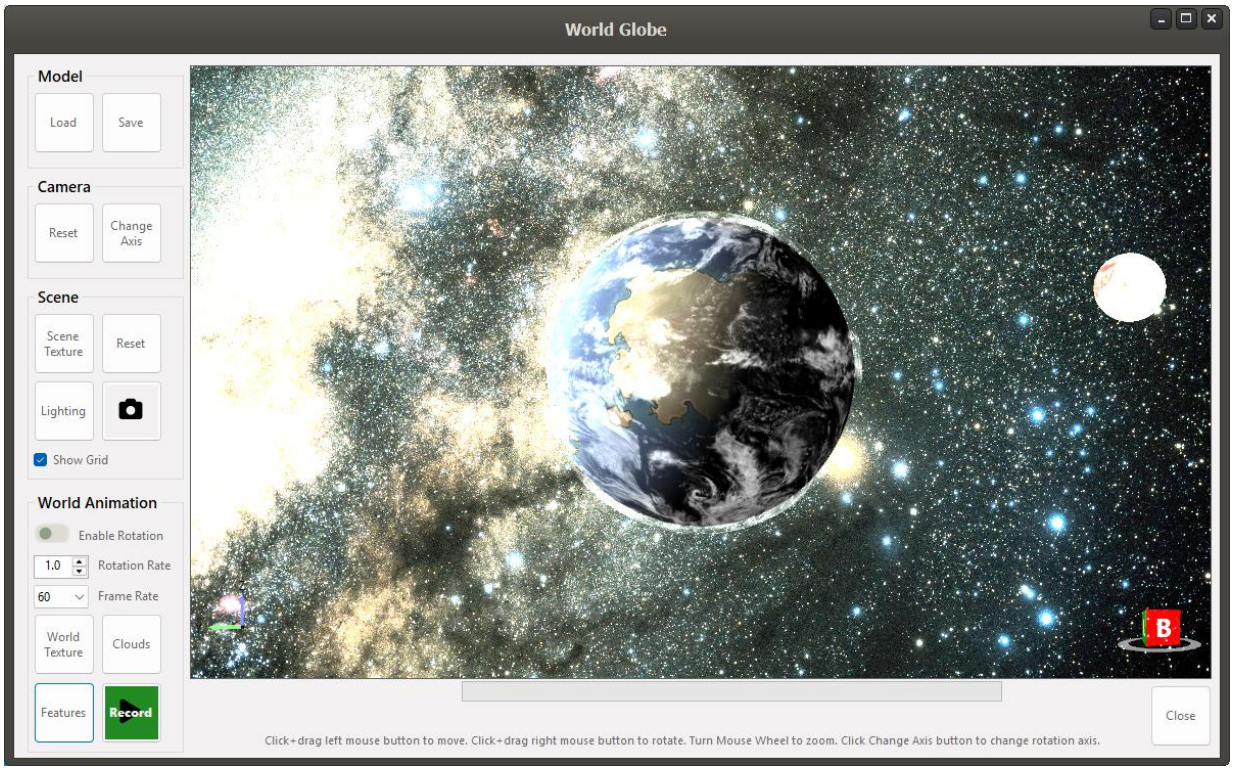
The scene with a blur effect:



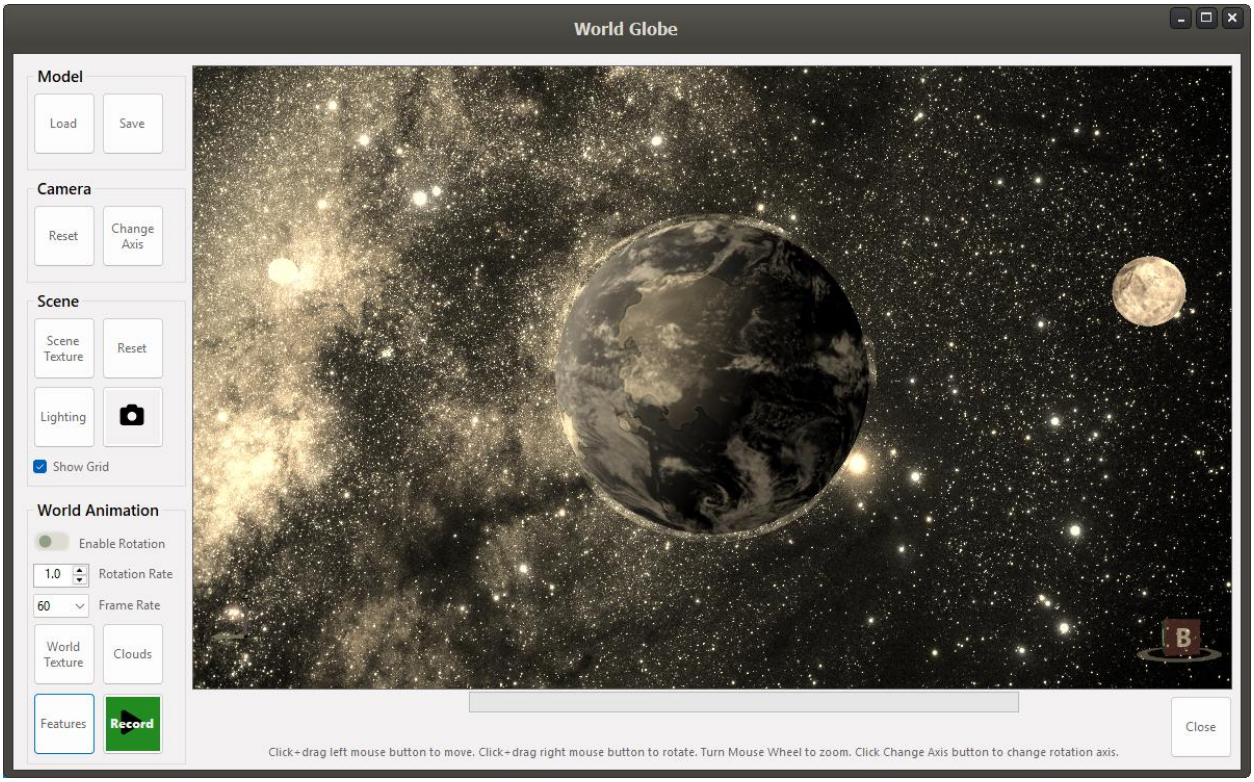
The scene with a cinematic bleach bypass effect applied:



The scene with a bloom effect:



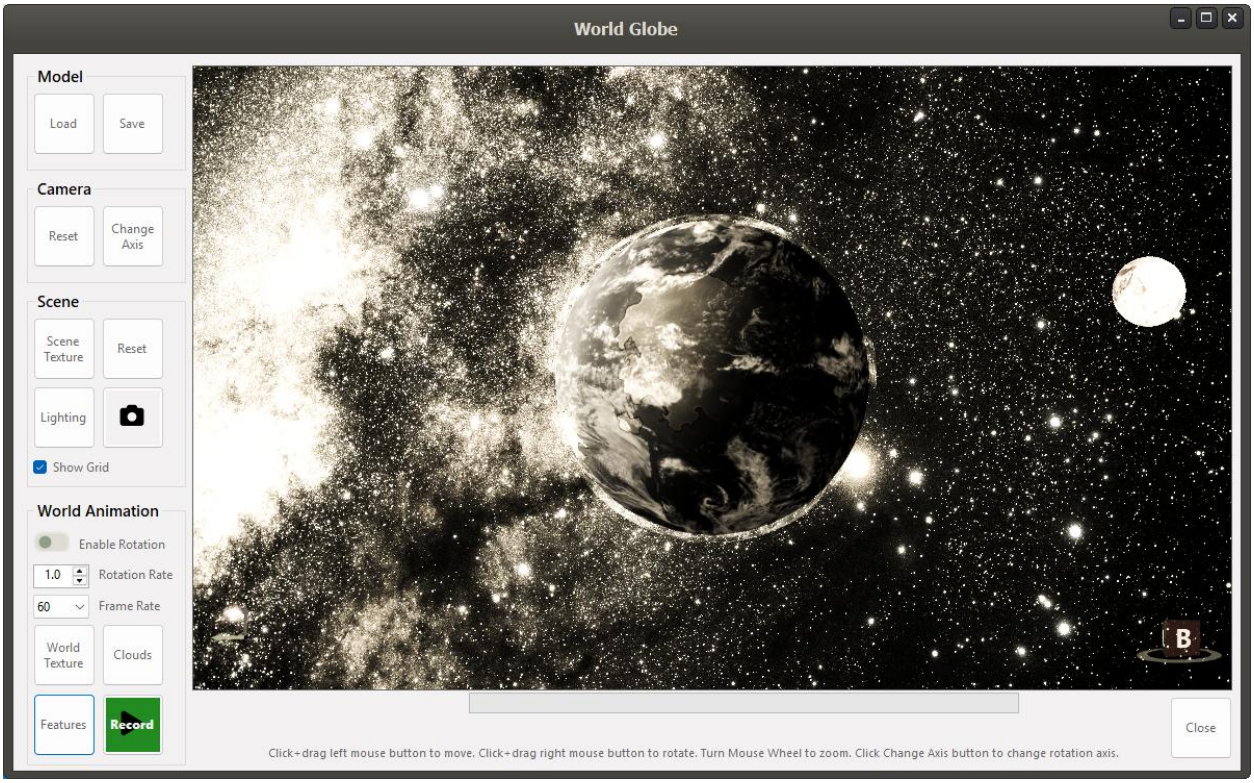
The scene with a sepia tone effect:



And finally, the scene with a grayscale (black and white) effect applied:



The various effects can be combined to achieve a unique look, like this one combining bleach bypass, bloom, and sepia toning:



You can experiment with the effects and combinations to achieve the look you want. Some combinations result in weird effects or for the scene to go completely white, so try different combinations and strengths (using the sliders) to find the right look.

Taking a Picture of your Scene

Use the Camera button to take and save a picture of your scene as it is currently shown. A file dialog will open, allowing you to choose the format and name of the picture. The scene will be saved in the format and with the name you choose. Here's the previous scene saved as a PNG file:



Recording AVI Video of the Scene

Realm Studio lets you create a video of your scene with your world globe rotating in space. Just click the green Record button, and an AVI video of your world rotating in space. All the effects that you've applied to the scene and other changes (local star, planetary ring, moons) are recorded in the video. The progress bar at the bottom of the view and a frame counter let you know the progress of the video.

When you click the Record button, the button turns red and changes to a Stop button. You can stop recording at any time, but there is no functionality to resume recording once stopped. While the video is being recorded, all the controls on the 3D World Globe View are disabled, other than the Stop button, so the scene cannot be changed while video is being recorded.

To preview what the video will look like, you can click the Enable Rotation switch. The scene will display the World Globe rotating without recording video.

By default, Realm Studio will record a video with one full rotation (360 degrees) of your world, at 60 frames per second (3600 total frames). You can change the rotation rate to speed up or slow down the rotation speed of the world and change the frame rate using the controls.

If you stop recording the video before all frames have been recorded, you can save the partial video. Realm Studio will present a message box asking if you want to save it. Choose Yes, and a File Save dialog will be shown allowing you to save it.

When all frames have been recorded, Realm Studio will present a File Save dialog allowing you to save the complete video. At this time, the video can only be saved in AVI format. Other options might be implemented (like MPEG or GIF) in the future. If you need the video in other formats, there are many tools available for converting from AVI to other formats.

Viewing Other Worlds

The primary purpose of the 3D World Globe View is to show your world map as a world globe; however, the 3D World Globe View can wrap any equirectangular texture onto the sphere, so you can view any planet or star in the viewer, if you have an equirectangular picture of it. Clicking the World Texture will open a File Open dialog to select the picture you want to show as a globe. You can download pictures, paint your own, or use one of many included in the Assets/Textures/Planet folder

A Final Note on the 3D World Globe View

There is no function currently to save the scene you have set up. If you close the 3D World Globe View before you have taken the snapshot(s) or recorded the video you want, you will have set up the view again. A function to save the scene may be implemented in a later release.

Copyright and Licensing Information

Realm Studio is licensed under the GNU Public License (GPL) 3.0. The full text of the license is available in the source code in the LICENSE.txt file and in the LICENSE.rtf file, and on the GNU website at https://www.gnu.org/licenses/.

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Technical Information

Realm Studio Implementation

Realm Studio is a Windows Forms (Winforms) application written in C# using Visual Studio 2022. The C# language version is 13.0. For graphics rendering, it uses the SkiaSharp implementation of the Skia graphics engine (version 3.119.1 as of this writing). Graphics rendering is accelerated by using the SkiaSharp SKGLControl, which is a Skia layer on top of the OpenTK.GLControl. OpenTK is an OpenGL library that allows for accelerated graphics rendering using the computer GPU.

Realm Studio also makes use of the AForge, Clipper2, and Delaunator libraries for some of the more complex geometry operations, as well as my own implementation of the Moore-Neighborhood algorithm for quickly finding the boundaries (contours) of landforms and water features. It uses the SimplexNoise library for creating simplex noise when generating random landform and water feature shapes.

Realm Studio uses icons from Font Awesome (via the FontAwesome.Sharp library) throughout the application, and libraries from Cyotek for creating the color selection dialog and Pixel Editor. It uses controls from Real.Taizor to create a more modern (and hopefully more attractive) appearance for the application than a typical Windows Forms application.

Logging is implemented using the Apache log4net library.

Realm Studio requires .NET 8.0 to run. The minimum Windows version is Windows 10 (version 10.0.26100.0). The Realm Studio installer will install the .NET libraries on your computer if they are not already installed.

The Layers of a Realm Studio Map

A Realm Studio map is constructed from 29 layers that are composited to create the final map. The layers are defined in code like this:

```
public const int BASELAYER = 0;
public const int OCEANTEXTURELAYER = 1;
public const int OCEANTEXTUREOVERLAYLAYER = 2;
public const int OCEANDRAWINGLAYER = 3;
public const int WINDROSELAYER = 4;
public const int ABOVEOCEANGRIDLAYER = 5;
public const int LANDCOASTLINELAYER = 6;
public const int LANDFORMLAYER = 7;
public const int LANDDRAWINGLAYER = 8;
public const int WATERLAYER = 9;
public const int WATERDRAWINGLAYER = 10;
public const int BELOWSYMBOLSGRIDLAYER = 11;
public const int PATHLOWERLAYER = 12;
public const int SYMBOLLAYER = 13;
public const int PATHUPPERLAYER = 14;
public const int REGIONLAYER = 15;
public const int REGIONOVERLAYLAYER = 16;
public const int DEFAULTGRIDLAYER = 17;
public const int BOXLAYER = 18;
public const int LABELLAYER = 19;
public const int OVERLAYLAYER = 20;
public const int FRAMELAYER = 21;
public const int MEASURELAYER = 22;
public const int DRAWINGLAYER = 23;
public const int VIGNETTELAYER = 24;
public const int SELECTIONLAYER = 25;
public const int HEIGHTMAPLAYER = 26;
public const int WORKLAYER = 27;
public const int WORKLAYER2 = 28;
```

File Formats

Realm Studio stores maps, themes, collections, and all its other data as XML. You can examine and edit the files using a text or XML editor, but it is not recommended that you directly edit any XML file created by Realm Studio, as irrecoverable damage can easily be caused. Realm Studio map files (those with a **`.rsmapx`** extension) are complex. Editing them with a text editor or an XML editor is likely to render the map unreadable by Realm Studio, potentially causing you to lose a lot of work.

Boxes

Boxes used as frames for labels have two parts: The **`.png`** file containing the image for the box and an XML file that includes the path to the.**`png`** file, a path to itself, and a series of tags defining the "center" of the box. The data for the center of the box is required, because boxes are drawn as "nine-patch" images, which allows them to be drawn at different sizes while minimizing the pixelation and distortion usually seen when bitmaps are drawn at different sizes and aspect ratios than for the original image. The box XML looks like this:

```
<?xml version="1.0" encoding="utf-8"?>
<mapbox xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns="RealmStudio">
  <BoxName>Avoro (1) Celtic</BoxName>
```

```
  <BoxBitmap>
    <Palette />
  </BoxBitmap>
  <BoxBitmapPath>RealmStudio\Assets\Boxes\Avoro (1) Celtic.png</BoxBitmapPath>
  <BoxXmlFilePath>RealmStudio\Assets\Boxes\Avoro (1) Celtic.xml</BoxXmlFilePath>
  <BoxCenterLeft>11</BoxCenterLeft>
  <BoxCenterTop>11</BoxCenterTop>
  <BoxCenterRight>11</BoxCenterRight>
  <BoxCenterBottom>11</BoxCenterBottom>
</mapbox>
```

When the box XML is first loaded by Realm Studio, the paths are rewritten to include the full path to the `.png` and XML files.

Frames

Frames, like boxes, are rendered as nine-patch images and are represented to Realm Studio in the same way as boxes. They also have two parts: The `.png` file containing the image for the frame and an XML file that includes the path to the.`png` file, a path to itself, and a series of tags defining the "center" of the frame.

```
<?xml version="1.0" encoding="utf-8"?>
<mapframe xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns="RealmStudio">
<FrameName>Arrow_frame_color</FrameName>
<FrameBitmapPath>RealmStudio\Assets\Frames\Arrow_frame_color.png</FrameBitmapPath>
<FrameXmlFilePath>RealmStudio\Assets\Frames\Arrow_frame_color.xml</FrameXmlFilePath>
  <FrameCenterLeft>511.5</FrameCenterLeft>
  <FrameCenterTop>511.5</FrameCenterTop>
  <FrameCenterRight>511.5</FrameCenterRight>
  <FrameCenterBottom>511.5</FrameCenterBottom>
</mapframe>
```

Label Presets

Label presets are XML files containing the data needed to set the font, font color, and other characteristics of a label. The XML file also contains the name of the Theme associated with the preset. Because different themes may contain label presets with the same name, the label preset XML files are named using a GUID. A label preset XML file looks like this:

```
<?xml version="1.0" encoding="utf-8"?>
<LabelPreset xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" IsDefault="true" xmlns="RealmStudio">
  <LabelFontString>Palatino Linotype, 36pt</LabelFontString>
  <PresetXmlFilePath>Realm Studio\Assets\LabelPresets\9bccb3a6-a5c2-446a-9420-
03463498b637.mclblprst</PresetXmlFilePath>
  <LabelPresetName>Region</LabelPresetName>
  <LabelPresetTheme>Medieval Quest</LabelPresetTheme>
  <LabelColor>-1158551863</LabelColor>
  <LabelOutlineColor>-2127155679</LabelOutlineColor>
  <LabelOutlineWidth>2</LabelOutlineWidth>
  <LabelGlowColor>-1</LabelGlowColor>
```

```
    <LabelGlowStrength>0</LabelGlowStrength>
  </LabelPreset>
```

Name Generators and Name Base Files

Name Generators are comma-separated values (`.csv`) files with two columns. When generating names, a randomly selected value from each of the two columns is combined to create the random name. Each entry in the first column of the contains a "%" character. When combining values from the first column and second column, Realm Studio replaces the "%" character in the selected entry from the first column with the random selection from the second column.

Realm Studio has extended this basic functionality by allowing Name Generator files with an empty second column. When the second column is empty, Realm Studio will generate a value from selected languages and replace the "%" character in the selected entry from the first column with the generated value from the selected languages. Generating names using the Bodies of Water name generator uses this method.

Name Base files are text (`.txt`) files containing one or more lists of names, each of which preceded by a header that describes the language of the names, the minimum and maximum name length, what characters are repeatable, and the proportion of single-word names. Realm Studio does not make use of the information from the Name Base file header for name generation. It simply selects a random name from the list based on the languages selected in the Name Generator Configuration dialog.